Business Statistics

J. Alejandro Gelves

2024-09-06

Table of contents

Introduction

"Whatever you would make habitual, practice it; and if you would not make a thing habitual, do not practice it, but accustom yourself to something else." – *Epictetus*

This course companion aims to build your mastery of statistics and its applications in R through consistent practice. Each chapter starts with key concepts to steer your learning, followed by problems designed to solidify these ideas through hands-on work. For extra R support, see Grolemund (2014). I thank my business statistics students and Haipei Liu for their valuable comments and reviews—all remaining errors are mine. Take your time, enjoy the journey, and make practice a habit!

Why R?

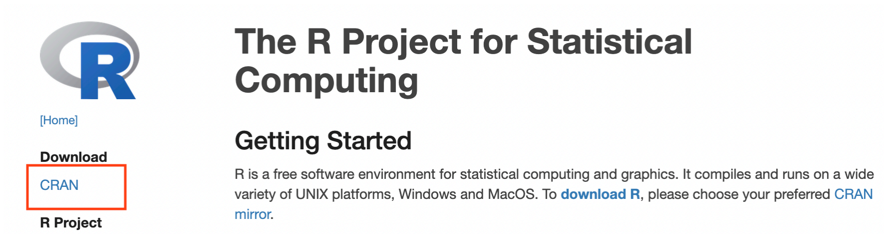
We will be using R to apply the lessons we learn in BUAD 231. R is a language and environment for statistical computing and graphics. There are several advantages to using the R software for statistical analysis and data science. Some of the main benefits include:

- R is a **powerful and flexible programming language** that allows users to manipulate and analyze data in many different ways.
- R has a large and **active community of users**, who have developed a wide range of packages and tools for data analysis and visualization.
- R is **free and open-source**, which makes it accessible to anyone who wants to use it.
- R is **widely used** in academia and industry, which means that there are many resources and tutorials available to help users learn how to use it.
- R is well-suited for working with **large and complex datasets**, and it can handle data from many different sources.
- R can be **easily integrated** with other tools and software, such as databases, visualization tools, and machine learning algorithms.

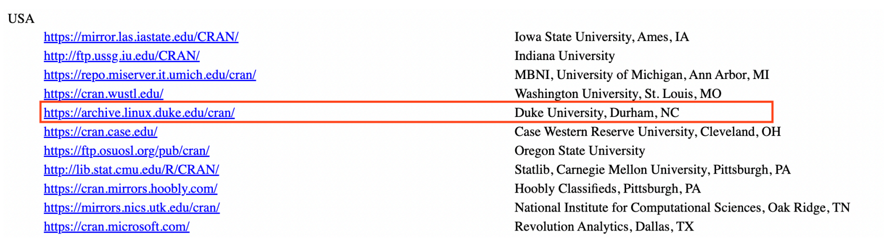
Overall, R is a powerful and versatile tool for data analysis and data science, and it offers many benefits to users who want to work with data.

Installing R.

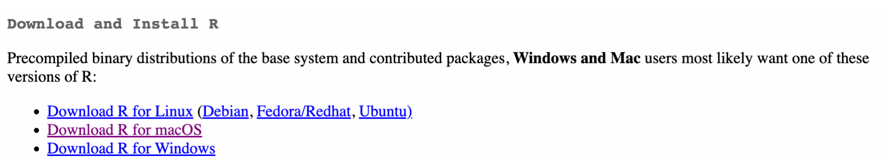
To install R, visit the R webpage at https://www.r-project.org/. Once in the website, click on the CRAN hyperlink.



Here you can select the CRAN mirror. Scroll down until you see USA. You are free to choose any mirror you like, I recommend using the Duke University mirror.



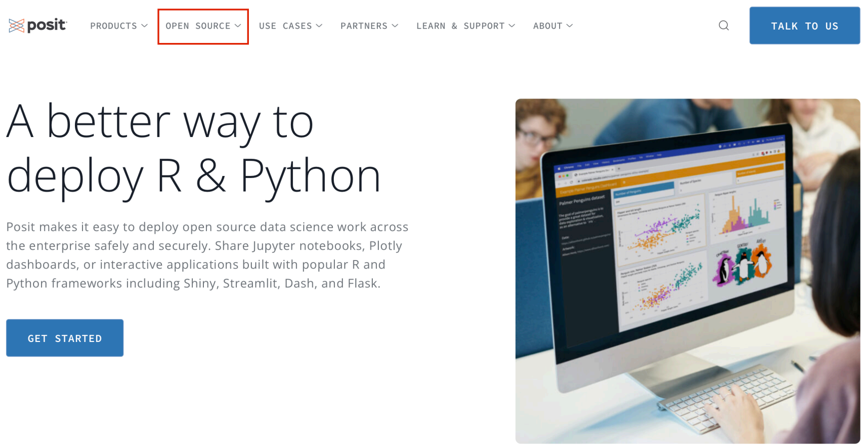
Once you click on the hyperlink, you will be prompted to choose the download for your operating system. Depending on your operating system, choose either a Windows or Macintosh download.



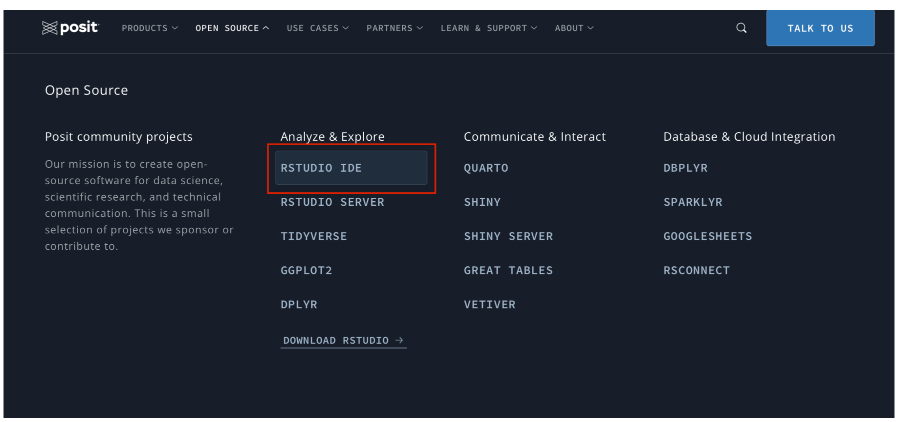
Follow all prompts and complete installation.

Installing RStudio

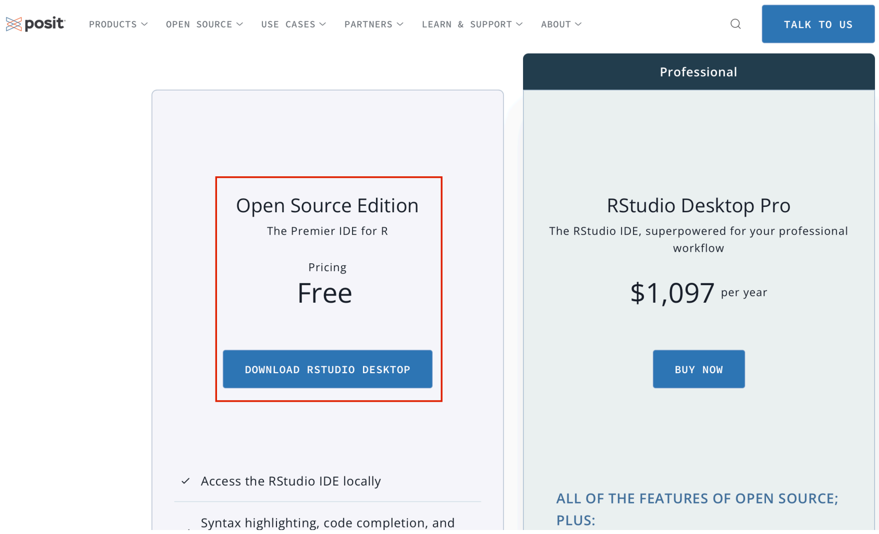
Visit the Posit website at https://posit.co. Once on the website, hover to the top of the screen and select "Open Source" from the drop down menus.



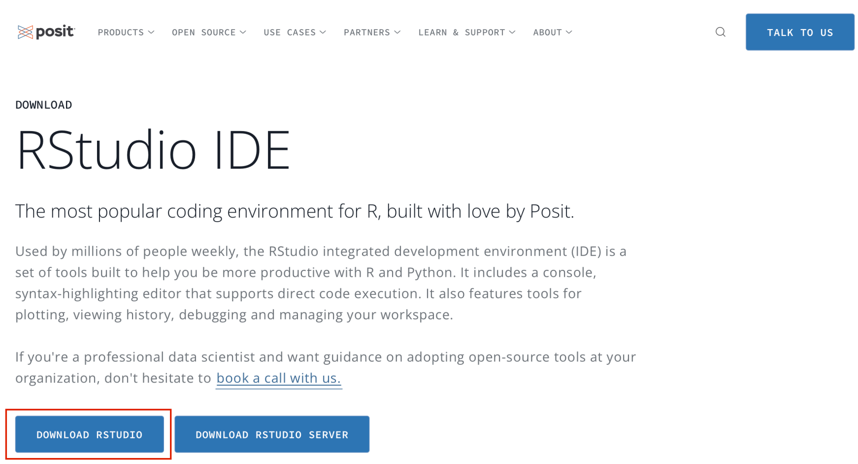
Next, choose "R Studio IDE".



Scroll down until you see the products. You want to download "RStudio Desktop" and make sure it is the free version.



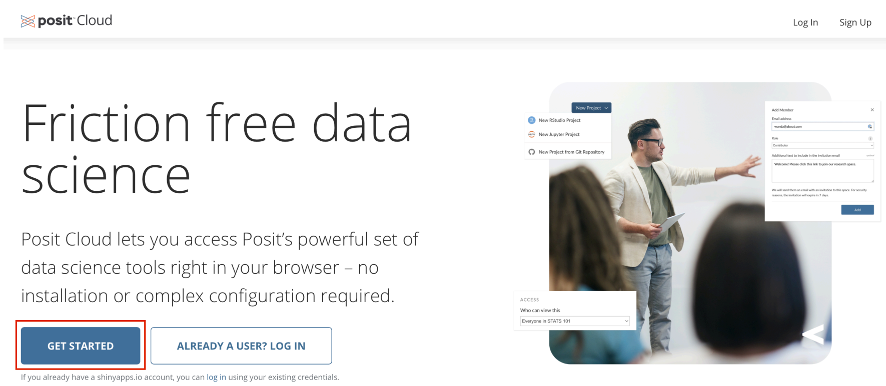
Finally, select "Download RStudio" and follow the instructions for installation.



It is important to note that RStudio will not work if R is not installed. You can think of R as the engine and RStudio as the interface.

Posit Cloud

If you do not wish to install R, you can always use the cloud version. To do this, visit https://posit.cloud/. On the main page click on the "Get Started" button.



Choose the "Cloud Free" option and log in using your Google credentials (if you have a Google account) or sign up if you want to create a new account.

1 Descriptive Stats I

Understanding the nature and classification of data is crucial for effective analysis and decision-making. Data are the building blocks of insights, providing a foundation for businesses, researchers, and policymakers to make informed choices. How we collect and categorize data significantly impacts how it is analyzed and interpreted. Below we highlight key types of data and their unique characteristics to help you get started with data analysis.

1.1 Data and Types of Data

Let's start by defining data and the types of data structures that you might encounter when conducting analysis. **Data** are facts and figures collected, analyzed and summarized for presentation and interpretation. Data can be classified as:

- **Cross Sectional Data** refers to data collected at the same (or approximately the same) point in time. *Ex: NFL standings in 1980 or Country GDP in 2015.*
- **Time Series Data** refers to data collected over several time periods. *Ex: U.S. inflation rate from 2000-2010 or Tesla deliveries from 2016-2022.*
- **Structured Data** resides in a predefined row-column format (tidy). *Ex: spreadsheet data.*
- **Unstructured Data** does not conform to a pre-defined row-column format. *Ex: Text, video, and other multimedia.*

Example: Consider a retail store analyzing its sales performance. If the store collects data on the total revenue generated *by each location* on Black Friday, it is cross-sectional data. On the other hand, if the store tracks *weekly sales for the past year* to observe trends, it is time series data. Structured data, like sales figures stored in *spreadsheets*, allows for easy comparison and analysis. Meanwhile, customer feedback gathered from *social media posts and video reviews* represents unstructured data, requiring advanced tools to extract meaningful insights.

1.2 Data Sets

A **data set** contains all data collected for a particular study. Understanding the structure the data set aids in communicating findings to stakeholders clearly and effectively. Data sets are composed of:

- **Elements** are the entities on which data are collected. *Ex: Football teams, countries, and individuals.*
- **Variables** are a set of characteristics collected for each element. *Ex: Goals scored, GDP, weight.*
- **Observations** are the set of measurements obtained for a particular element. *Ex: Salah, 20 (goals), 15 (assists). US, 2.3 (inflation), 4.5% (federal interest rate).*

Elements	Variable 1	Variable 2
Element 1	#	#
Element 2	#	#
Element 3	#	#
...	...	...

Example: Consider the dataset on electric vehicles (EVs) displayed below:

Make	Model	Range_km	TopSpeed_kmh	Price_pounds	Charge_kmh
Tesla	Model 3	415	201	39990	690
BYD	ATTO 3	330	160	37195	370
Tesla	Model 3 Long Range Dual Motor	500	201	49990	770
Tesla	Model Y Long Range Dual Motor	435	217	52990	670
BYD	SEAL 82.5 kWh AWD Excellence	490	180	48695	540
Tesla	Model Y	350	217	44990	580
MG	MG4 Electric 64 kWh	360	160	29495	630
Renault	Scenic E-Tech EV87 220hp	490	170	40995	510
BYD	DOLPHIN 60.4 kWh	340	160	30195	340
BMW	i4 eDrive40	515	190	57890	800

In this data set, each row represents an electric vehicle model, making the elements the specific EV models rather than the manufacturers. The variables collected for each model include:

- Make: The manufacturer of the EV.
- Model: The specific name of the EV model.
- Range_km: Driving range in kilometers on a full charge.
- TopSpeed_kmh: Maximum speed in km/h.
- Price_pounds: Price in pounds (£).
- Charge_kmh: Charging speed in kilometers per hour.

An example observation is “Tesla Model 3,” with the following data: Make: Tesla, Model: Model 3, Range_km: 415, TopSpeed_kmh: 201, Price_pounds: 39,990, Charge_kmh: 690.

1.3 Scales of Measurement

Understanding scales of measurement is crucial for analyzing and interpreting data effectively in business. By distinguishing between categorical (e.g., marital status, satisfaction ratings) and numerical data (e.g., profits, prices), you'll know what methods to use for analysis. Knowing whether data is nominal, ordinal, interval, or ratio ensures your analysis and conclusions are accurate and relevant.

The **scales of measurements** determine the amount and type of information contained in each variable. In general, variables can be classified as **categorical** or **numerical**.

- **Categorical** (qualitative) data includes labels or names to identify an attribute of each element. Categorical data can be **nominal** or **ordinal**.
  - With **nominal** data, the order of the categories is arbitrary. Numbers (if used) serve only as identifiers. Common statistics used are based on counting frequencies and the mode (most common category) *Ex: Marital Status, Race/Ethnicity, or NFL division.*
  - With **ordinal** data, the order or rank of the categories is meaningful, but intervals between ranks are not necessarily equal or quantifiable. Common statistics are based on median, percentiles, or rank-order correlations. *Ex: Rating, Difficulty Level, or Spice Level.*
- **Numerical** (quantitative) include numerical values that indicate how many (discrete) or how much (continuous). The data can be either **interval** or **ratio**.
  - With **interval** data, the distance between values is expressed in terms of a fixed unit of measure. The zero value is arbitrary and does not represent the absence of the characteristic. Ratios are not meaningful. Common statistics are the arithmetic mean, standard deviation, or the Pearson correlation, *Ex: Temperature or Dates.*
  - With **ratio** data, the ratio between values is meaningful. The zero value is not arbitrary and represents the absence of the characteristic. All statistics are permissable including geometric mean, and the coefficient of variation. *Ex: Prices, Profits, Wins.*

Example: Let's keep using the EV example. Consider the new data set below:

Car	Brand	Range	Rating	Year
Mustang Mach-E	Ford	217	4	2021
E-Tron GT	Audi	250	3	2020
...	...	...	...	
Volt EV	Chevrolet	124	2	2021

The variables can be classified as follows:

- Car (Categorical - Nominal), consists of names of cars, which are labels used to identify each row. The order of these names does not matter, making it nominal data.
- Brand (Categorical - Nominal) represents the manufacturer of the car (e.g., Ford, Audi). These are labels with no inherent order, making it nominal data.
- Range (Numerical - Ratio), refers to the car's driving range in miles. It is numerical and ratio because it has a meaningful zero (a car with zero range cannot move), and ratios are meaningful (e.g., a car with 250 miles range has double the range of one with 125 miles).
- Rating (Categorical - Ordinal) represents a rank or score (e.g., 4, 3, 2). The order matters, as higher ratings indicate better performance. However, the intervals between ratings are not consistent, so it is ordinal data.
- Year (Numerical - Interval) represents a point in time. While numerical, it is interval data because the zero point is arbitrary (e.g., year 0 does not indicate the "absence" of time), and ratios are not meaningful (e.g., 2020 is not "twice as late" as 1010).

1.4 Exercises

The following exercises will help you test your knowledge on the Scales of Measurement. In these exercises you will:

- Identify numerical and categorical data.
- Classify data according to their scale of measurement.

Answers are provided below. Try not to peek until you formulate your own answer and double checked your work for any mistakes.

Exercise 1

A bookstore has compiled data set on their current inventory. A portion of the data is shown below:

Title	Price	Year Published	Rating
Frankenstein	5.49	1818	4.2
Dracula	7.60	1897	4.0
...	...	...	...
Sleepy Hollow	6.95	1820	3.8

1. Which of the above variables are categorical and which are numerical?

Answer

- *The "Title" variable represents the names of books. Therefore, this is a categorical variable.*
- *"Price" represents the cost of each book in a numeric format, making it a numerical variable.*
- *"Year Published" indicates the publication year of each book. It is numerical.*
- *If "Rating" represents a numerical score based on a continuous scale (e.g., average user ratings on a platform like Goodreads), it is numerical because arithmetic operations like averaging or comparing differences are meaningful. If "Rating" represents predefined categories (e.g., "Excellent," "Good," "Fair," "Poor") or is interpreted as ranks without meaningful differences between values, it would be categorical.*

2. What is the measurement scale of each of the above variable?

Answer

- *The measurement scale is nominal for "Title" since these are labels used to identify each book and do not have a numerical meaning or order.*
- *If "Rating" represents a score (e.g., 4.2, 4.0) given to each book, it is numerical and could be considered interval data because the scale represents a meaningful difference, but it may not have an absolute zero or meaningful ratios (e.g., a book rated 4.0 is not "twice as good" as one rated 2.0).*
- *"Price" is a measurable quantity with a meaningful zero (e.g., a book priced at $0 means it is free), making it ratio data.*
- *"Year" is interval data because the zero point is arbitrary (year 0 does not represent the absence of time) and differences between years are meaningful (e.g., 1897 - 1818 = 79 years).*

Exercise 2

A car company tracks the number of deliveries every quarter. A portion of the data is shown below:

Year	Quarter	Deliveries
2016	1	14800
2016	2	14400
...	...	...
2022	3	343840

1. What is the measurement scale of the Year variable? What are the strengths and weaknesses of this type of measurement scale?

Answer

The variable Year is measured on the interval scale because the observations can be ranked, categorized and measured when using this kind of scale. However, there is no true zero point so we cannot calculate meaningful ratios between years.

2. What is the measurement scale for the Quarter variable? What is the weakness of this type of measurement scale?

Answer

The variable Quarter is measured on the ordinal scale, even though it contains numbers.

3. What is the measurement scale for the Deliveries variable? What are the strengths of this type of measurement scale?

Answer

The variable Deliveries is measured on the ratio scale. It is the strongest level of measurement because it allows us to categorize and rank the data as well as find meaningful differences between observations. Also, with a true zero point, we can interpret the ratios between observations.

Exercise 3

A coffee shop chain collects customer feedback data for analysis. For each visit/transaction recorded in their database, the following variables are stored:

- Customer ID: A unique identifier assigned to each registered customer (e.g., C001234, C005678).
- Satisfaction Level: Customers rate their overall experience as 1 (Very Dissatisfied), 2 (Dissatisfied), 3 (Neutral), 4 (Satisfied), or 5 (Very Satisfied).

- Visit Duration: The time spent in the store in minutes (e.g., 15, 42, 8).
- Beverage Type: The main drink ordered (e.g., Latte, Cappuccino, Americano, Tea, Other).

Answer

- *Customer ID: Categorical, Nominal. These are arbitrary labels used only for identification. There is no order or numerical meaning; you can only determine if two IDs are the same or different.*
- *Satisfaction Level: Categorical, Ordinal. The ratings have a clear order (higher numbers indicate greater satisfaction), but the intervals between levels are not necessarily equal (the difference between 4 and 5 may not be perceived the same as between 1 and 2). Ratios are not meaningful.*
- *Visit Duration: Numerical, Ratio. Time in minutes has a true zero (0 minutes means no time spent), equal intervals, and meaningful ratios (e.g., 60 minutes is twice as long as 30 minutes). All arithmetic operations are valid.*
- *Beverage Type: Categorical, Nominal. These are unordered categories/labels for different drinks. You can group or count them, but there is no natural order or numerical relationship between categories like "Latte" and "Tea".*

2 Descriptive Stats II

Understanding and visualizing data distributions is a fundamental step in data analysis. It provides critical insights into the underlying characteristics of the data, which directly impact decision-making, model performance, and interpretation. Below, we introduce tabular and visual techniques to describe your data.

2.1 Frequency Distributions (Categorical)

A **frequency distribution** is perhaps the most valuable tool for summarizing categorical data. It illustrates with a table the number of items within distinct, non-overlapping categories. An alternative known as the **relative frequency** quantifies the proportion of items in each category relative to the total number of observations. You can calculate it by taking the frequency of a particular class (f_i), and dividing it by the number of observations n. Relative frequency helps contextualize the data by highlighting the significance of each category compared to the whole.

Example: Consider data on students' answers to the question, what is your favorite food? You can see the data below:

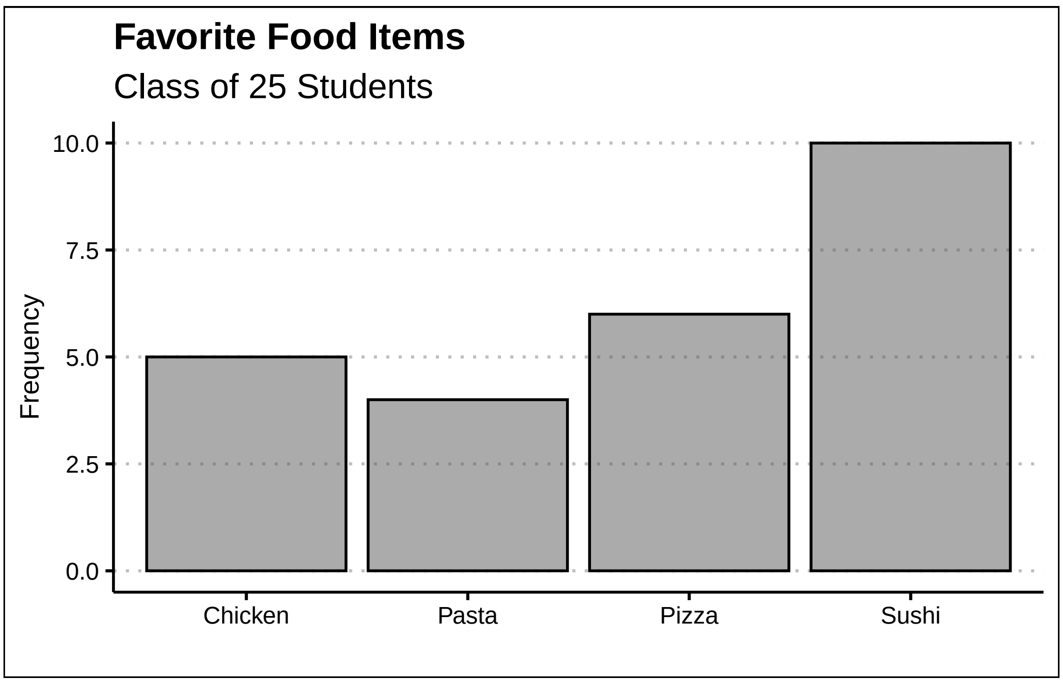
PIZZA	SUSHI	SUSHI	CHICKEN	CHICKEN
PASTA	PASTA	PASTA	SUSHI	PASTA
CHICKEN	PIZZA	CHICKEN	SUSHI	PIZZA
SUSHI	SUSHI	SUSHI	SUSHI	PIZZA
PIZZA	CHICKEN	SUSHI	PIZZA	SUSHI

Simply observing raw data can make identifying the most and least popular items challenging. A frequency distribution organizes this information into a clear table, showcasing the popularity of each item. The frequency distribution of the table is displayed below:

Each food item is tallied up, and the result is shown in the frequency column. Alternately, we can show the tally as a proportion of the total (i.e., 25). For example, five students liked chicken; out of the 25 students surveyed, this represents 0.2 or 20%. This calculation is shown for each food item in the relative frequency column.

Food	Frequency	Relative
Chicken	5	0.20
Pasta	4	0.16
Pizza	6	0.24
Sushi	10	0.40

Below, you can see the bar graph showing the frequency distribution of the food items data. Note that the visualization is constructed by showing each food item as a bar with a height equal to the frequency.



In sum, the **bar plot** illustrates the frequency distribution of categorical data. It includes the classes in the horizontal axis and frequencies or relative frequencies in the vertical axis and has gaps between each bar.

2.2 Frequency Distributions (Numerical)

When working with numerical data, building a frequency distributions requires additional steps compared to categorical data. The challenge lies in the absence of predefined categories or classes. To construct a frequency distribution for numerical data, it is essential to determine the number, width, and limits of the classes. Here are the steps to create a frequency distribution when data is numerical:

1. Determine the Number of Classes: The number of classes can be estimated using the 2^k rule, where k is the smallest integer such that 2^k exceeds the total number of observations by the least amount. This ensures the chosen number of classes provides a reasonable level of granularity for summarizing the data.

Ex1: If a data set has 50 observations, we would choose six classes since $2^6 = 64$ is greater than 50 by the least amount.

Alternatively, you can use Sturges' rule to estimate the number of classes. This rule suggests using $k = \lceil 1+3.322 \log_{10}(n) \rceil$ where n is the number of observations and $\lceil \cdot \rceil$ denotes the ceiling function (round up to the nearest integer).

Ex2: If a data set has 50 observations, $k = \lceil 1 + 3.322 \times 1.699 \approx 6.64 \rceil = 7$ so we would choose seven classes.

2. Calculate the Width of Each Class: The width of a class is determined using the formula:

$$Width = \frac{Max - Min}{Number\ of\ Classes}$$

Ex: If the data set has 50 observations and the minimum value 20 and the maximum is 78, then the width of each class is $58/6 \approx 9.7$. Hence, we can round up and use a class width of 10. It is important to note to always round up, as this ensures that all data points are included in an class.

3. Establish Class Limits: The class limits define the range of values in each class. These limits should be chosen such that each data point belongs to only one class.

Ex: Consider a data set of 50 observations where each class has a width of 10. Set the class limits of the first class to [20,30). Note that the square bracket indicates that the point should be included in the class, whereas) indicates that the point should not be included in the class. The six classes would be [20,30), [30,40), [40,50), [50,60), [60,70), and [70,80). By choosing these classes, each point belongs to only one class.

Example: Let's look at a snapshot of the Dow Jones Industrial 30 stock prices. Below you can see the data:

$277	$175	$202	$383	$358
$188	$294	$157	$212	$42
$149	$410	$303	$165	$203
$104	$23	$60	$287	$121
$312	$52	$54	$158	$501
$96	$95	$43	$189	$201

Let's follow the steps to build the frequency distribution.

1. Determine the Number of Classes: Here we choose five classes since $2^5 = 32$ is greater than 30 by the least amount.

2. Calculate the Width of Each Class: The smallest values in the data set is 23 and the maximum is 501. This gives us a range of 478. Now we can just take the range and divide by five to get 95.6. To make things simple we can round to 100 and use a class width of 100.

3. Establish Class Limits: Since we have rounded up we can be flexible with our class limits. The following class limits are suggested [20,120), [120,220), [220,320), [320,420), and [420,520). Note that each class has a width of 100, and that each data point belongs to one single class.

2.3 Frequency Distributions in R (Categorical)

The process of constructing frequency distributions in R is straightforward. We will be mainly using the `table()` function. Let's start by saving the data in a vector:

```
food<-c("Pizza","Sushi","Sushi","Chicken",
        "Chicken","Pasta","Pasta","Pasta",
        "Sushi","Pasta","Chicken","Pizza",
        "Chicken","Sushi","Pizza","Sushi",
        "Sushi","Sushi","Sushi","Pizza",
        "Pizza","Chicken","Sushi","Pizza",
        "Sushi")
```

Here we define a vector called food by assigning (`<-`) the combination (`c`) of all the food items. To generate the frequency distribution we simply pass the food vector into the `table()` command.

```
table(food)
```

```
food
Chicken   Pasta   Pizza   Sushi
      5       4       6      10
```

As you can see this tallies all the instances for each item. If instead we wanted to obtain the relative frequency we can use the `prop.table()` function. This function requires the frequency distribution created by the `table()` function. Hence, we can first create the frequency distribution, save it into an object, and then generate the relative frequency. The code is below:

```
freq<-table(food)
prop.table(freq)
```

```
food
Chicken   Pasta   Pizza   Sushi 
   0.20    0.16    0.24    0.40 
```

As a last modification. If you want percent frequencies, you can multiply the `prop.table()` function by 100, as shown below:

```
prop.table(freq)*100
```

```
food
Chicken   Pasta   Pizza   Sushi 
     20      16      24      40 
```

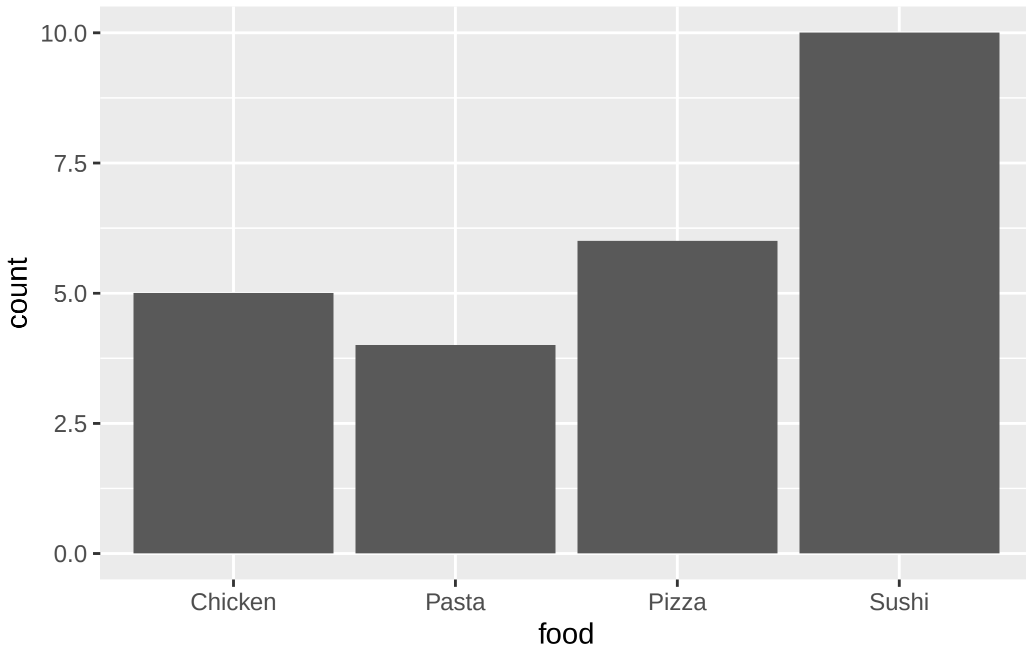
2.4 Bar Plot in R

To create the bar plot we will be using the `geom_bar()` function within the `tidyverse` package. We start by loading the package:

```
library(tidyverse)
```

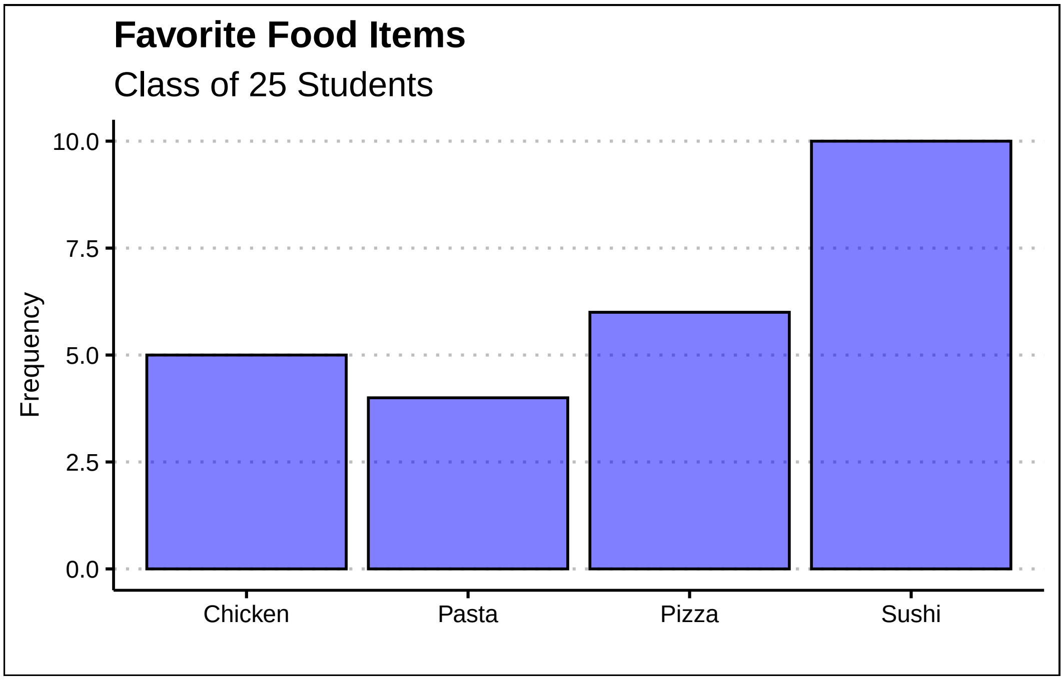
Now R will identify the functions `ggplot()` and `geom_bar()` from the `ggplot` library. To construct the plot we will first call on `ggplot()` and then specify the type of graph we want by calling on `geom_bar()`. In the `aes()` function we will specify which variable (or vector) we want to plot.

```
ggplot() + geom_bar(aes(food))
```



We can enhance the visualization by adding a title and changing the theme. The `labs()` function allows us to change the titles and the `ggthemes` package allows us to choose from a variety of themes.

```
ggplot() + geom_bar(aes(food), col="black", alpha=0.5, bg="blue") +
  labs(title="Favorite Food Items",
       subtitle="Class of 25 Students",
       x="", y="Frequency") +
  theme_clean()
```



A few arguments are worth explaining in the code above. The arguments in the `geom_bar()` function change the background color (bg), the transparency of the color (alpha), and the color of the outline for the bars (col). Title and subtitles are added within the `labs()` function. To omit labels we can just open and close quotations. Hence, `x=""` omits the x label.

2.5 Frequency Distribution in R (Numerical)

To construct the frequency distribution in R we will be first generating the classes and then using the `table()` function as we did in the categorical case. Let's first get the data into R:

```
dow<-c(277,174,202,383,358,188,293,156,
       212,42,149,410,303,165,203,103,
       22,59,287,121,312,52,53,158,500,
       96,95,43,188,200)
```

To generate the bins we will use the example and procedure found in 2.2. That is, we will be using five classes, of width 100. Below is the code to do this:

```
intervals<-cut_width(dow, boundary=20, width=100)
(dowfreq<-table(intervals))
```

```
intervals
 [20,120] (120,220] (220,320] (320,420] (420,520]
        9        12         5         3         1
```

The process involves two steps. First we place each observation in the dow, into the bins by using the `cut_width()` function. This involves specifying where we want to start our first bin (*boundary=20*) and the width of each bin (*width 100*). The function creates the bins and places each observation in the respective bin. The last step is to tally the results with the `table()` function.

To obtain the cumulative distribution, we can use the `cumsum()` function. Below we just wrap the frequency distribution (freq) into the `cumsum()` function.

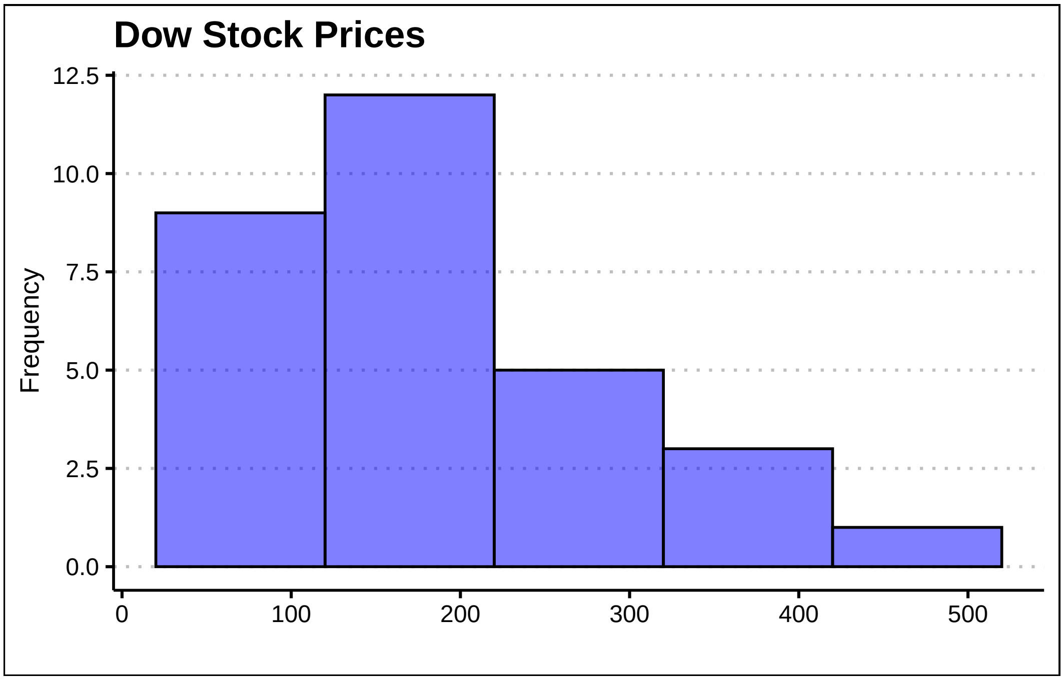
```r
cumsum(dowfreq)
```

```
 [20,120] (120,220] (220,320] (320,420] (420,520]
        9        21        26        29        30
```

2.6 Histograms in R

To generate the histogram in R we will use once again the `tidyverse` package. This time we will use the `geom_histogram()` function. Below is the code to generate the histogram for the dow data:

```r
ggplot() +
  geom_histogram(aes(dow), bg="blue", alpha=0.5, col="black",
                 bins=5,
                 binwidth = 100,
                 boundary=20) +
  labs(title="Dow Stock Prices",
       y="Frequency", x="") +
  theme_clean()
```



Within the `geom_histogram()` command we can set the classes (or bins) for the histogram. The bins argument specifies the number of bins, bin.width specifies the bin width, and the boundary specifies where should the histogram starts. This histogram allows us to observe quickly that most stocks in the dow are priced between 20 and 220 dollars.

2.7 Exercises

The following exercises will help you practice summarizing data with tables and simple graphs. In particular, the exercises work on:

- Developing frequency distributions for both categorical and numerical data.
- Constructing bar charts, histograms, and line charts.
- Creating contingency tables.

Answers are provided below. Try not to peek until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

Install the `ISLR2` package in R. You will need the **BrainCancer** data set to answer this question.

1. Construct a frequency and relative frequency table of the *Diagnosis* variable. What was the most common diagnosis? What percentage of the sample had this diagnosis?

Answer

The most common diagnosis is Meningioma, a slow-growing tumor that forms from the membranous layers surrounding the brain and spinal cord. The diagnosis represents about 48.28% of the sample.

Start by loading the ***ISLR2*** *package. To construct the frequency distribution table, use the* ***table()*** *function.*

```
library(ISLR2)
table(BrainCancer$diagnosis)
```

```

Meningioma  LG glioma  HG glioma      Other 
        42          9         22         14 
```

The relative frequency distribution can be easily retrieved by saving the frequency table in an object and then using the ***prop.table()*** *function.*

```
freq<-table(BrainCancer$diagnosis)
prop.table(freq)
```

```

Meningioma  LG glioma  HG glioma      Other 
 0.4827586  0.1034483  0.2528736  0.1609195 
```

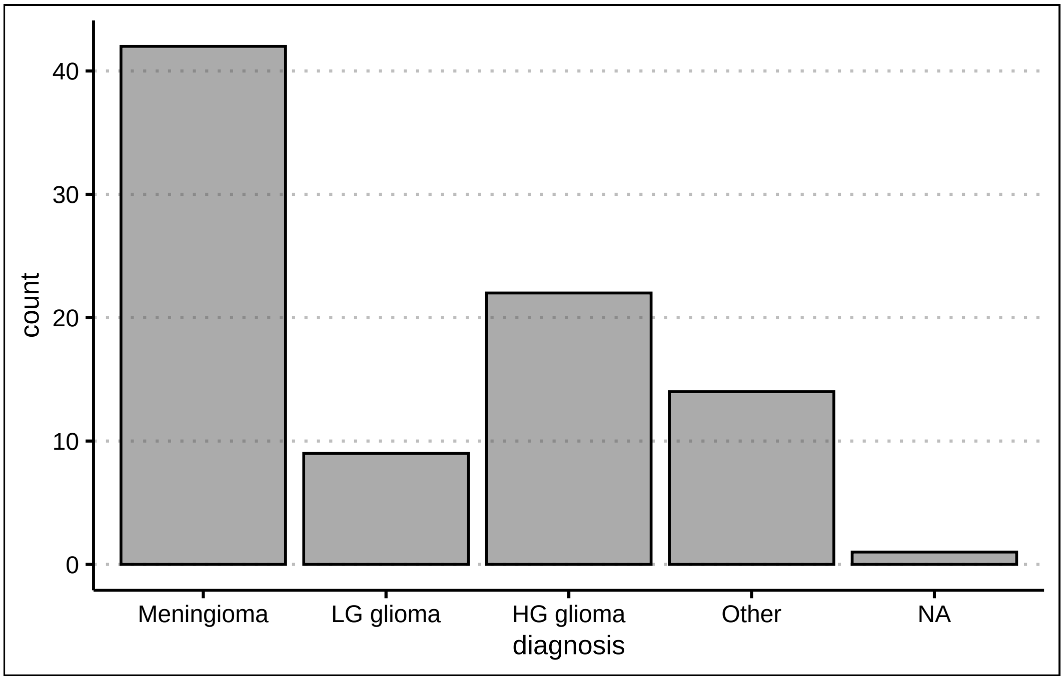
2. Construct a bar chart. Summarize the findings.

Answer

The majority of diagnosis are Meningioma. Low grade glioma is the least common of diagnosis. High grade glioma and other diagnosis have about the same frequency.

To construct the bar chart use the `geom_bar()` function from `tidyverse`.

```
library(tidyverse)
library(ggthemes)
ggplot(data=BrainCancer) +
  geom_bar(aes(diagnosis), alpha=0.5, col="black") +
  theme_clean()
```



3. Construct a contingency table that shows the *Diagnosis* along with the *Status*. Which diagnosis had the highest number of non-survivals (0)? What was the survival rate of this diagnosis?

Answer

33 people did not survive Meningioma. The survival rate of Meningioma is only 21.43%.

Use the `table()` function one more time to create the contingency table for the two variables.

```
(freq2<-table(BrainCancer$status,BrainCancer$diagnosis))
```

```
   Meningioma LG glioma HG glioma Other
 0         33         5         5     9
 1          9         4        17     5
```

To get the survival rates, we can use the `prop.table()` function once again.

```
prop.table(freq2,margin = 2)
```

```
   Meningioma LG glioma HG glioma     Other
 0  0.7857143 0.5555556 0.2272727 0.6428571
 1  0.2142857 0.4444444 0.7727273 0.3571429
```

Exercise 2

You will need the **airquality** data set (in base R) to answer this question.

1. Construct a frequency distribution for *Temp*. Use five classes with widths of $50 < x \leq 60$; $60 < x \leq 70$; etc. Which interval had the highest frequency? How many times was the temperature between 50 and 60 degrees?

Answer

The highest frequency is in the $80 < x \leq 90$ bin. 8 temperatures were between $50 < x \leq 60$ degrees.

We can create the intervals using the **`cut_width()`** *function:*

```
intervals <- cut_width(airquality$Temp, width = 10,
                       boundary=50)
```

The frequency distribution can be obtained by using the **`table()`** *function on the interval.cut object created above.*

```
table(intervals)
```

```
intervals
 [50,60]  (60,70]  (70,80]  (80,90] (90,100]
       8       25       52       54       14
```

2. Construct a relative frequency, cumulative frequency and the relative cumulative frequency distributions. What proportion of the time was *Temp* between 50 and 60 degrees? How many times was the *Temp* 70 degrees or less? What proportion of the time was *Temp* more than 70 degrees?

Answer

The temperature was 5.22% of the time between 50 and 60; The temperature was 70 or less 33 times; The temperature was above 70, 78.43% of the time.

To get the relative frequency table, start by saving the proportion table into an object.Then you can use the **`prop.table()`** *function.*

```
freq<-table(intervals)
prop.table(freq)
```

```
intervals
   [50,60]    (60,70]    (70,80]    (80,90]   (90,100]
0.05228758 0.16339869 0.33986928 0.35294118 0.09150327
```

For the cumulative distribution you can use the **`cumsum()`** *function on the frequency distribution.*

```
cumulfreq<-cumsum(freq)
cumulfreq
```

```
 [50,60]  (60,70]  (70,80]  (80,90] (90,100]
       8       33       85      139      153
```

Lastly, for the relative cumulative distribution table, you can use the **`cumsum()`** *function on the relative frequency table.*

```
cumsum(prop.table(freq))
```

```
   [50,60]    (60,70]    (70,80]    (80,90]   (90,100]
0.05228758 0.21568627 0.55555556 0.90849673 1.00000000
```

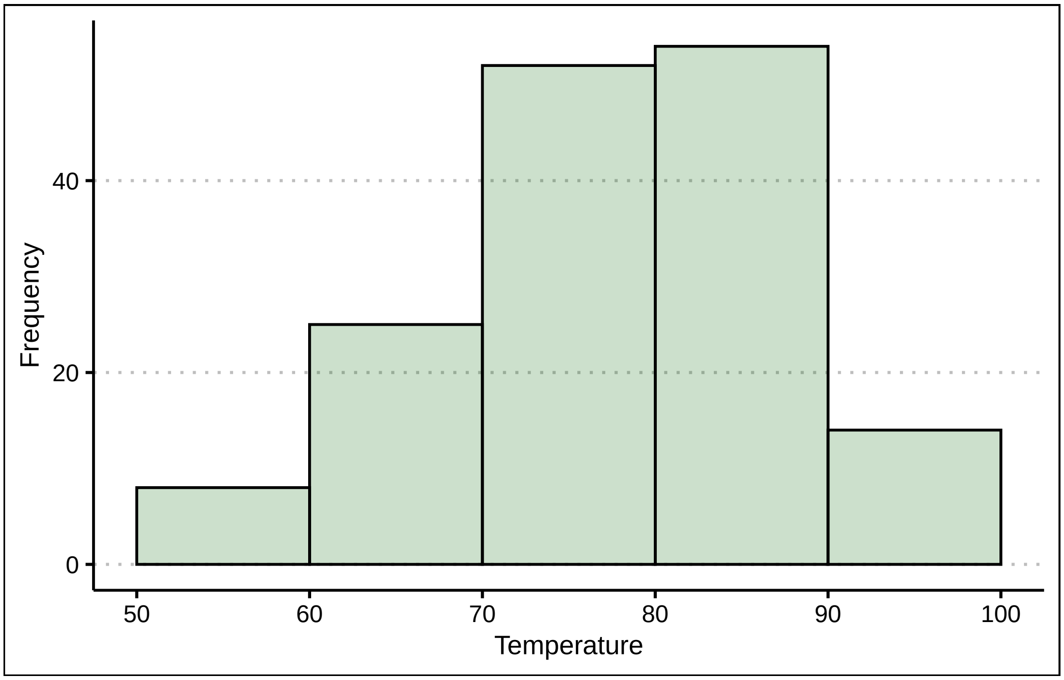
3. Construct the histogram. Is the distribution symmetric? If not, is it skewed to the left or right?

Answer

The distribution is not perfectly symmetric. It is skewed slightly to the left (see histogram.)

Use the `geom_histogram()` function to create the histogram.

```
ggplot() +
  geom_histogram(aes(airquality$Temp), col="black",
                 bg="darkgreen",alpha=0.2,
                 bins=5,
                 binwidth = 10,
                 boundary=0) + theme_clean() +
  labs(x="Temperature", y="Frequency")
```



Exercise 3

You will need the **Portfolio** data set from the `ISLR2` package to answer this question.

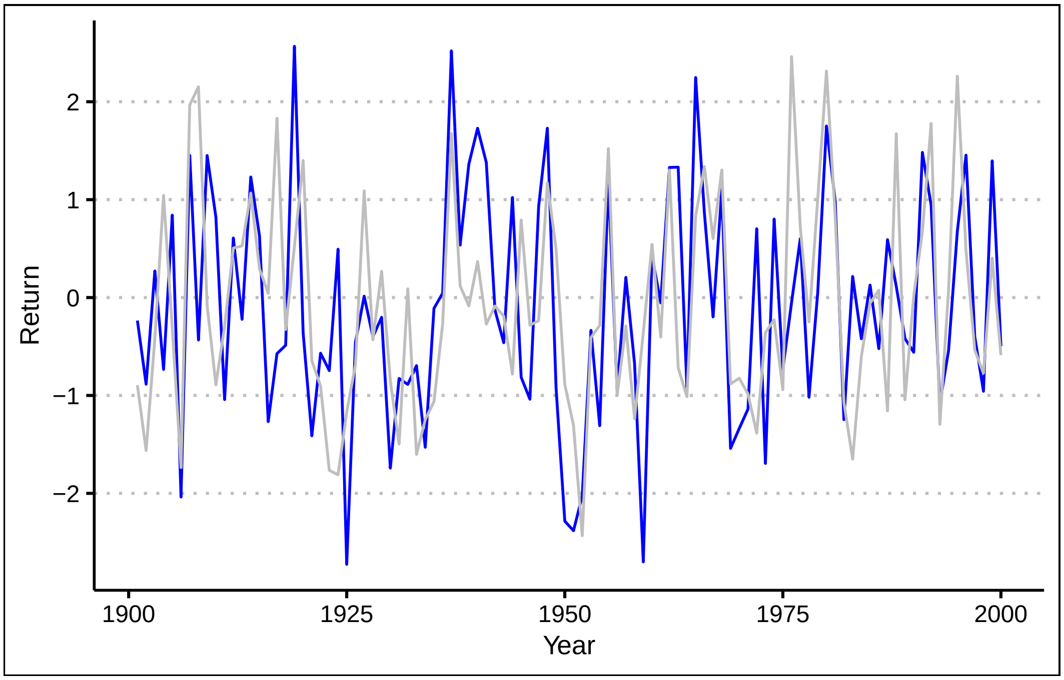
1. Construct a line chart that shows the returns over time for each portfolio (X and Y) by using two lines each with a unique color. Assume the data is for the period 1901 to 2000. Include also a legend that matches colors to portfolios.

Answer

From 1901 through 2000, both portfolios have behaved very similarly. Returns are between -3% and 3%, there is no trend, and positive (negative) returns for X seem to match with positive (negative) returns of Y.

You can use the `geom_line()` function to create a line for each portfolio.

```
ggplot() +
  geom_line(aes(y=Portfolio$Y,x=seq(1901,2000)), col="blue") +
  geom_line(aes(y=Portfolio$X,x=seq(1901,2000)), col="grey") +
  theme_clean() +
  labs(x="Year", y="Return")
```



Exercise 4

A company surveyed 40 employees about their primary department. The responses are:

```
dept <- c("Marketing", "Sales", "Finance", "Operations", "Marketing", "HR", "IT", "Finance",
```

1. Use R to create a frequency distribution table for these departments.

Answer

```
table(dept)
```

```
dept
Customer Service          Finance               HR               IT
               6                8                3                2
       Marketing       Operations              R&D            Sales
               7                6                3                5
```

2. Compute the relative frequencies (as percentages).

Answer

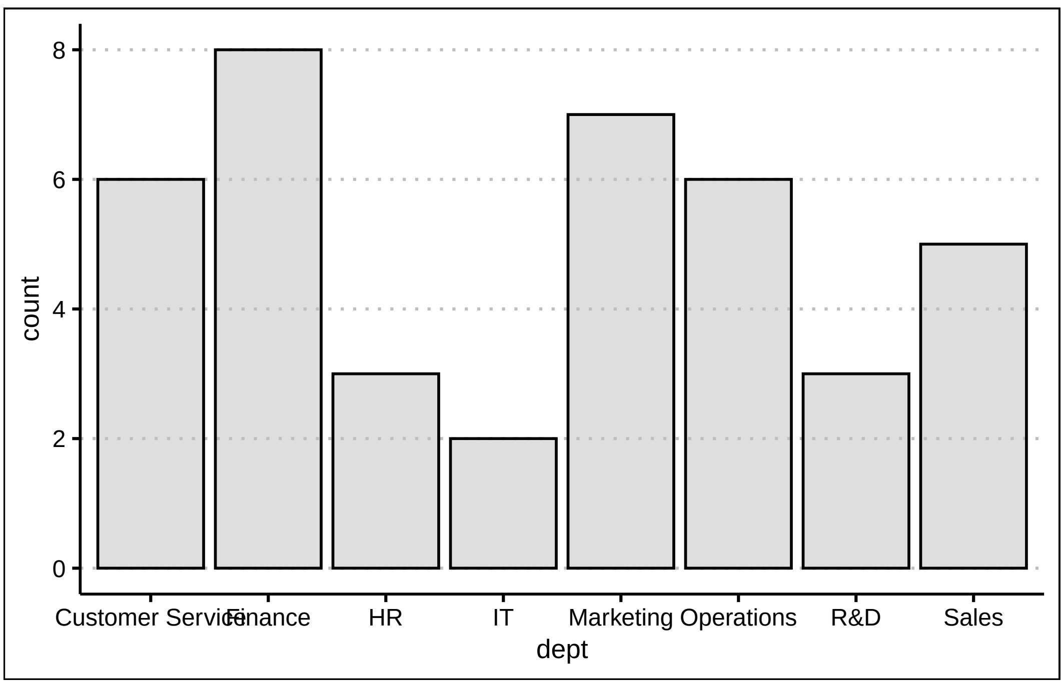
```r
prop.table(table(dept)) * 100
```

```
dept
Customer Service          Finance               HR               IT
            15.0             20.0              7.5              5.0
       Marketing       Operations              R&D            Sales
            17.5             15.0              7.5             12.5
```

3. Create a bar plot of the frequencies. Which department has the highest frequency? What is its relative frequency (percentage)?

Answer

```r
ggplot() +
  geom_bar(aes(x=dept), bg="grey",
           col="black", alpha=0.5) +
  theme_clean()
```



The highest frequency is in the Finance department with 8 employees. The relative frequency is 20%.

Exercise 5

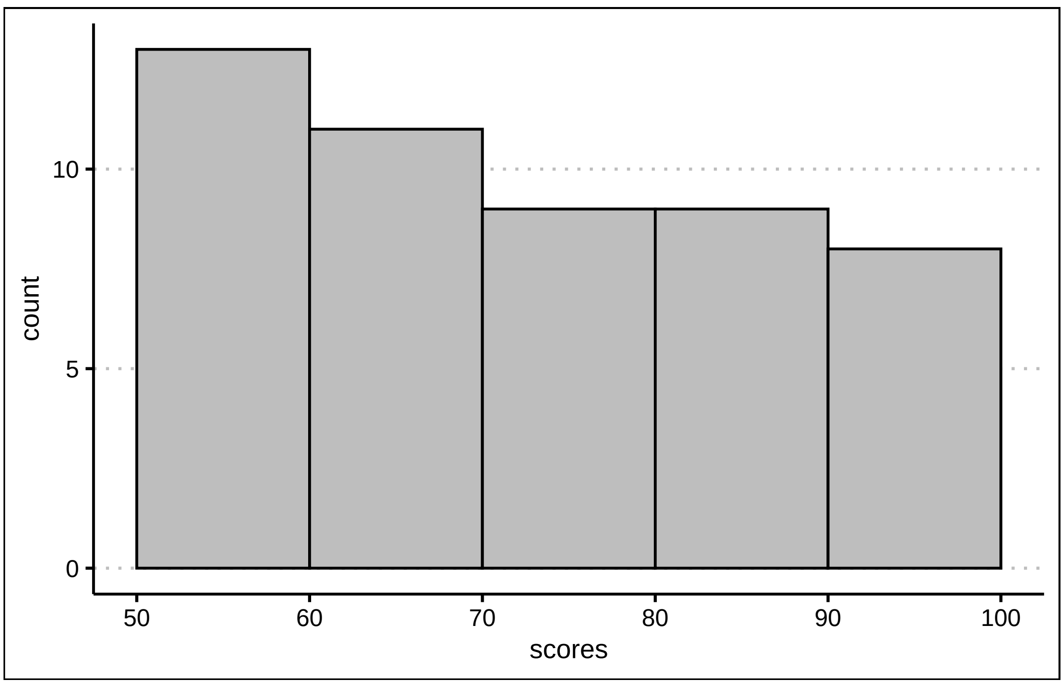
A class of 50 students took a business statistics exam (out of 100 points). Their scores were:

```
scores <- c(75.6, 66.9, 83.4, 67.2, 95.3, 90.5, 54.4, 76.8, 88.5, 63.3, 95.1, 69.9, 80.4, 91
```

1. Construct a histogram using 7 classes with width 10, starting at 50 (bins: [50,60), [60,70), etc.). Use `geom_histogram()` with appropriate parameters. In which bin is the highest frequency? How many students scored in that range?

Answer

```
ggplot() +
  geom_histogram(aes(x=scores),
                 breaks = seq(50, 100, by = 10),
                 closed = "left",
                 col="black", bg="grey") +
  theme_clean()
```



The highest frequency is in the bin (60,70] with 9 students.

3 Descriptive Statistics III

Understanding where the "center" of a dataset lies is a fundamental step when interpreting and analyzing data. Measures of central location provide different ways to identify the "typical" value in a dataset, each with its unique strengths and limitations.

3.1 The Mean

The **mean** is the average value for a numerical variable. It is a widely understood and straightforward measure to calculate. It incorporates all data points, providing a comprehensive representation of the data set. However, its reliance on every value also makes it sensitive to outliers or skewed distributions, which may cause it to not accurately reflect the true center of the data.

The sample statistic is estimated by:

$$\bar{x} = \frac{\sum x_i}{n}$$

where x_i is observation i, and n is the number of observations in the sample. The formula instructs you to add up all of the values in a variable and then divide by the sample size. If you wish to estimate the population parameter you can use:

$$\mu = \frac{\sum x_i}{N}$$

where N is the population size.

Ex 1: Consider the following sample of numbers $x = \{1, 4, 2, 1\}$*. The mean would be equal to* $\bar{x} = \frac{1+4+2+1}{4}$ *or* $\bar{x} = 2$.

Ex 2: Consider the following sample of numbers $x = \{1, 4, 2, 1, 100\}$. The mean is $\bar{x} = 21.6$. Although most of the numbers are in the range $(1, 4)$, the 100 biases the mean to 21.6.

3.2 The Median

The **median** is the value in the middle when data is organized in ascending order. When n is even, the median is the average between the two middle values. The median is resistant to outliers, making it an ideal measure for skewed data or data sets with irregular distributions. It is particularly useful for ordinal data, where precise ranking is important. However, the median does not utilize all data points, which can lead to less precise comparisons compared to other measures like the mean.

Ex 1: Consider the following numbers $x = \{1, 4, 2, 1\}$. We first sort the data to obtain $x_{sorted} = \{1, 1, 2, 4\}$. Since the number of observations is even ($n = 4$), the median is the average of the two middle numbers $(1, 2)$. $median_x = \frac{1+2}{2}$ or 1.5.

Ex 2: Consider the following numbers $x = \{1, 4, 2, 1, 100\}$. We once again sort the number obtaining $x_{sorted} = \{1, 1, 2, 4, 100\}$. Since $n = 5$ in this case we can identify the median as the third value in x_{sorted}. $median_x = 2$. Note that the inclusion of 100 in the data did not change much the measure of central location.

3.3 The Mode

The **mode** is the value with highest frequency from a set of observations. This measure is particularly useful for categorical data as it helps determine popularity of values. It can be applied to both numerical and non-numerical data sets. The mode has its limitations; it may not exist in cases where all values occur with equal frequency, and there may be multiple modes, which can complicate interpretation. Additionally, since the mode focuses only on the most frequent value, it does not account for other data points, limiting its overall utility as a comprehensive measure.

Ex 1: Consider the following numbers $x = \{1, 4, 2, 1\}$. Since 1 is repeated twice and all other numbers just repeated once, $x_{mode} = 1$.

Ex 2: Consider the following numbers $x = \{1, 4, 2, 1, 4\}$. Now 4 is also repeated twice. The variable has two modes $x_{mode} = \{1, 2\}$. x is said to be bimodal.

3.4 The Weighted Mean

The weighted mean is useful in scenarios where some data points are more significant than others, such as in financial portfolios, grade point averages, or survey results, as it accounts for variability in importance across observations. This measure requires additional information in the form of weights (w_i), which may not always be available or accurate. It is calculated the sum product of values (x_i) and weights (w_i) and then dividing by the sum of weights. Mathematically, the weighted average is:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Ex: Consider three different stocks $S = \{T, C, X\}$ with stock returns of $R = \{2, 4, 10\}$. Each stock has a weight in the portfolio of $W = \{0.3, 0.2, 0.5\}$. The average return of the portfolio is $\bar{x}_{weighted} = \frac{0.6+0.8+5}{1}$ or $\bar{x}_{weighted} = 6.4$.

3.5 The Geometric Mean

The **geometric mean** is a multiplicative average that is less sensitive to outliers relative to the arithmetic mean. It is useful when averaging growth rates or rates of return. It is calculated by:

$$\bar{x}_g = \sqrt[n]{(1 + r_1) * (1 + r_2)...(1 + r_n)} - 1$$

where $\sqrt[n]{}$ is the n_{th} root, and r_i are the returns or growth rates. When working with growth rates or rates of return, you add 1 to each rate because these metrics represent changes relative to a base value.

When dealing with proportions, these already reflect a standalone quantity, not a relative change. As a consequence, there is no need to add 1. In this case the geometric mean simplifies to:

$$\bar{x}_g = \sqrt[n]{(x_1) \times (x_2)...(x_n)}$$

Ex: Consider the variable $x = \{0.2, 0.3, 0.1, 0.1\}$. The geometric mean is equal to $\bar{x}_g = \sqrt[4]{(1.2 \times 1.3 \times 1.1 \times 1.1)} - 1$ or $\bar{x}_g = 0.17$ if x represents growth rates. When x represents ratios from a whole, the geometric mean is $\bar{x}_g = \sqrt[4]{0.2 \times 0.3 \times 0.1 \times 0.1)}$ or 0.156.

3.6 Measures of Central Location in R

Base R has a collection of functions that calculate measures of central location. Let's consider the following data on approval ratings:

```
library(tidyverse)
(poll<-tibble(date=c("01/01/24", "02/01/24",
                     "03/01/24", "04/01/24"),
              people=c(50,100,30,250),
              approval=c(0.25,0.25,0.7,0.85)))
```

```
# A tibble: 4 x 3
  date     people approval
  <chr>     <dbl>    <dbl>
1 01/01/24     50     0.25
2 02/01/24    100     0.25
3 03/01/24     30     0.7
4 04/01/24    250     0.85
```

To calculate the mean we can just pass a vector into the `mean()` function. Hence, the mean approval is:

```r
mean(poll$approval)
```

```
[1] 0.5125
```

To calculate the mode we will use the `table()` function, as there is no mode function in base R.

```r
table(poll$approval)
```

```

0.25  0.7 0.85
   2    1    1
```

For the median we will use the `median()` function.

```r
median(poll$approval)
```

```
[1] 0.475
```

The weighted average can be calculated using the `weighted.mean()` function. Let the approval be the value and number of people surveyed the weight.

```r
weighted.mean(x=poll$approval,w = poll$people)
```

```
[1] 0.6302326
```

Lastly, the geometric mean has no built in function in base R. However, we can easily calculate it with the command:

```r
geometric_mean <- prod(poll$approval)^(1/length(poll$approval))
```

Since the approval rating is a percentage of the total people polled there is no need to add one to these numbers. The `summary()` function calculates a collection of summary statistics for a vector or data frame. Below we apply it to the entire data set:

```r
summary(poll)
```

```
     date               people          approval     
 Length:4           Min.   : 30.0   Min.   :0.2500  
 Class :character   1st Qu.: 45.0   1st Qu.:0.2500  
 Mode  :character   Median : 75.0   Median :0.4750  
                    Mean   :107.5   Mean   :0.5125  
                    3rd Qu.:137.5   3rd Qu.:0.7375  
                    Max.   :250.0   Max.   :0.8500  
```

Below a list of the functions used:

- The `mean()` function calculates the average for a vector of numbers.
- The `median()` function calculates the median for a vector of numbers.
- The `table()` function generates the frequency distribution so that we can identify the mode or modes.
- The `weighted.mean()` function calculates the weighted mean for a vector of numbers, and corresponding vector of weights.
- The `length()` function calculates the length of a vector.
- The `summary()` function provides measures of location for a vector of numbers.

3.7 Exercises

The following exercises will help you practice the measures of central location. In particular, the exercises work on:

- Calculating the mean, median, and the mode.
- Calculating the weighted average.
- Applying the geometric mean for growth rates and returns.

Answers are provided below. Try not to peek until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using R functions when possible.

1. Use the following observations to calculate the mean, the median, and the mode.

$$8 \quad 10 \quad 9 \quad 12 \quad 12$$

Answer

To find the mean we will use the following formula ($\frac{1}{n}\sum_{i=i}^{n} x_i$). The summation of the values is 51 and the number of observations is 5. The mean is $51/5 = 10.2$.

The median is found by locating the middle value when data is sorted in ascending order. The median in this example is 10.

The mode is the value with the highest frequency. In this example the mode is 12 since it is repeated twice and all other numbers appear only once.

The mean can be easily verified in R by using the `mean()` function:

```r
mean(c(8,10,9,12,12))
```

```
[1] 10.2
```

Similarly, the median is easily verified by using the `median()` function:

```r
median(c(8,10,9,12,12))
```

```
[1] 10
```

We can use the `table()` function to calculate frequencies and easily identify the mode.

```r
table(c(8,10,9,12,12))
```

```

 8  9 10 12 
 1  1  1  2 
```

2. Use following observations to calculate the mean, the median, and the mode.

-4 0 -6 1 -3 -4

Answer

The mean is -2.67, the median is -3.5, the mode is -4.

The mean is verified in R:

```
mean(c(-4,0,-6,1,-3,-4))
```

```
[1] -2.666667
```

The median in R:

```
median(c(-4,0,-6,1,-3,-4))
```

```
[1] -3.5
```

Finally, the mode in R:

```
table(c(-4,0,-6,1,-3,-4))
```

```

-6 -4 -3  0  1 
 1  2  1  1  1 
```

3. Use the following observations, calculate the mean, the median, and the mode.

20 15 25 20 10 15 25 20 15

Answer

The mean is 18.33, the median is 20, the data is bimodal with both 15 and 20 being modes.

The mean is verified in R:

```
mean(c(20,15,25,20,10,15,25,20,15))
```

```
[1] 18.33333
```

The median in R:

```
median(c(20,15,25,20,10,15,25,20,15))
```

```
[1] 20
```

The frequency distribution identifies the modes:

```
table(c(20,15,25,20,10,15,25,20,15))
```

```

10 15 20 25
 1  3  3  2
```

Exercise 2

Download the `ISLR2` package. You will need the **OJ** data set to answer this question.

1. Find the mean price for Country Hill (*PriceCH*) and Minute Maid (*PriceMM*).

Answer

The mean price for Country Hill is 1.87. The mean price for Minute Maid is 2.09.

The means can be easily found with the `mean()` *function:*

```
library(ISLR2)
OJ=OJ
mean(OJ$PriceCH)
```

```
[1] 1.867421
```

```
mean(OJ$PriceMM)
```

```
[1] 2.085411
```

2. Find the mean price of Country Hill (*PriceCH*) at each store (*StoreID*). Which store provides the better price?

Answer

The mean price at store 1 for Country Hill is 1.80. The juice is cheaper at store 1.

The means for each store can be found by using ***group_by()*** *and* ***summarise()****.The mean price at each store is:*

```
OJ %>% group_by(StoreID) %>% summarise(MeanCH=mean(PriceCH))
```

```
# A tibble: 5 x 2
  StoreID MeanCH
    <dbl>  <dbl>
1       1   1.80
2       2   1.84
3       3   1.93
4       4   1.95
5       7   1.84
```

3. Find the median price paid by Country Hill (*PriceCH*) purchasers (*Purchase*) in all stores? Which store had the better median price?

Answer

Purchasers of Country Hill at store 1 paid a median price of 1.76 for Country Hill juice. This once again was the lowest price.

The median price for Country Hill purchasers at each store is given by:

```
OJ %>% filter(Purchase=="CH") %>% group_by(StoreID) %>% summarise(MedianCH=median(PriceCH))
```

```
# A tibble: 5 x 2
  StoreID MedianCH
    <dbl>    <dbl>
1       1     1.76
2       2     1.86
3       3     1.99
4       4     1.99
5       7     1.86
```

Exercise 3

1. Over the past year an investor bought TSLA. She made these purchases on three occasions at the prices shown in the table below. Calculate the average price per share.

Date	Price Per Share	Number of Shares
February	250.34	80
April	234.59	120
Aug	270.45	50

Answer

The average price of sale is found by using the weighted average formula. $\frac{\sum w_i x_i}{\sum w_i}$ The weights (w_i) are given by the number of shares bought and the values (x_i) are the prices. The weighted average is 246.802.

In R you can create two vectors. One holds the share price and the other one the number of shares bought.

```
PricePerShare<-c(250.34,234.59,270.45)
NumberOfShares<-c(80,120,50)
```

Next, can use the `weighted.mean()` *function in R, with PricePerShare as the value and NumberOfShares as the weights. The weighted average is:*

```
(WeightedAverage<-weighted.mean(PricePerShare,NumberOfShares))
```

```
[1] 246.802
```

2. What would have been the average price per share if the investor would have bought equal amounts of shares each month?

Answer

The average if equal shares were bought would be 251.7933.

In R you can use the `mean()` *function on the PricePerShare vector.*

```
(Average<-mean(PricePerShare))
```

```
[1] 251.7933
```

Exercise 4

1. Consider the following observations for the consumer price index (CPI). Calculate the inflation rate (Growth Rate of the CPI) for each period.

1.0	1.3	1.6	1.8	2.1

Answer

The inflation rate is the percentage change in the CPI. The inflation rate for each period is shown in the table below:

30%	23.08%	12.5%	16.67%

In R create an object to store the values of the CPI:

```
CPI<-c(1,1.3,1.6,1.8,2.1)
```

Next use the **`diff()`** *function to find the difference between the end value and start value. Divide the result by a vector of starting value and multiply times 100.*

```
(Inflation<-100*diff(CPI)/CPI[1:4])
```

```
[1] 30.00000 23.07692 12.50000 16.66667
```

2. What is the average growth rate for the inflation rate?

Answer

The average growth rate is 31.61

We can use the geometric mean formula with compounding. In R:

```
(Growth<-prod(1.3,1.2308,1.125,1.1667)^(1/4)-1)
```

```
[1] 0.2038175
```

2. Suppose that you want to invest \$1000 dollars in a stock that is predicted to yield the following returns in the next four years. Calculate both the arithmetic mean and the geometric mean. Use the geometric mean to estimate how much money you would have by the end of year 4.

Year	Annual Return
1	17.3
2	19.6
3	6.8
4	8.2

Answer

At the end of 4 years it is predicted that you would have 1621.17 dollars. Each year you would have gained 12.84% on average.

In R include the annual rates in a vector:

```
growth<-c(0.173,0.196,0.068,0.082)
```

The arithmetic mean is:

```
100*mean(growth)
```

```
[1] 12.975
```

The geometric mean is:

```
(geom<-((prod(1+growth))^(1/length(growth))-1)*100)
```

```
[1] 12.8384
```

At the end of the four years we would have:

```
1000*(1+geom/100)^4
```

```
[1] 1621.167
```

Exercise 5

The ISLR2 package includes a dataset called **Hitters** containing data on Major League Baseball players, including their salaries (in thousands of dollars) from 1987.

1. Load the Hitters dataset using data(Hitters) from the ISLR2 package. Compute the overall mean and median salary for all players. Which measure (mean or median) better represents the typical salary?

Answer

Start by loading the libraries and previewing the data.

```r
library(ISLR2)
library(dplyr)
data(Hitters)

glimpse(Hitters)
```

```
Rows: 322
Columns: 20
$ AtBat     <int> 293, 315, 479, 496, 321, 594, 185, 298, 323, 401, 574, 202, ~
$ Hits      <int> 66, 81, 130, 141, 87, 169, 37, 73, 81, 92, 159, 53, 113, 60,~
$ HmRun     <int> 1, 7, 18, 20, 10, 4, 1, 0, 6, 17, 21, 4, 13, 0, 7, 3, 20, 2,~
$ Runs      <int> 30, 24, 66, 65, 39, 74, 23, 24, 26, 49, 107, 31, 48, 30, 29,~
$ RBI       <int> 29, 38, 72, 78, 42, 51, 8, 24, 32, 66, 75, 26, 61, 11, 27, 1~
$ Walks     <int> 14, 39, 76, 37, 30, 35, 21, 7, 8, 65, 59, 27, 47, 22, 30, 11~
$ Years     <int> 1, 14, 3, 11, 2, 11, 2, 3, 2, 13, 10, 9, 4, 6, 13, 3, 15, 5,~
$ CAtBat    <int> 293, 3449, 1624, 5628, 396, 4408, 214, 509, 341, 5206, 4631,~
$ CHits     <int> 66, 835, 457, 1575, 101, 1133, 42, 108, 86, 1332, 1300, 467,~
$ CHmRun    <int> 1, 69, 63, 225, 12, 19, 1, 0, 6, 253, 90, 15, 41, 4, 36, 3, ~
$ CRuns     <int> 30, 321, 224, 828, 48, 501, 30, 41, 32, 784, 702, 192, 205, ~
$ CRBI      <int> 29, 414, 266, 838, 46, 336, 9, 37, 34, 890, 504, 186, 204, 1~
$ CWalks    <int> 14, 375, 263, 354, 33, 194, 24, 12, 8, 866, 488, 161, 203, 2~
$ League    <fct> A, N, A, N, N, A, N, A, N, A, A, N, N, A, N, A, N, A, A, N, ~
$ Division  <fct> E, W, W, E, E, W, E, W, W, E, E, W, E, E, E, W, W, W, W, W, ~
$ PutOuts   <int> 446, 632, 880, 200, 805, 282, 76, 121, 143, 0, 238, 304, 211~
$ Assists   <int> 33, 43, 82, 11, 40, 421, 127, 283, 290, 0, 445, 45, 11, 151,~
$ Errors    <int> 20, 10, 14, 3, 4, 25, 7, 9, 19, 0, 22, 11, 7, 6, 8, 0, 10, 1~
$ Salary    <dbl> NA, 475.000, 480.000, 500.000, 91.500, 750.000, 70.000, 100.~
$ NewLeague <fct> A, N, A, N, N, A, A, A, N, A, A, N, N, A, N, A, N, A, A, N, ~
```

We can now find the mean and the median.

```r
mean(Hitters$Salary, na.rm = TRUE)
```

```
[1] 535.9259
```

```r
median(Hitters$Salary, na.rm = TRUE)
```

```
[1] 425
```

Overall: mean 536 (thousand dollars), median = 425. Due to right-skew from superstar salaries, the median better represents the typical player.

2. filter to only players with at least one home run (HmRun >= 1) and compute the mean and median salary for this subgroup.

Answer

*Filter the data using **`dplyr`**:*

```r
Hitters %>%
  filter(HmRun >= 1) %>%
  summarise(mean_salary = mean(Salary, na.rm = TRUE),
            median_salary = median(Salary, na.rm = TRUE))
```

```
  mean_salary median_salary
1    531.9612           425
```

For players with 1 home run: mean higher (around 600+), median still lower than mean — skewness persists, median again more representative.

4 Descriptive Stats IV

Measures of dispersion provide insights into how much the values in a data set deviate from the central tendency. In a business context, understanding variability is essential for assessing risk, as it helps decision-makers anticipate fluctuations in revenue, costs, and market conditions. Below, we explore the main statistics that help us quantify variability and use them to make better business decisions.

4.1 Sample Vs. Population Statistics

In this book you will find formulas for calculating sample statistics and population parameters. In general, you will use population formulas when you have data for the entire group you care about or when you are working with a theoretical distribution and the parameter is known; population formulas use N in denominators and estimate parameters. Use sample formulas when your data are a subset drawn from a larger population and you want unbiased estimators of the population parameters; sample formulas use n and the degrees of freedom correction. In practice most empirical work uses sample formulas because we rarely observe entire populations; below you will see both population and sample versions so you can choose the correct one depending on whether your data represent the full population or a sample.

4.2 The Range

This is the simplest measure of dispersion, defined as the difference between the largest and smallest values in a variable. While straightforward, the range only accounts for the extremes and does not reflect the variability within the variable. The **range** is calculated by:

$$Range = Maximum - Minimum$$

Ex: Consider the data $x = \{480, 1050, 1400, 2500, 3200\}$. The range is given by $Range = 3200 - 480 = 2720$.

4.3 The Variance

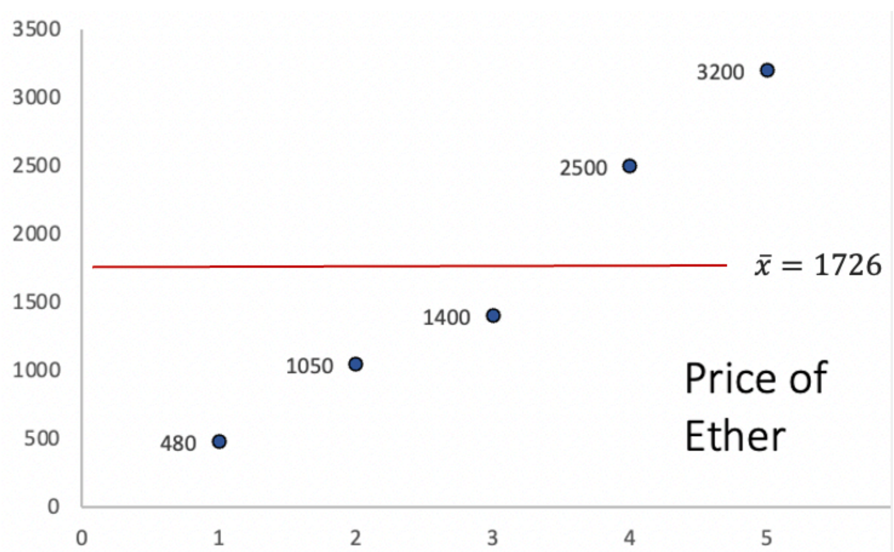
The variance gives us a more comprehensive look at dispersion by considering how each data point deviates from the mean. It summarizes the squared deviations from the mean by finding their average.The population parameter is given by:

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$$

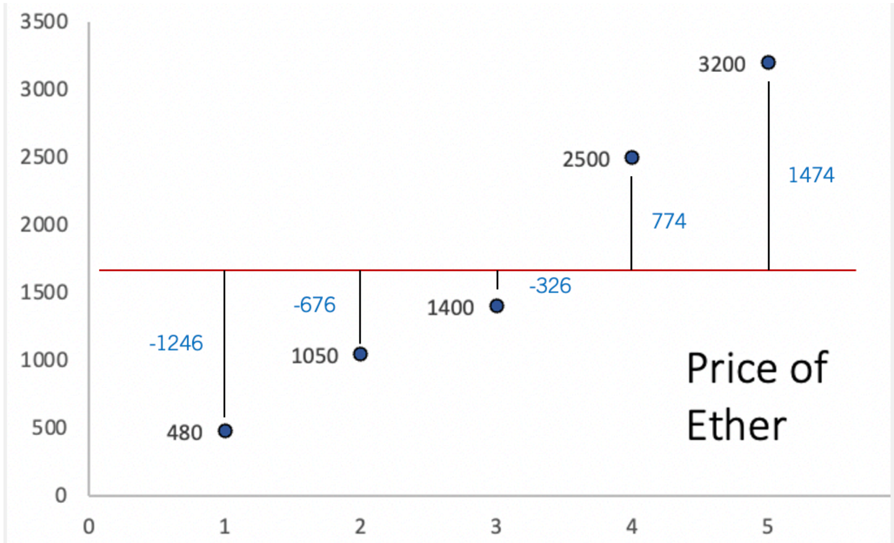
while the sample statistic is

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

Example: Let's consider a sample of the price of Ether (a famous cryptocurrency). Below is a graph of the prices:



The y axis represents the price, while the x axis represents the period of time. The red line is the average price of Ether. The way the variance calculates dispersion, is by first finding the distance between each point and the average. The image below illustrates these distances:



Ideally, our measure of dispersion would simply average the distances from each data point to the mean, giving us a clear indication of how much the data varies around this central point. Since the mean is a measure that balances the distances from the mean, the sum of all deviations is equal to zero. This "balancing" effect is visually apparent in the graph, where positive deviations are exactly offset by negative ones.

To address this issue, the variance squares each deviation from the mean before finding their average. This squaring eliminates negative values, ensuring a non-zero measure of spread. The table below illustrates these calculations:

Price	Deviations $(x_i - \bar{x})$	Squared Deviations $(x_i - \bar{x})^2$
480	-1246	1552516
1050	-676	456976
1400	-326	106276
2500	774	599076
3200	1474	2172676

The second column shows the deviations from the mean. You can convince yourself that these add up to zero. The third column squares the deviations. The sample variance, averages the numbers in the third column using the degrees of freedom $(n-1)$. The result is $s^2 = 1221880$, which means that the price of Ether varies on average 1221880 squared deviations from the mean. The result by itself is not very intuitive as it is measured in squared dollars. We can, however, compare the variances from different variables to assess which variable has the most dispersion.

4.4 The Standard Deviation

The standard deviation is derived by taking the square root of the variance. Remember, the variance employs squared deviations to eliminate negative values and calculate spread. By taking the square root, the standard deviation reverts the measure of dispersion back to the original units of the variable. This transformation makes the standard deviation more intuitive; it directly quantifies how much each data point deviates from the mean in the same units as the variable itself. Consequently, the standard deviation is a clear and intuitive measure of variability.

To find the standard deviation, take the square root of the variance. For the population parameter use:

$$\sigma = \sqrt{\sigma^2}$$

For the sample standard deviation use:

$$s = \sqrt{s^2}$$

Ex: Consider once more the price of Ether. That is. $x = \{480, 1050, 1400, 2500, 3200\}$. In the previous section we found that the variance $s^2 = 1221880$. Using the square root we find that $s = \sqrt{1221880} = 1105.387$. If the price of Ether is in dollars, then, the price varies from the mean 1105.39 dollars on average.

4.5 Mean Absolute Deviation

Another measure of data variability is the Mean Absolute Deviation (MAD), which calculates variability using absolute deviations from the mean. This approach not only keeps the measure in the same units as the data but also makes it less sensitive to large deviations. Practically, the **Mean Absolute Deviation** measures the average deviation from the mean, by using absolute deviations. It is calculated by:

$$MAD = \frac{\sum |x_i - \mu|}{N}$$

where the | | operator indicates absolute value. You can estimates the mad for a sample with:

$$mad = \frac{\sum |x_i - \bar{x}|}{n}$$

Example: Let's consider once more the price of Ether. Recall, that if we calculate the deviations from the mean we obtain the second column in the table below:

Price	Deviations $(x_i - \bar{x})$	Squared Deviations $(x_i - \bar{x})^2$	Absolute Deviations $\lvert x_i - \bar{x} \rvert$
480	-1246	1552516	1246
1050	-676	456976	676
1400	-326	106276	326
2500	774	599076	774
3200	1474	2172676	1474

For the MAD, focus on the fourth column which lists the absolute deviations from the mean. Averaging these gives us the Mean Absolute Deviation (MAD), which equals $mad = 899.2$. This result states that, on average, each data point is roughly 900 dollars away from the mean. Note that the standard deviation is higher (988.73) since the squaring of deviations disproportionately amplifies larger ones, pushing the standard deviation above the MAD.

4.6 Coefficient of Variation

The **Coefficient of Variation** simplifies comparisons of variability across variables with different units or scales, by dividing the standard deviation by the mean. It is calculated by:

$$CV = \frac{s}{\bar{x}}$$

Example: Consider the table below, that shows information on two different stocks:

Stock	Average Price	s
A	1	1
B	100	1

The third column shows the standard deviation of each stock. One could conclude that both stocks vary the same as they have the same standard deviation of one. Recall that the coefficient of variation considers the variable's scale by incorporating the mean into its calculation. Since one stock is centered around 1 dollar and the other around 100 dollars (second column), it is clear that they do not vary similarly percentage-wise. Stock A varies 100% from the mean, whereas Stock B only varies 1%. Hence, the coefficient of variation would identify Stock A as the more variable stock.

4.7 The Sharpe Ratio

The Sharpe Ratio measures how much excess return an investor receives for the extra volatility (risk) taken on beyond the risk-free rate. It essentially uses the same principle of normalization as the CV but adds the dimension of risk-adjusted performance. If the CV tells you the variability of returns in relation to their mean, the Sharpe Ratio tells you if that variability is worth it by comparing it against what you could safely earn without risk. In essence, while the CV highlights variability, the Sharpe Ratio motivates investment choices by rewarding higher returns per unit of risk taken.

In sum, the **Sharpe ratio** quantifies the excess return of an investment over the risk free return. It is calculated by:

$$\frac{\bar{R_p} - R_f}{s}$$

where $\bar{R_p}$ is the mean return of the portfolio, R_f is the risk free return, and s is the standard deviation.

Example: Consider the table below that includes a collection of investments.

				STANDARD &POOR'S
Daily Return	0.2%	0.4%	2%	0.1%
Std. deviation	2%	4%	22%	0.9%
Sharpe Ratio	0.1	0.1	0.09	0.11

The table shows four investments (Apple, Bitcoin, Shiba, and S&P). Just looking at the daily return, it is clear that Shiba provides the best returns. However, the cost for that high return is reflected in variability (22% for Shiba). The S&P is clearly the safest investment with a standard deviation of only 0.9%. The Sharpe Ratio, marries these two metrics showing the return per unit of risk taken over the free rate. If we assume a 0% risk free rate, the investment with the highest Sharpe Ratio is the S&P at 0.11.

4.8 Measures of Dispersion in R

Let's use some stock returns to explore R functions that allow us to calculate measures of dispersion. You can run the following command to get the data into R:

```r
library(tidyverse)
returns<-read_csv("https://jagelves.github.io/Data/returns.csv")
```

Use the `glimpse()` function to view the data:

```r
glimpse(returns)
```

```
Rows: 1,182
Columns: 3
$ date   <chr> "1/3/20", "1/6/20", "1/7/20", "1/8/20", "1/9/20", "1/10/20", "1~
$ Stock  <chr> "AAPL", "AAPL", "AAPL", "AAPL", "AAPL", "AAPL", "AAPL", "AAPL",~
$ Return <dbl> -0.009769540, 0.007936685, -0.004714194, 0.015958317, 0.0210183~
```

It seems like the data is stock returns for different companies. We can start by finding the average return of each stock. To do this, let's use the `group_by()` and `summarise()` functions.

```r
returns %>% group_by(Stock) %>%
  summarise(Mean=mean(Return))
```

```
# A tibble: 3 x 2
  Stock     Mean
  <chr>    <dbl>
1 AAPL  0.00173
2 SPY   0.000828
3 TSLA  0.00511
```

The `group_by()` function makes a group out of every unique entry in the Stock variable. Since, there are three unique entries (AAPL, SPY and TSLA), it created three groups. The `summarise()` function allows us to combine (or use) all of the entries of a particular stock to find a summary statistic. Since we specified `mean()`, the command returns the mean of the particular stock returns. It seems like TSLA has the highest mean return at 0.5%.

We can keep adding to the table by specifying other measures. To calculate the variance we can use the `var()` function, for the standard deviation we can use the `sd()` function, and for the coefficient of variation we can find the ratio between the standard deviation and the mean.

```r
returns %>% group_by(Stock) %>%
  summarise(Mean=mean(Return),
            Variance=var(Return),
            SD=sd(Return),
            CV=SD/Mean*100)
```

```
# A tibble: 3 x 5
  Stock     Mean Variance     SD    CV
  <chr>    <dbl>    <dbl>  <dbl> <dbl>
1 AAPL  0.00173  0.000659 0.0257 1483.
2 SPY   0.000828 0.000313 0.0177 2138.
3 TSLA  0.00511  0.00252  0.0502  983.
```

TSLA seems to have the highest variation when following the standard deviation. However, if we look at the variation as a percentage of the mean, SPY seems to have the highest variation for the period considered. In the table below we once more use the `group_by()` and `summarise()` functions to calculate the other measures of dispersion.

```r
returns %>% group_by(Stock) %>%
  summarise(Range=diff(range(Return)),
            MAD=mean(abs(Return-mean(Return))),
            Sharpe=(mean(Return)-0.0001)/sd(Return))
```

```
# A tibble: 3 x 4
  Stock Range    MAD Sharpe
  <chr> <dbl>  <dbl>  <dbl>
1 AAPL  0.251 0.0179 0.0635
2 SPY   0.203 0.0107 0.0411
3 TSLA  0.418 0.0354 0.0998
```

In this second table the range is calculated by finding the difference between the minimum and the maximum. We can retrieve the minimum and the maximum by using the `range()` function. The `diff()` function finds the difference (or range) between these two numbers. The MAD is calculated straight from the formula. Mainly, finding the absolute deviations from the mean and the averaging them out. Lastly, the Sharpe ratio assumes a risk free daily rate of 0.01%. Once again, TSLA seems like the investment that yields the highest return despite its higher variability.

Below is a list of the functions used:

- The `range()` function returns the maximum and minimum of a vector of values.
- The `diff()` function finds the first difference of a vector. Position 2 minus position 1, position 3 minus position 2, position 4 minus position 3, etc.
- The `var()` function calculates the sample variance for a vector of values. To calculate the population variance, adjust the result by a factor of $(n-1)/n$.
- The `sd()` function calculates the sample standard deviation.

4.9 Exercises

The following exercises will help you practice the measures of dispersion. In particular, the exercises work on:

- Calculating the range, MAD, variance, and the standard deviation.
- Using R to calculate measures of dispersion.
- Calculating and using the Sharpe ratio to select investments.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using R functions when possible. Make sure to calculate the deviations from the mean.

1. Use the following observations to calculate the Range, MAD, Variance and Standard Deviation. Assume that the data below is the entire population.

$$70 \quad 68 \quad 4 \quad 98$$

Answer

The mean is 60, the Range is 94, the MAD is 28, the variance is 1186 and the variance is 34.44.

Start by creating a vector to hold the values:

```
Ex1<-c(70,68,4,98)
```

The range can be calculated by using the `range()` *and* `diff()` *functions in R.*

```
(Range<-diff(range(Ex1)))
```

```
[1] 94
```

Next, we can create a table by hand that captures the deviations from the mean. Let's calculate the mean first:

```r
(Average1<-mean(Ex1))
```

```
[1] 60
```

Now we can use the mean to fill out a table of deviations:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x}\rvert$
70	10	100	10
68	8	64	8
4	-56	3136	56
98	38	1444	38

The variance averages out the squared deviations $(x_i - \bar{x})^2$, the MAD averages out the absolute deviations $|x_i - \bar{x}|$, and the standard deviation is the square root of the variance.

Let's verify the variance in R:

```r
SquaredDeviations1<-(Ex1-Average1)^2
AverageDeviations1<-mean(SquaredDeviations1)
var(Ex1)*3/4
```

```
[1] 1186
```

Note that R calculates the sample variance. Hence, we must multiply the result by 3/4 to get the population variance. The standard deviation is just the square root of the variance:

```r
sqrt(AverageDeviations1)
```

```
[1] 34.43835
```

Lastly, the MAD is calculated by averaging the absolute deviations $|x_i - \bar{x}|$.

```r
AbsoluteDeviations1<-abs(Ex1-Average1)
mean(AbsoluteDeviations1)
```

```
[1] 28
```

2. Use the following observations to calculate the Range, MAD, Variance and Standard Deviation. Assume that the data below is a sample from the population.

-4 0 -6 1 -3 0

Answer

The mean is -2, Range is 7, the MAD is 2.33, the variance is 7.6 and the standard deviation is 2.76.

Here is the table of deviations from the mean:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x} \rvert$
-4	-2	4	2
0	2	4	2
-6	-4	16	4
1	3	9	3
-3	-1	1	1
0	2	4	2

We can check the results in R. Let's start with the variance:

```
Ex2<-c(-4,0,-6,1,-3,0)
var(Ex2)
```

```
[1] 7.6
```

The standard deviation can be found with the `sd()` *function:*

```
sd(Ex2)
```

```
[1] 2.75681
```

The MAD is given by:

```
(MAD<-mean(abs(Ex2-mean(Ex2))))
```

```
[1] 2.333333
```

Lastly, the range:

```
diff(range(Ex2))
```

```
[1] 7
```

Exercise 2

You will need the **Stocks** data set to answer this question. You can find this data at https://jagelves.github.io/Data/Stocks.csv The data is a sample of daily stock prices for ticker symbols TSLA (Tesla), VTI (S&P 500) and GBTC (Bitcoin).

1. Calculate the standard deviations for each stock. Which stock had the lowest standard deviation?

Answer

For the sample taken, GBTC has the less variation. The standard deviation of GBTC is 9.43, which is less than 16.57 for VTI or 50.38 for TSLA.

Start by loading the data set from the website. Since the file is in csv format, we will use the ***`read.csv()`*** *function.*

```
StockPrices<-read.csv("https://jagelves.github.io/Data/Stocks.csv")
```

Let's start with the standard deviation of the Tesla stock. The standard deviation is given by:

```
sd(StockPrices$TSLA)
```

```
[1] 50.38092
```

Next, let's find the standard deviation for the S&P 500 or VTI. The standard deviation is given by:

```
sd(StockPrices$VTI)
```

```
[1] 16.5731
```

Finally, let's calculate the standard deviation for GBTC or Bitcoin.

```r
sd(StockPrices$GBTC)
```

```
[1] 9.434213
```

2. Calculate the MAD. Does your answer in 1. remain the same?

Answer

The answer is the same, since the MAD for GBTC is 8.46 which is lower than 14.27 for VTI or 41.67 for TSLA.

To calculate the MAD for TSLA we can use the following command:

```r
(MADTSLA<-mean(abs(StockPrices$TSLA-mean(StockPrices$TSLA))))
```

```
[1] 41.67163
```

The MAD for VTI is:

```r
(MADVTI<-mean(abs(StockPrices$VTI-mean(StockPrices$VTI))))
```

```
[1] 14.27169
```

The MAD for GBTC is:

```r
(MADGBTC<-mean(abs(StockPrices$GBTC-mean(StockPrices$GBTC))))
```

```
[1] 8.458029
```

3. Finally, calculate the coefficient of variation. Any changes to your conclusions?

Answer

By considering the magnitudes of the stock prices, it seems like VTI is the less volatile stock. VTI has a CV of 0.08 which is lower than 0.44 for GBTC or 0.18 for TSLA. In fact, by CV Bitcoin seems to be the most risky asset.

The coefficients of variations are as follows. For TSLA the CV is:

```r
(CVTSLA<-sd(StockPrices$TSLA)/mean(StockPrices$TSLA))
```

```
[1] 0.1793755
```

For VTI the CV is:

```
(CVVTI<-sd(StockPrices$VTI)/mean(StockPrices$VTI))
```

```
[1] 0.07970004
```

For GBTC we get:

```
(CVGBTC<-sd(StockPrices$GBTC)/mean(StockPrices$GBTC))
```

```
[1] 0.4442497
```

Exercise 3

Install the `ISLR2` package. You will need the **Portfolio** data set to answer this question. The data has 100 records of the returns of two stocks.

1. Calculate the mean and standard deviation for each stock. Which investment has higher returns on average? Which investment is safest as measured by the standard deviation?

Answer

The best performing stock on average is stock X. It has an average return of -0.078% vs. 0.097% for stock Y. The safest stock is stock X as well, since the standard deviation is 1.062 percentage points vs. 1.14 percentage points for stock Y.

Start by loading the `ISLR2` package:

```
library(ISLR2)
```

Next, calculate the mean for stock X:

```
mean(Portfolio$X)
```

```
[1] -0.07713211
```

and stock Y.

```
mean(Portfolio$Y)
```

```
[1] -0.09694472
```

Then, calculate the standard deviation for stock X

```r
sd(Portfolio$X)
```

```
[1] 1.062376
```

and stock Y.

```r
sd(Portfolio$Y)
```

```
[1] 1.143782
```

2. Use a Risk Free rate of return of 3.5% to calculate the Sharpe ratio for each stock. Which stock would you recommend?

Answer

The Sharpe Ratio measures the excess return per unit of risk taken. Stock X has the better Sharpe Ratio. -0.106 vs. -0.115. Stock X is recommended since it provides a higher excess return per unit of risk taken.

To calculate Sharpe Ratios use both the average return, and the standard deviation. For stock X, the Sharpe Ratio is:

```r
(mean(Portfolio$X)-0.035)/sd(Portfolio$X)
```

```
[1] -0.1055484
```

The Sharpe Ratio for stock Y:

```r
(mean(Portfolio$Y)-0.035)/sd(Portfolio$Y)
```

```
[1] -0.1153583
```

3. Calculate the average return for a portfolio that has 30% of stock X and 70% of stock Y. What is the standard deviation of the portfolio?

Answer

The portfolio has an average return of -0.091 which is worse than stock X but better than stock Y. The standard deviation is 1.00. This is better than stock X and Y separately. The Sharpe ratio of -0.091 is also better for the portfolio than for each stock individually.

The mean of the portfolio is given by:

```r
(mean_return=0.3*mean(Portfolio$X)+0.7*mean(Portfolio$Y))
```

```
[1] -0.09100094
```

The covariance matrix is given by:

```r
(risk<-cov(Portfolio))
```

```
          X         Y
X 1.1286424 0.6263583
Y 0.6263583 1.3082375
```

Using the matrix we can now calculate the standard deviation:

```r
(standard<-sqrt(t(c(0.3,0.7)) %*% (risk %*% c(0.3,0.7))))
```

```
         [,1]
[1,] 1.002838
```

Finally, the Sharpe ratio for the portfolio is:

```r
mean_return/standard[1]
```

```
[1] -0.09074338
```

Exercise 4

The monthly returns (in percent) for a small project are:{2.1,-0.5,3.0,1.2,−1.8,0.7,2.5,−0.3}. Compute the sample mean, sample variance, sample standard deviation, coefficient of variation (CV) (in percent), and the mean absolute deviation (MAD). Briefly interpret the CV: what does it tell you about the returns relative to their mean?

Answer

You can use the `summarise()` *function in r to get the list of sample statistics.*

```r
library(tidyverse)

returns <- tibble(month = 1:8,
                  ret = c(2.1, -0.5, 3.0, 1.2, -1.8, 0.7, 2.5, -0.3))

returns %>%summarise(n = n(),
                     Mean = mean(ret),
                     Var_sample = var(ret),
                     SD = sd(ret),
                     CV = sd(ret) / mean(ret) * 100,
                     MAD = mean(abs(ret - mean(ret)))
                     )
```

```
# A tibble: 1 x 6
      n  Mean Var_sample    SD    CV   MAD
  <int> <dbl>      <dbl> <dbl> <dbl> <dbl>
1     8 0.862       2.75  1.66  192.  1.34
```

The CV is about 192%, which is large: the standard deviation is almost twice the mean. That indicates the returns are highly variable relative to their average — the mean is small compared with typical month-to-month fluctuations.

5 Descriptive Stats V

There are statistical measures that describe the shape and distribution of the data beyond simple measures of central location or dispersion. They provide a view of how data is spread out, where it concentrates, and how it deviates from what might be expected. The tools shown below will help you describe the data's shape, symmetry, and anomalies.

5.1 Quantiles and Percentiles

A **quantile** is a location within a set of ranked numbers (or distribution), below which a certain proportion, q, of that set lie. If we instead express quantiles as a percentage, they are referred to as **percentiles**.

Ex: Imagine all your data points lined up from smallest to largest. If you say you're looking at the 25th percentile, it means you're finding the value below which 25% of your data falls. If you had 100 test scores, the 25th percentile would be the score where 25 students scored lower than that, and 75 scored higher or equal. It's a way to see where a value stands in relation to the rest of the data in terms of percentage.

To calculate a percentile we follow the steps below:

1. Sort the data in ascending order. Each point has now a location. The smallest number will be first, the second smallest number will be second, etc.
2. Compute the location of the percentile desired using $L_p = \frac{(n+1)P}{100}$ where L_p is the location (in the sorted data) of the P_{th} percentile, and P is the percentile desired.
3. The data point at location L_p, is the the P_{th} percentile.

Example: Let's use the IQ scores for a group of students to find the 25th percentile. $IQ = \{80, 100, 110, 75, 130, 90\}$.

1. We sort the data: $IQ_{sorted} = \{75, 80, 90, 100, 110, 130\}$
2. Find the location of the 25th percentile: $L_{25} = 7 \times 0.25 = 1.75$. The 25th percentile is in the position 1.75 of the sorted data.
3. Retrieve the 25th percentile: Since position 1 is 75 and position 2 is 80, the 25th percentile lies 0.75 of the way between position 1 and 2. Hence, the 25th percentile is $P_{25} = 75 + 0.75(80 - 75) = 78.75$.

5.2 Chebyshev's Theorem

Chebyshev's Theorem is an important theorem, as it helps you form an expectation of the proportion of data that must lie between a given standard deviation from the mean. This offers a baseline to understanding the range and distribution of your data, and aids in detecting outliers. Formally, **Chebyshev's Theorem** states that regardless of the shape of the distribution, at least $(1 - 1/z^2)\%$ of the data lies between z standard deviations from the mean.

Ex: For a given data set, we want to know at least how much of the data is between two standard deviations. Substituting 2 into Chebyshev's formula yields $1-1/4=0.75. Hence, 75% of the data lies between two standard deviations from the mean.

5.3 The Empirical Rule

Whereas Chebyshev's theorem holds for any data distribution, the empirical rule is a bit more precise when looking at "bell shaped" data. Formally, the **Empirical Rule** or (68,95,99.7 rule) states that 68%, 95%, and 99.7% of the data lies between 1, 2, and 3 standard deviations from the mean respectively. The rule requires that the data be bell shape (normally) distributed.

5.4 Outliers Z-Scores

Given the boundaries set by both the empirical rule and Chebyshev's theorem, we can classify points as being common (normal) and not common (outliers). Specifically, **outliers** are extreme deviations from the mean. They are values that are not "common" or rarely occurring. Since both the empirical rule and Chebyshev's theorem state that a large proportion of the data is between three standard deviations, it would be uncommon to have a data point that is more that three standard deviations away from the mean.

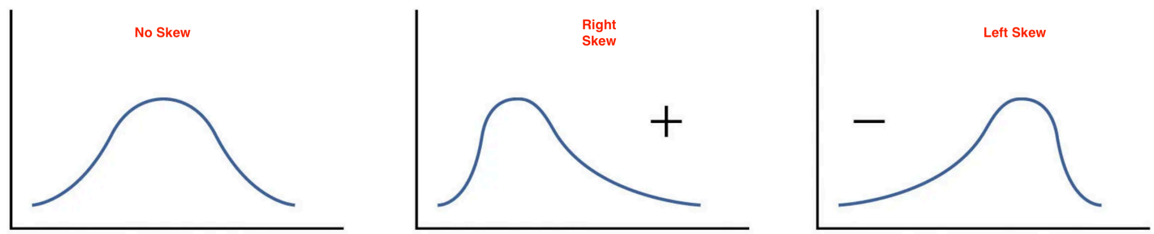
To identify outliers we use a z-score, which is a measure of distance from the mean in units of standard deviations. It can be calculated for any data point in your variable by using the formula $z_i = \frac{x_i - \bar{x}}{s_x}$. By definition, z-scores above 3 are suspected to be outliers.

Ex: On Jan 22, 2006 Kobe Bryant scored 81 points against the Toronto Raptors. He had averaged 30 point per game with a standard deviation of 4 points. If we calculate the z-score we get: $z_{81} = \frac{81-30}{4} = 12.5$. This mean that 81 is 12.5 standard deviations away from the mean, making this value extremely rare.

5.5 Skew

A measurement of skew, identifies asymmetry in the distribution of data. If most of the data leans towards one side, it's skewed. If it leans to the left, it's left-skewed or negatively skewed, meaning the tail on the left side is longer. If it leans to the right, it's right-skewed or positively skewed, with a longer tail on the right. If the data is evenly distributed, it's not skewed at all, it's symmetric. To determine if the data is skewed, calculate the **Pearson's Coefficient of Skew**. $Sk = \frac{3(\bar{x} - Median)}{s_x}$. The distribution is skewed to the left if $Sk < 0$, skewed to the right is $Sk > 0$, and symmetric if $Sk = 0$.

The image below shows the different types of skew:



Ex: Assume that for a variable the mean is 10, the median is 8, and the standard deviation is 3. The Pearson coefficient of skew is equal to $Sk = 3(10-8)/3 = 2$. Since the skew is positive, we expect the distribution to be right skewed.

5.6 Five Point Summary

A popular way to summarize data is by calculating the minimum, first quartile, median, third quartile and maximum (five point summary). This gives us a good idea of how data is distributed. We can additionally inquire how the middle 50% of the data varies. Recall, that we can use a range to assess dispersion. The **interquartile range (IQR)** quantifies the dispersion of the middle 50% of the data. Formally, the IQR is the difference between the third quartile (75th percentile) and the first quartile (25th percentile).

Example: Let's use the IQ scores for a group of students once more. Recall that the data is given by $IQ = \{80, 100, 110, 75, 130, 90\}$. The minimum and the maximum are easily identified as $Max = 130$ and $Min = 75$. The first quartile (P_{25}) was calculated in 1.1 as 78.75. Using the same steps the third quartile (P_{75}) is 115. The median is the average between the third and fourth numbers $Median = 190/2 = 95$. The five point summary is given in the table below:

Min	P_{25}	$Median$	P_{75}	Max
75	78.75	95	115	130

To calculate the interquartile range we find the difference between the 75th and 25th percentiles. $IQR = 115 - 78.75 = 36.25$ which means that the middle 50% of the data has a range of 36.25.

5.7 Outliers IQR

An alternate way to identify outliers is by using the interquartile range. This measure and the z-score method can disagree. To calculate the outliers using the IQR, we first calculate $Q_1 - 1.5(IQR)$ and $Q_3 + 1.5(IQR)$, where Q_1 is the first quartile, Q_3 is the third quartile, and IQR is the interquartile range. If the observation (x_i) is less than $Q_1 - 1.5(IQR)$ or greater than $Q_3 + 1.5(IQR)$, then it is considered an outlier.

Ex: Consider once more the IQ data. $IQ = \{80, 100, 110, 75, 130, 90\}$. The lower limit for on outlier is given by $LL = Q_1 - 1.5(IQR) = 78.75 - 1.5(36.25)$ or $LL = 24.375$. The upper limit is given by $UL = Q_3 + 1.5(IQR) = 115 + 1.5(36.25)$ or $UL = 169.375$. Any data point outside the range [24.375,169.375] is considered an outlier. In other words, 200 and 20 would be outliers, but 100 would not.

5.8 Quantiles and Quartiles in R

R quickly calculates quantiles for a given variable (vector) using the `quantile()` function. Below we use the IQ example once more.

```r
IQ <- c(80,100,110,75,130,90)
quantile(IQ, type=6)
```

```
    0%     25%     50%     75%    100% 
 75.00   78.75   95.00  115.00  130.00 
```

You will notice that an extra argument *type* has been been passed into the quantile function. Since, there are several ways to calculate quantiles, R allows you to identify which method you want to use. In sec 1.1 we explained method 6.

5.9 Outliers in R

To identify outliers we can us the `scale()` function. We'll consider the first five observations in the *faithful* data set.

```
head(scale(faithful),5)
```

```
    eruptions    waiting
1  0.09831763  0.5960248
2 -1.47873278 -1.2428901
3 -0.13561152  0.2282418
4 -1.05555759 -0.6544374
5  0.91575542  1.0373644
```

The data shown above are z-scores for the first five observations of the faithful data set. As you can see none of the observations are outliers, as they are all less that 3 standard deviations away from the mean.

If we want to filter all observations that are say 1.3 standard deviations away from the mean, we can use the following command from `tidyverse`:

```
library(tidyverse)
faithful %>% mutate(z_eruptions=scale(eruptions)) %>%
                      filter(scale(eruptions)>1.3)
```

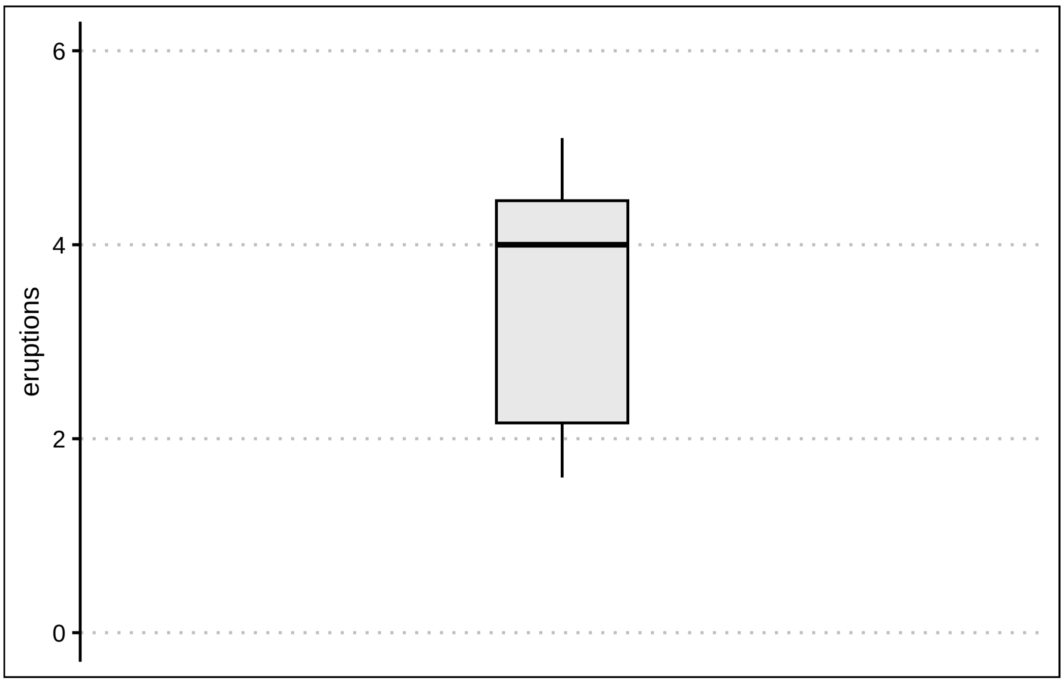
```
  eruptions waiting z_eruptions
1     5.067      76    1.383614
2     5.100      96    1.412526
3     5.033      77    1.353825
4     5.000      88    1.324912
```

This confirms that there are no outliers in the eruptions variable.

5.10 Box Plots in R

A **box plot** is a graph that shows the five point summary, outliers (if any), and the distribution of data. It can be easily constructed using `geom_boxplot()` in R. Let's use the eruptions variable once more.

```
library(ggthemes)
faithful %>%
  ggplot() +
  geom_boxplot(aes(y=eruptions),
               fill="lightgrey", alpha=0.5,
               col="black", width=0.3) +
  theme_clean() +
  scale_x_continuous(breaks = NULL, limits=c(-1,1))+
  ylim(limits=c(0,6))
```

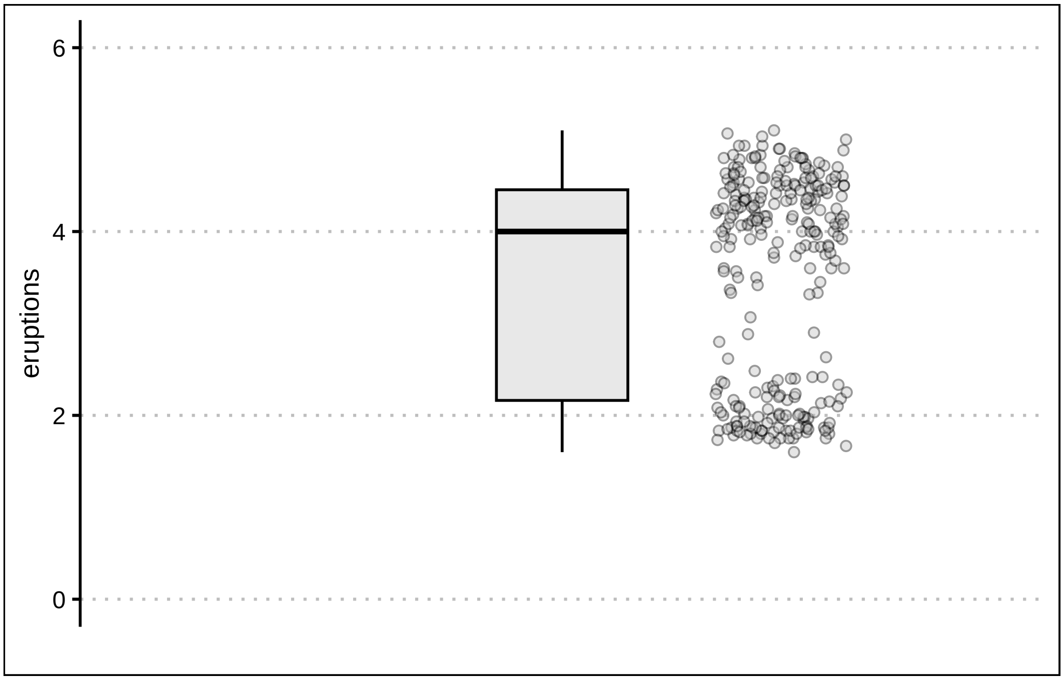


The boxplot highlights the minimum just below 2, the maximum around 5, the first quartile just above 2, the median at 4, and the third quartile just above 4. Any outlier would be shown as a point beyond the whiskers of the box plot.

We can also overlay the data to the boxplot by using the `geom_jitter()` function. The data points allow us to see how the data is distributed. The code is below:

```
faithful %>%
  ggplot() +
  geom_boxplot(aes(y=eruptions),
               fill="lightgrey", alpha=0.5,
               col="black", width=0.3) +
  theme_clean() +
```

```
  scale_x_continuous(breaks = NULL, limits=c(-1,1))+
  ylim(limits=c(0,6)) + labs(x="") +
  geom_jitter(aes(y=eruptions,x=0.5),
              width=0.15, pch=21,
              alpha=0.4, bg="grey")
```



5.11 Skewness in R

Unfortunately, base R does not have a function to calculate the Pearson Coefficient of skewness. Hence, we will have to install the `e1071` package. The package contains the function `skewness()` which makes the estimation of skew simple. Below is the code to calculate the skewness for the eruptions variable.

```
library(e1071)
skewness(faithful$eruptions)
```

```
[1] -0.4135498
```

The result indicates that the distribution is mildly skewed to the left.

Here is a summary of the functions used in this section:

- The `quantile()` function returns the five point summary when no arguments are specified. For a specific quantile, specify the *probs* argument.
- The `scale()` function calculates the z-scores for a vector of values.
- The `geom_boxplot()` command returns a box plot for a vector of values.
- The `skewness()` function is found in the `e1071` package. It takes as an input a vector of values and returns the Pearson Coefficient of Skew.

5.12 Exercises

The following exercises will help you practice other statistical measures. In particular, the exercises work on:

- Constructing a five point summary and a boxplot.
- Applying Chebyshev's Theorem.
- Identifying skewness.
- Identifying outliers.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using R functions when possible.

1. Use the following observations to calculate the minimum, the first, second and third quartiles, and the maximum. Are there any outliers? Find the IQR to answer the question.

$$3 \quad 10 \quad 4 \quad 1 \quad 0 \quad 30 \quad 6$$

Answer

The minimum is 0, the first quartile is 2, second quartile is 4, third quartile is 8, and maximum is 30. 30 is an outlier since it is beyond $Q_3 + 1.5 \times IQR$.

Quartiles are calculated using the percentile formula $(n+1)P/100$. The data set has seven numbers. The first quartile's location is $8/4 = 2$, the second quartile's location is $8/2 = 4$ and the third quartile's location is $24/4 = 6$. The values at these location, when data is organized in ascending order, are 1, 4, and 10.

*In R we can get the five number summary by using the **`quantile()`** function. Since there are various rules that can be used to calculate percentiles, we specify type* 6 to match our rules.

```
Ex1<-c(3,10,4,1,0,30,6)
quantile(Ex1,type = 6)
```

```
  0%  25%  50%  75% 100% 
   0    1    4   10   30 
```

The interquartile range is needed to determine if there are any outliers. The IQR for this data set is $Q_3 - Q_1 = 9$. This reveals that 30 is and outlier, since $10 + 1.5 \times 9 = 23.5$. Everything beyond 23.5 is an outlier.

2. Confirm your finding of an outlier by calculating the z-score. Is 30 an outlier when using a z-Score?

Answer

If we use the z-score instead we find that 30 is not an outlier since the z-score is $Z_{30} = 2.15$. This observation is only 2.15 standard deviations away from the mean.

In R we can make a quick calculation of the z-Score to confirm our results. The z-score is given by $Z_i = \frac{x_{30} - \mu}{\sigma}$.

```
(Z30<-(30-mean(Ex1))/sd(Ex1))
```

```
[1] 2.148711
```

3. Use Chebyshev's theorem to determine what percent of the data falls between the z-score found in 2.

Answer

Chebyshev's theorem states that $1 - \frac{1}{z_2}$ of the data lies between z standard deviation from the mean.

Substituting the z-score found in 2. we get 78.34% of the data lies between the standard deviation calculated. In R:

```
1-1/(Z30)^2
```

```
[1] 0.7834073
```

Exercise 2

You will need the **Stocks** data set to answer this question. You can find this data at https://jagelves.github.io/Data/Stocks.csv The data is a sample of daily stock prices for ticker symbols TSLA (Tesla), VTI (S&P 500) and GBTC (Bitcoin).

1. Construct a boxplot for Stock A. Is the data skewed or symmetric?

Answer

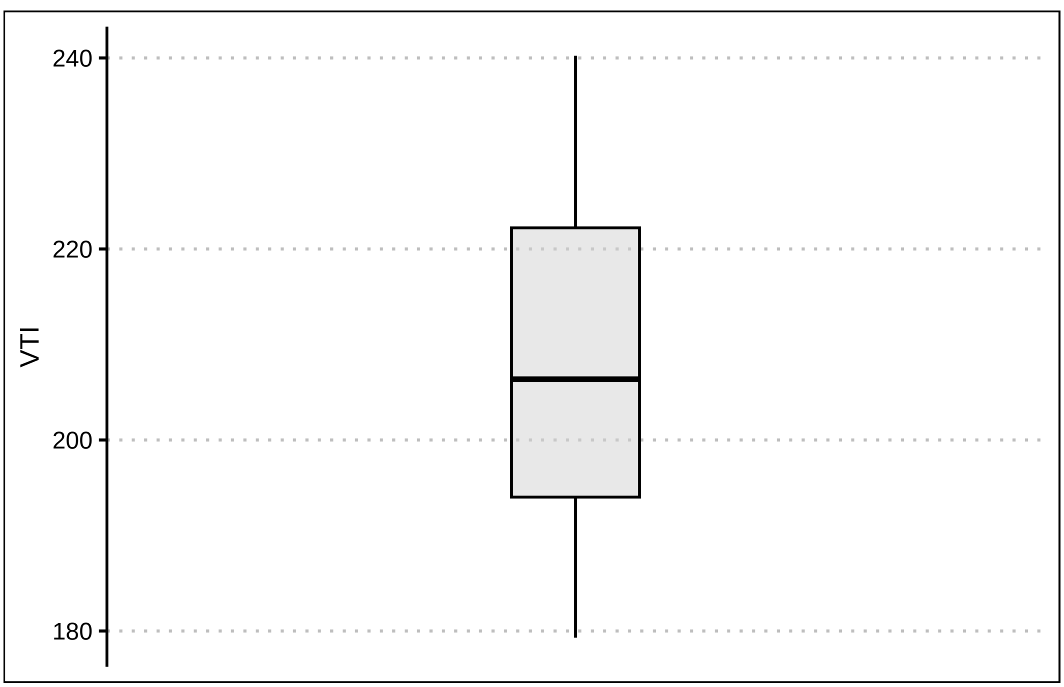
The data is skewed to the right.

Start by loading the data set:

```
StockPrices<-read.csv("https://jagelves.github.io/Data/Stocks.csv")
```

To construct the boxplot in R, use the ***boxplot()*** *command.*

```
StockPrices %>%
  ggplot() +
  geom_boxplot(aes(y=VTI),
               fill="lightgrey", alpha=0.5,
               col="black", width=0.3) +
  theme_clean() +
  scale_x_continuous(breaks = NULL, limits=c(-1,1))
```



The boxplot shows that there are no outliers. The data also looks like it has a slight skew to the right.

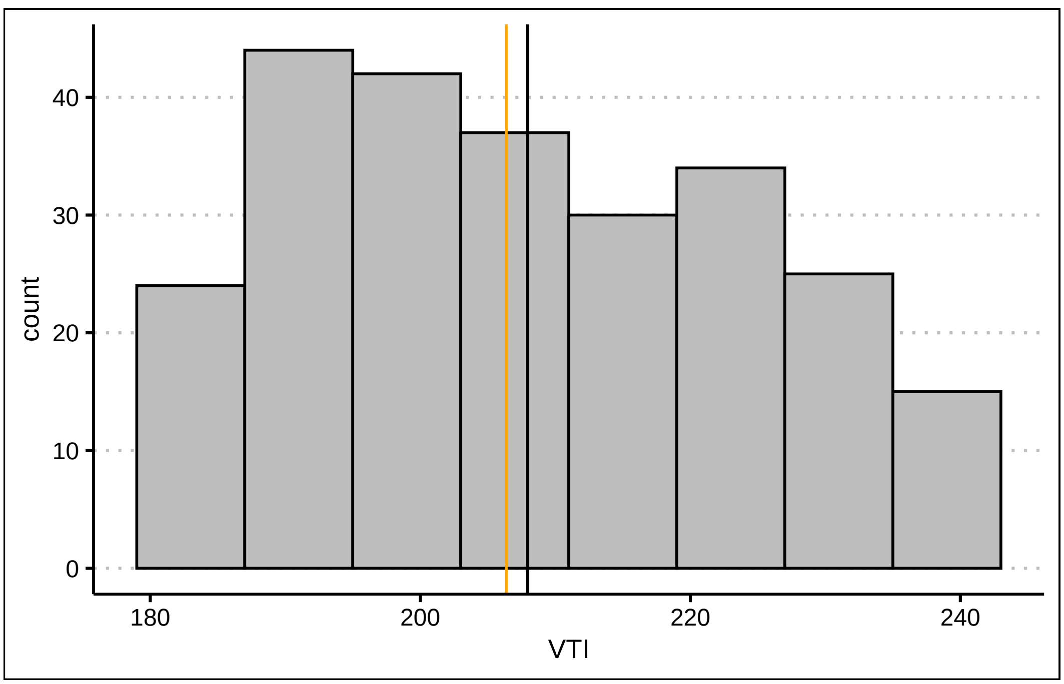
2. Create a histogram of the data. Include a vertical line for the mean and median. Explain how the mean and median indicates a skew in the data. Calculate the skewness statistic to confirm your result.

Answer

The mean is more sensitive to outliers than the median. Hence, when the data is skewed to the right we expect that the mean is larger than the median.

Let's construct a histogram in R to search for skewness.

```
StockPrices %>% ggplot() +
  geom_histogram(aes(VTI), bins = 8,
                 binwidth = 8,
                 col="black", bg="grey",
                 boundary=179) +
  theme_clean() +
  geom_vline(xintercept = mean(StockPrices$VTI), col="black")+
  geom_vline(xintercept = median(StockPrices$VTI), col="orange")
```



The lines are close to each other but the mean is slightltly larger than the median. Let's confirm with the skewness statistic $3(mean - median)/sd$.

```
(skew<-3*(mean(StockPrices$VTI-median(StockPrices$VTI))/sd(StockPrices$VTI)))
```

```
[1] 0.2856304
```

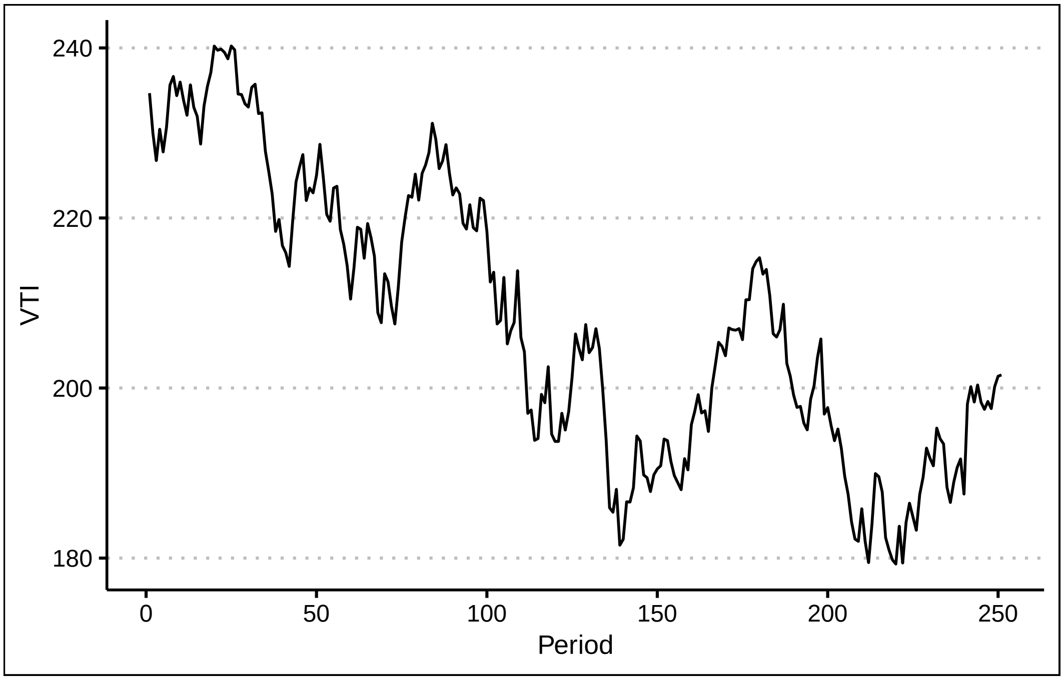
This indicates that there is a slight skew to the right of the data.

3. Use a line chart to plot your data. Can you explain why the data has a skew?

Answer

The line chart indicates that the data has a downward trend in the early periods. This creates a few points that are large. In later periods the stock price stabilizes to lower levels.

```
StockPrices %>% ggplot() +
  geom_line(aes(y=VTI, x=seq(1,length(VTI)))) + theme_clean() +
  labs(x="Period")
```



Exercise 3

You will need the **mtcars** data set to answer this question. This data set is part of R. You don't need to download any files to access it.

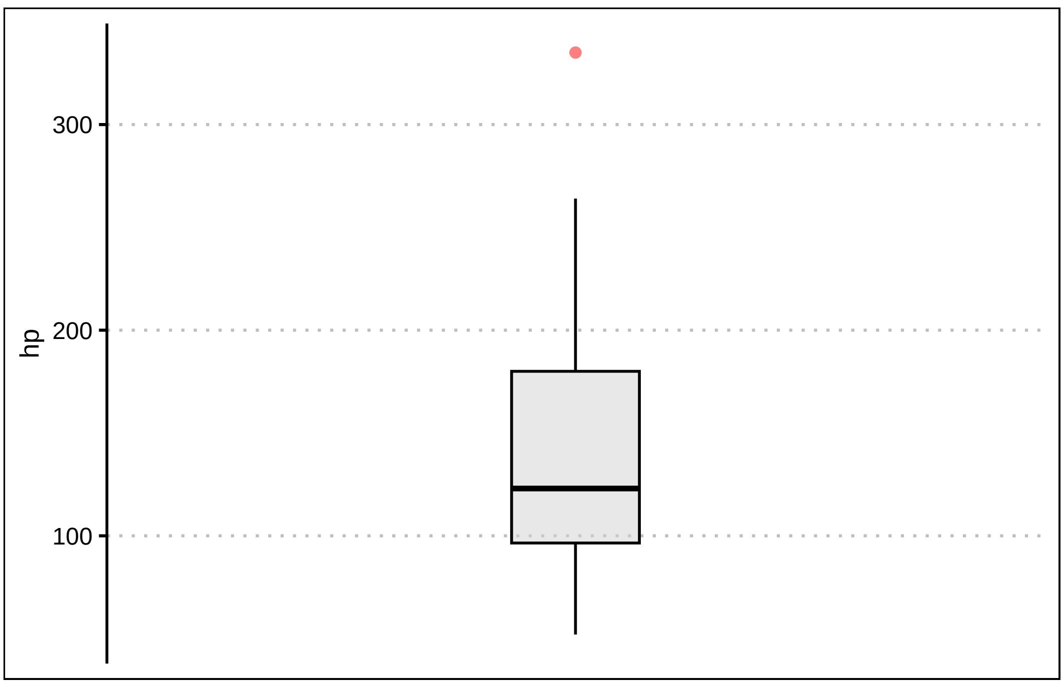
1. Construct a boxplot for the *hp* variable. Write a command in R that retrieves the outlier. Which car is the outlier?

Answer

The outlier is the Masserati Bora. The horse power is 335.

In R we can construct a boxplot with the following command:

```
mtcars %>%
  ggplot() +
  geom_boxplot(aes(y=hp),
               fill="lightgrey", alpha=0.5,
               col="black", width=0.3,
               outlier.colour = "red") +
  theme_clean() +
  scale_x_continuous(breaks = NULL, limits=c(-1,1))
```



From the graph it seems like the outlier is beyond a horsepower of 275. Let's write an R command to retrieve the car.

```
mtcars %>% filter(hp>300)
```

```
              mpg cyl disp  hp drat   wt qsec vs am gear carb
Maserati Bora  15   8  301 335 3.54 3.57 14.6  0  1    5    8
```

It's the Masserati Bora!

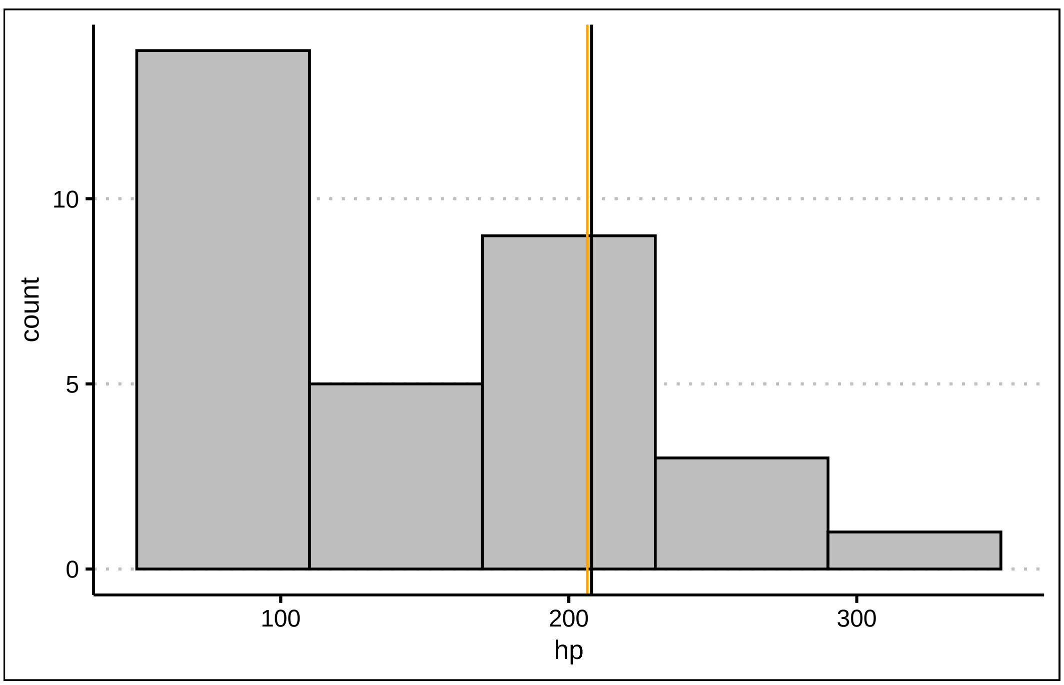
2. Create a histogram of the data. Is the data skewed? Include a vertical line for the mean and median. Calculate the skewness statistic to confirm your result.

Answer

The histogram looks skewed to the right. This is confirmed by the estimation of a Pearson coefficient fo skewness of 1.04.

In R we can construct a histogram with vertical lines for the mean and median with the following code:

```
mtcars %>% ggplot() +
  geom_histogram(aes(hp), bins = 5,
                 binwidth = 60,
                 col="black", bg="grey",
                 boundary=50) +
  theme_clean() +
  geom_vline(xintercept = mean(StockPrices$VTI), col="black")+
  geom_vline(xintercept = median(StockPrices$VTI), col="orange")
```



The histogram looks skewed to the right. Pearson's Coefficient of Skewness is:

```
(SkewHP<-3*(mean(mtcars$hp)-median(mtcars$hp))/sd(mtcars$hp))
```

```
[1] 1.036458
```

3. Transform the data by taking a natural logarithm. Specifically, create a new variable called *Loghp*. Repeat the procedure in 2. Is the skew still there?

Answer

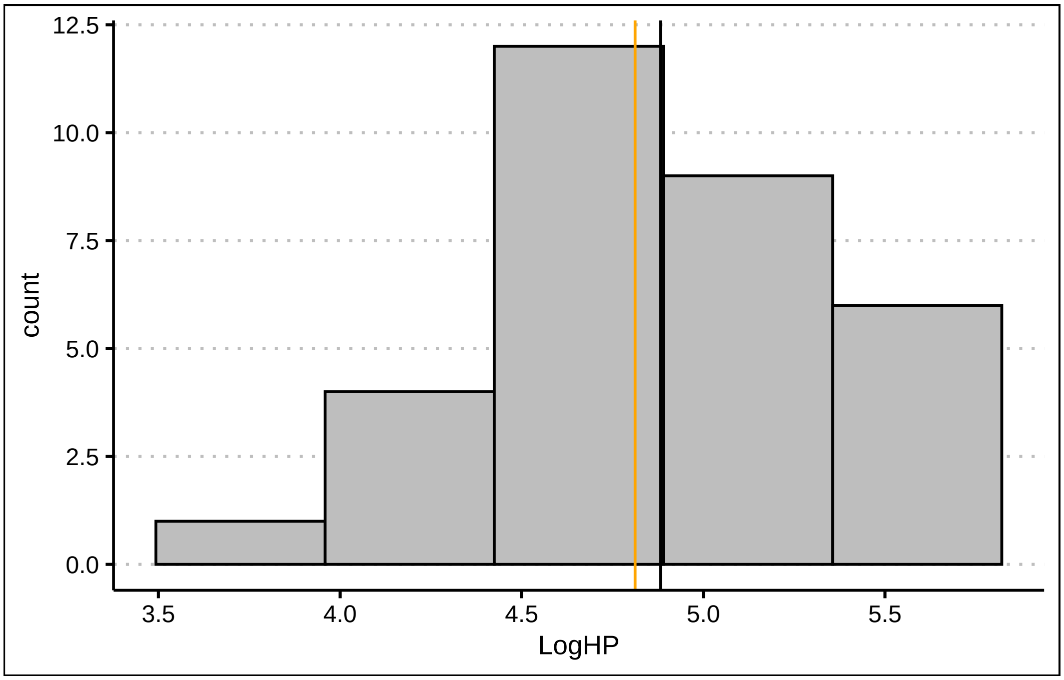
The skew is still there, but the distribution now look more symmetrical and the Skew coefficient has decreased to 0.44.

In R we can create an new variable that captures the log transformation. The `log()` *function takes the natural logarithm of a number or vector.*

```
LogHP<-log(mtcars$hp)
```

Let's use this new variable to create our histogram:

```
ggplot() +
  geom_histogram(aes(LogHP), bins = 5,
                 col="black", bg="grey") +
  theme_clean() +
  geom_vline(xintercept = mean(LogHP), col="black")+
  geom_vline(xintercept = median(LogHP), col="orange")
```



The mean and the variance now look closer together. The tail of the distribution (skew) now also looks diminished. The Skewness coefficient has decreased significantly:

```
(SkewLogHP<-3*(mean(LogHP)-median(LogHP))/sd(LogHP))
```

```
[1] 0.4402212
```

6 Regression I

Measures of association are essential tools in business statistics for analyzing relationships between variables. They help determine whether changes in one variable are linked to changes in another, providing valuable insights into patterns and dependencies. Understanding these relationships is critical for making informed decisions. By quantifying the strength and direction of associations, businesses can better interpret data, optimize strategies, and drive meaningful outcomes. Below we study important measures of association.

6.1 The Covariance

The **covariance** is a measure that determines the direction of the relationship between two variables. It is calculated by:

$$s_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

The result of the covariance indicates the relationship between the two variables. If $s_{xy} > 0$ there is a direct relationship, if $s_{xy} < 0$ there is an inverse relationship, and if $s_{xy} = 0$ there is no relationship.

Example: Let's consider the following data that captures the price of stocks (SPY) and bonds (BND):

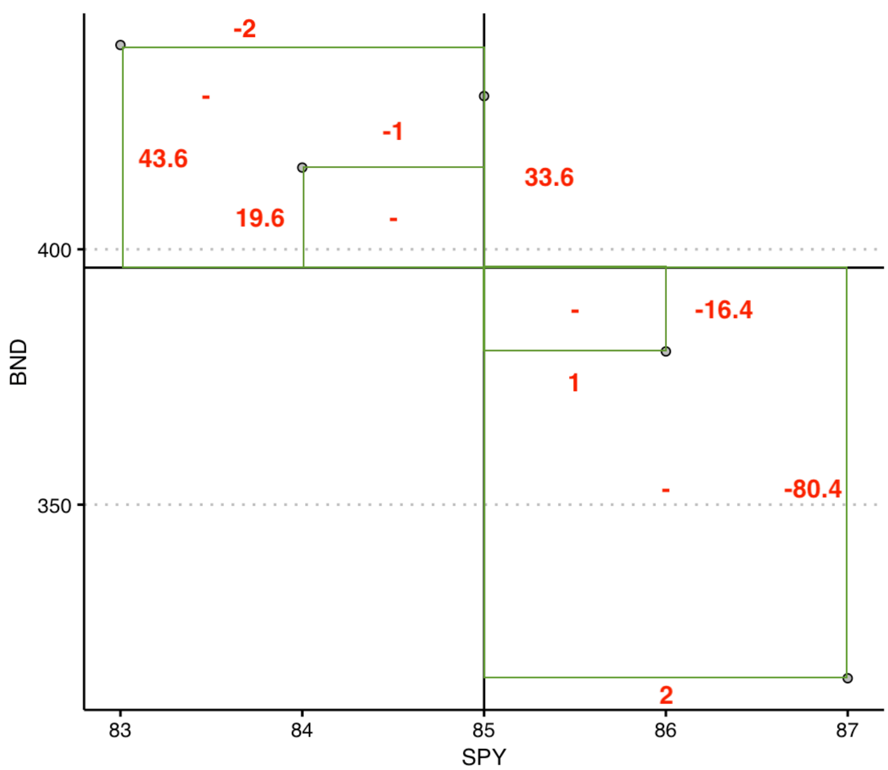
SPY	BND
87	316
86	380
84	416
85	430
83	440

The idea behind calculating a covariance is to determine whether there is a relationship between the two variables. Let's start by calculating the formula. We can do this by finding deviations from the mean for both SPY and BND. The table below shows the deviations from the mean for each variable in columns 3 and 4.

SPY	BND	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
87	316	2	-80.4	-160.8
86	380	1	-16.4	-16.4
84	416	-1	19.6	-19.6
85	430	0	33.6	0
83	440	-2	43.6	-87.2

Column 5 calculates the product between column 3 and column 4. The covariance is simply the average of the numbers of column 5 by using the degrees of freedom $n-1$. Hence, $s_{xy} = -71$. Since s_{xy} is negative we can establish that there is an inverse relationship between SPY and BND.

Intuitively, the covariance checks whether on average the products of the deviations from the mean is positive or negative. The image below explains the intuition.



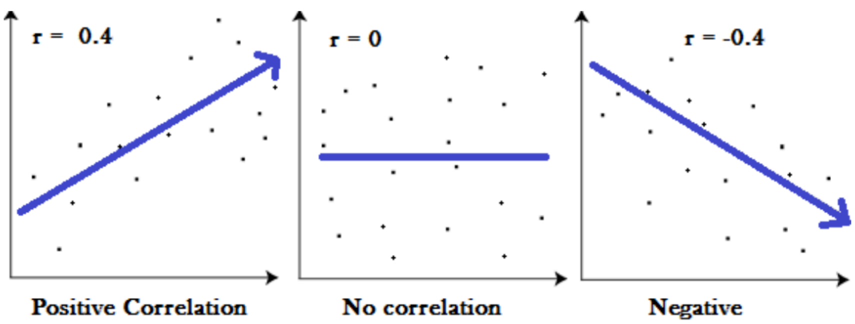
The image plots the SPY and BDN in a scatter plot. You can see that there is an inverse relationship. The vertical and horizontal lines represents the mean of SPY and the mean of BND, respectively. The red numbers are the deviations from the mean (you can compare with the table). So whenever x lies below the mean and y lies above its mean, the product is negative. This happens as well when x lies above the mean and y underneath its mean. As a result, the points are aligned so that an inverse relationship is reflected. You'll notice that the product of the deviations in these cases is negative, and ultimately measured with the covariance.

6.2 The Correlation

The **correlation** measures the strength of the linear relationship between two variables. It is calculated by

$$r = \frac{s_{xy}}{s_x s_y}$$

The correlation coefficient is between $[-1, 1]$. When the correlation coefficient is 1 (-1), there is a perfect direct (inverse) relationship between the two variables.



The image above shows three examples of correlation coefficients. When the correlation coefficient is -0.4 or 0.4 the direction of the relationship depends on the sign of the coefficient. The magnitude of 0.4 indicates that the strength of the relationship is moderate (i.e., a general linear pattern is observed but many points do not lie on the trend line). When the correlation coefficient is zero, there is no linear pattern in the relationship. When the absolute value of the correlation is 1, then there is a perfect linear relationship between the two variables.

Ex: Consider the SPY and BND data. The standard deviation of SPY is 1.58 and that of BDN is 50.37. Substituting into the correlation formula we get a correlation coefficient of $r = \frac{-71}{1.58 \times 50.37} = -0.89$. This indicates that the relationship between SPY and BND is inverse and strong.

6.3 The Coefficient of Determination (R^2)

The **coefficient of determination** or R^2, measures the percent of variation in y explained by variations in x. It is calculated by:

$$R^2 = r^2$$

The number that we get from the R^2 tells us how well a variable (x) explains the variation in the another variable (y). The number could be anywhere from zero (indicating that the two variables are unrelated), to one (indicating that one variable explains entirely the variation of the other variable).

Ex: Consider once more the SPY and BND example. The correlation coefficient is given by $r = -0.89$. The $R^2 = (-0.89)^2 = 0.79$. This indicates that about 80% of the variation in the BND can be explained by the changes in SPY. Hence, these two variables are closely related.

6.4 Measures of Association in R

R makes it very convenient to retrieve measures of association. Let's get the data from SPY and BND example into R:

```r
library(tidyverse)
library(ggthemes)

data<- tibble(SPY=c(87,86,84,85,83), BND=c(316,380,416,430,440))
```

Now we can retrieve the covariance by using the `cov()` command in R. Below is the code:

```r
cov(data$SPY,data$BND)
```

```
[1] -71
```

We confirm that the covariance is -71 and that there is an inverse relationship between the variables. To verify the correlation coefficient, we use the code below:

```r
cor(data$SPY,data$BND)
```

```
[1] -0.891549
```

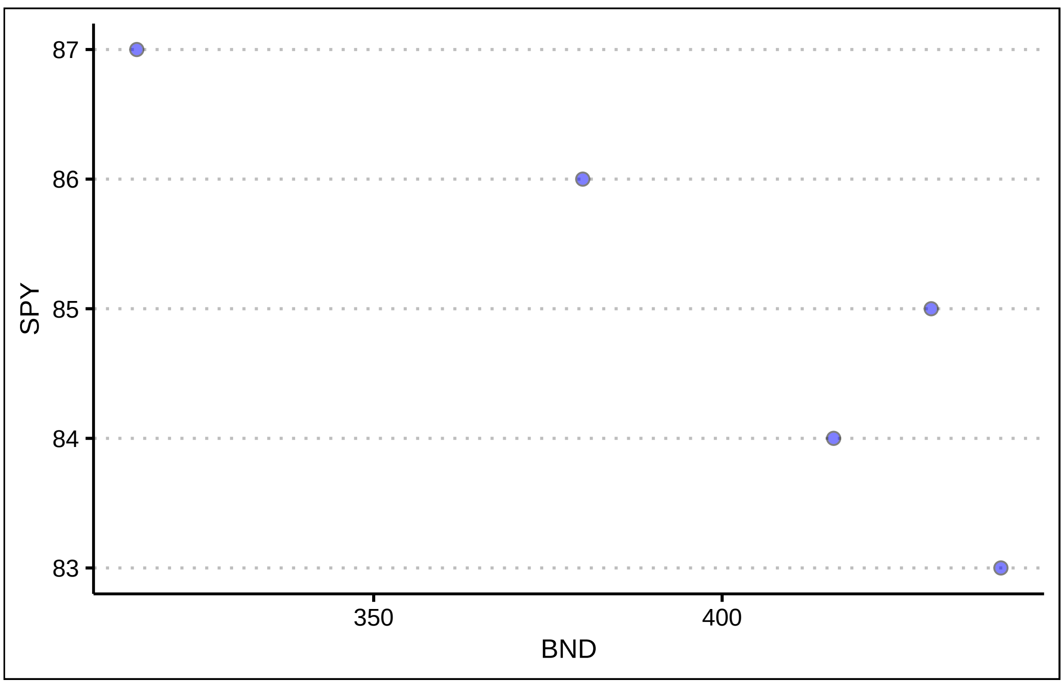
The correlation coefficient of -0.89 indicates a strong inverse linear relationship between the two variables. Lastly, the coefficient of determination is calculated below:

```
cor(data$SPY,data$BND)^2
```

```
[1] 0.7948597
```

It seems that about 80% of the variation in the price of bonds (BND) is explained by the variation in the price of stocks (SPY). We can create a visual of the relationship between the two variables by using the `geom_point()` function in R.

```
data %>% ggplot() +
  geom_point(aes(y=SPY,x=BND), col="black",
             cex=2, bg="blue", alpha=0.5, pch=21) +
  theme_clean()
```



The visualization above is called a scatter plot. A **scatter plot** displays pairs of $[x,y]$ as points on the Cartesian plane. The plot acts as a visual aid to determine the relationship between two variables. We can see that the points are inversely related to each other.

Below is a list of the functions used:

- To calculate the covariance use the `cov()` function. The input must be two vectors (variables).
- The correlation coefficient is calculated with the `cor()` function. The input must be two vectors (variables).
- The `geom_point()` function will create scatter plots. Make sure you include two variables in the `aes()` function. The argument *cex* increases the size of the points, *pch* changes the point character, *bg* selects the color, and *alpha* adjusts the transparency of the background color (bg).

6.5 Exercises

The following exercises will help you understand statistical measures that establish the relationship between two variables. In particular, the exercises work on:

- Calculating covariance and correlation.
- Using R to plot scatter diagrams.
- Calculating the coefficient of determination.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using R functions when possible.

1. Consider the data below. Calculate the covariance and correlation coefficient by finding deviations from the mean. Use R to verify your result. Is there a direct or inverse relationship between the two variables? How strong is the relationship?

x	20	21	15	18	25
y	17	19	12	13	22

Answer

The covariance is 14.9 and the correlation is 0.96. The results indicate that there is a strong direct relationship between the two variables.

Let's start by finding the deviations from the mean for the x variable in R.

```
x<-c(20,21,15,18,25)
(devx<-x-mean(x))
```

```
[1]  0.2  1.2 -4.8 -1.8  5.2
```

We will do the same with y:

```
y<-c(17,19,12,13,22)
(devy<-y-mean(y))
```

```
[1]  0.4  2.4 -4.6 -3.6  5.4
```

Note that when the deviations in x are negative (positive), they are also negative (positive) in y. This is indicative of a direct relationship between the two variables. The covariance is given by:

```
(Ex1Cov<-sum(devx*devy)/(length(devx)-1))
```

```
[1] 14.9
```

We can verify this by using `cov()` *function in R.*

```
cov(x,y)
```

```
[1] 14.9
```

The correlation coefficient is found by dividing the covariance over the product of standard deviations. In R:

```
(Ex1Cor<-Ex1Cov/(sd(x)*sd(y)))
```

```
[1] 0.9678386
```

We can once more verify the result in R with the built in function `cor()`*.*

```
cor(x,y)
```

```
[1] 0.9678386
```

2. Consider the data below. Calculate the covariance and correlation coefficient by finding deviations from the mean. Use R to verify your result. Is there a direct or inverse relationship between the two variables? How strong is the relationship?

w	19	16	14	11	18
z	17	20	20	16	18

Answer

The covariance is 0.85 and the correlation is 0.148. The results indicate that there is a very weak direct relationship between the two variables. They might be unrelated.

Let's start with w and finding the deviations from the mean in R.

```
w<-c(19,16,14,11,18)
(devw<-w-mean(w))
```

```
[1]  3.4  0.4 -1.6 -4.6  2.4
```

We will do the same with z:

```
z<-c(17,20,20,16,18)
(devz<-z-mean(z))
```

```
[1] -1.2  1.8  1.8 -2.2 -0.2
```

The covariance is given by:

```
(Ex2Cov<-sum(devw*devz)/(length(devz)-1))
```

```
[1] 0.85
```

We can verify this with the `cov()` *function in R.*

```
cov(w,z)
```

```
[1] 0.85
```

The correlation coefficient is found by dividing the covariance over the product of standard deviations. In R:

```
(Ex2Cor<-Ex2Cov/(sd(z)*sd(w)))
```

```
[1] 0.1480558
```

We can once more verify the result in R with the built in function `cor()`.

```
cor(w,z)
```

```
[1] 0.1480558
```

Exercise 2

You will need the **mtcars** data set to answer this question. This data set is part of R. You don't need to download any files to access it.

1. Calculate the correlation coefficient between *hp* and *mpg*. Explain the results. Specifically, the direction of the relationship and the strength given the context of the problem.

Answer

The correlation coefficient is -0.78. This is indicative of a moderately strong inverse relationship between mpg and mp.

In R we can easily calculate the correlation coefficient with the `cor()` function.

```
cor(mtcars$mpg,mtcars$hp)
```

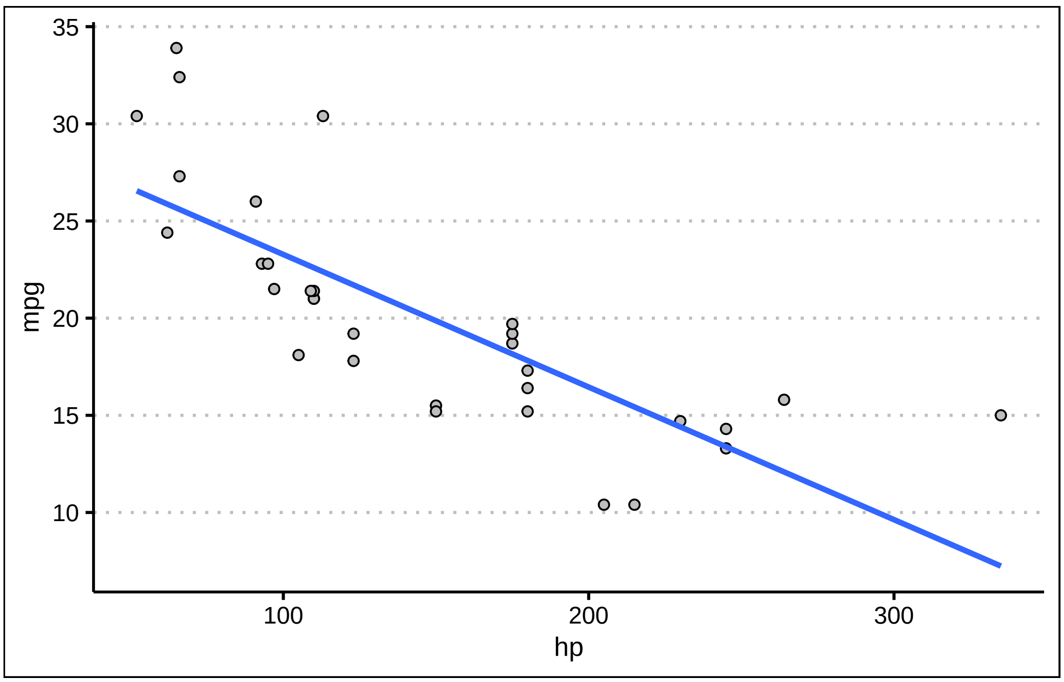
```
[1] -0.7761684
```

2. Create a scatter diagram of the two variables. Is the scatter diagram what you expected after you calculated the correlation coefficient?

Answer

The scatter diagram is downward sloping. Most points are close to the trend line. It is what was expected from a correlation coefficient of -0.78.

```
library(tidyverse)
library(ggthemes)
mtcars %>% ggplot() +
  geom_point(aes(y=mpg,x=hp), col="black",
             bg="grey", pch=21) +
  geom_smooth(aes(y=mpg,x=hp), formula=y~x,
              method="lm", se=F) +
  theme_clean()
```



3. Calculate the coefficient of determination. How close is it to one? What else could be explaining the variation in the *mpg*? Let your dependent variable be *mpg*.

Answer

The coefficient of determination is 0.6. This value is not very close to one. This is expected since miles per gallon can also vary because of the cars weight, and fuel efficiency. It makes sense that the hp only explains 60% of the total variation.

In R we can calculate the coefficient of determination by squaring the correlation coefficient.

```
cor(mtcars$mpg,mtcars$hp)^2
```

```
[1] 0.6024373
```

Exercise 3

You will need the **College** data set to answer this question. You can find this data set here: https://jagelves.github.io/Data/College.csv

1. Create a scatter diagram between *GRAD_DEBT_MDN* (Median Debt) and *MD_EARN_WNE_P10* (Median Earnings). What type of relationship do you observe between the variables?

Answer

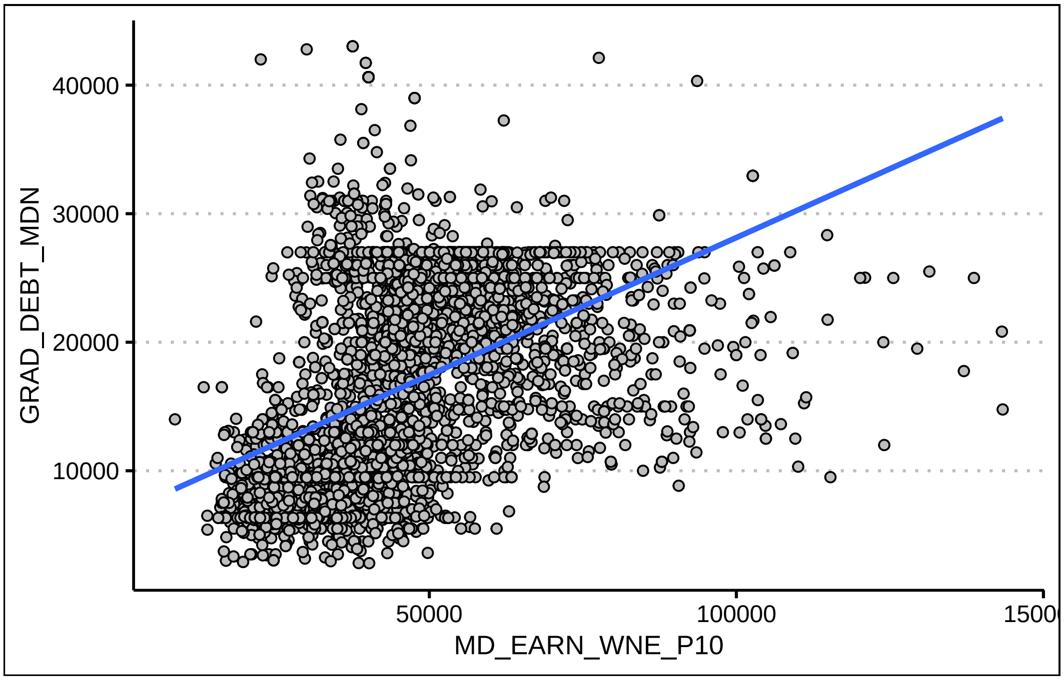
It seems like there is a direct relationship between both variables. The more debt you take, the higher the salary.

Start by loading the data. We'll use the `read_csv()` function:

```
library(tidyverse)
College<-read_csv("https://jagelves.github.io/Data/College.csv")
```

The two variables of interest are *GRAD_DEBT_MDN* and *MD_EARN_WNE_P10*. The following code creates the scatter plot:

```
College %>% ggplot() +
  geom_point(aes(y=GRAD_DEBT_MDN,x=MD_EARN_WNE_P10), col="black",
             bg="grey", pch=21) +
  geom_smooth(aes(y=GRAD_DEBT_MDN,x=MD_EARN_WNE_P10), formula=y~x,
              method="lm", se=F) +
  theme_clean()
```



2. Calculate the correlation coefficient and the coefficient of determination. According to the data, are higher debts correlated with higher earnings?

Answer

The correlation coefficient shows a moderate direct relationship between earnings and debt 0.46. The coefficient of determination indicates that only 21% of the variation in earnings can be explained by debt.

In R we can start with the correlation coefficient:

```
(Correlation<-cor(College$GRAD_DEBT_MDN,
                  College$MD_EARN_WNE_P10,"complete.obs"))
```

```
[1] 0.4615361
```

We can simply square the correlation to obtain the coefficient of determination:

```
Correlation^2
```

```
[1] 0.2130155
```

7 Regression II

Quantifying relationships between variables is an important skill in business analytics. The regression line, representing the best linear fit between two variables, enables businesses to make predictions, identify trends, and optimize strategies using data-driven insights. Understanding how to calculate and interpret a regression line provides the foundation for analyzing business relationships, such as the effect of advertising on sales or customer satisfaction on revenue. Below, we delve into generating and analyzing a regression line.

7.1 The Regression Line

The regression line is calculated to minimize the average distance (or errors) between the line and the observed data points. It is defined by two key components: a **slope** (β) and an **intercept** (α). Mathematically, the regression line is expressed as:

$$\hat{y_i} = \hat{\alpha} + \hat{\beta} x_i$$

where $\hat{y_i}$ are the predicted values of y given the x's. The **slope** determines the steepness of the line. The estimate quantifies how much a unit increase in x changes y. We can easily calculate the slope by using the covariance between y and x, and the variance of x. Mathematically we have:

$$\hat{\beta} = \frac{s_{xy}}{s_x^2}$$

The **intercept** determines where the line crosses the y axis. In other words it returns the value of y when x is zero. Once we have the slope of the regression line we can estimate the intercept by:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Example: Let's examine a data set on Price and Advertisement. In general, one expects that when a company advertises, it can convince consumers to pay more for their product. Below is the data used to calculate the regression line, followed by its interpretation.

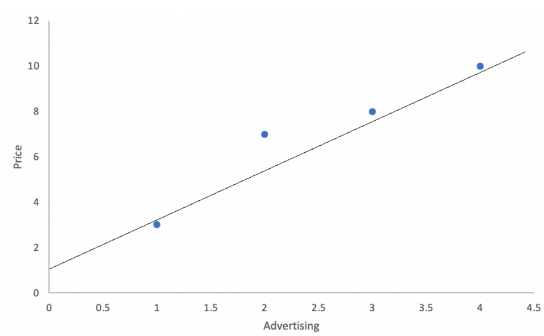
Advertisement (x)	Price (y)
2	7
1	3
3	8
4	10

The data shows a clear relationship between advertisement and price: when advertisement spending is high, price also tends to be high. This suggests a direct relationship between the two variables. In a previous chapter, we learned to quantify such relationships using covariance and the correlation coefficient.

In this section, we aim to answer two key questions:

1. Effectiveness: How much can we increase the price for every additional dollar spent on advertisement?
2. Prediction: What is the predicted price if we have a budget of 6 for advertisement?

The regression line allows us to answer these questions by quantifying the relationship between the two variables. Specifically, if we can estimate the regression line, as shown below, we can answer these questions.



Let's start by calculating the slope of the regression line, as this will help us answer the effectiveness question. The slope measures how much the price increases for every additional dollar spent on advertisement. It is calculated using the formula: $\widehat{\beta} = \frac{s_{xy}}{s_x^2}$. Given that the covariance is 3.67 and the variance of x is 1.67, the slope of the regression line is $\widehat{\beta} = \frac{3.67}{1.67} = 2.2$. This tells us that for every additional dollar spent on advertisement, price increases by 2.2 on average. Thus, we have answered the effectiveness question.

Next, we calculate the intercept to complete the regression line and answer the prediction question. The intercept represents the predicted price when advertisement spending is zero. It is calculated as: $\hat{\alpha} = \bar{y} - \widehat{\beta}\bar{x}$. Since the mean of advertisement is 2.5 and the mean of price is 7, the intercept is $\hat{\alpha} = 7 - 2.2(2.5) = 1.5$. This means that if we do not advertise, the predicted price is 1.5.

The regression line can be completed by using the intercept and slope that we have estimated above. In particular, the regression line is $\hat{y}_i = 1.5 + 2.2x_i$. With this equation we can now establish that if we had a budget of 6 for advertisement, our predicted price would be $\hat{y}_i = 1.5 + 2.2(6) = 14.7$. This answers our prediction question.

In conclusion regression line has allowed us to answer both questions: 1. The effectiveness of advertisement: For every dollar spent on advertisment, the price increases by 2.2. 2. The predicted price: With an advertisement budget of 6, the predicted price is 14.7.

This demonstrates the value of regression analysis in quantifying relationships and making informed predictions.

7.2 Measures of Goodness of Fit

When analyzing the effectiveness of a regression model, it is crucial to assess how well the model fits the data. This is where measures of goodness of fit come into play. Below we revisit the coefficient of determination or R^2.

Coefficient of Determination

The **coefficient of determination** or R^2 is the percent of the variation in y that is explained by changes in x. The higher the R^2 the better the explanatory power of the model. The R^2 is always between [0,1]. To calculate use $R^2 = SSR/SST$. Below you can find how to calculate each component of the R^2.

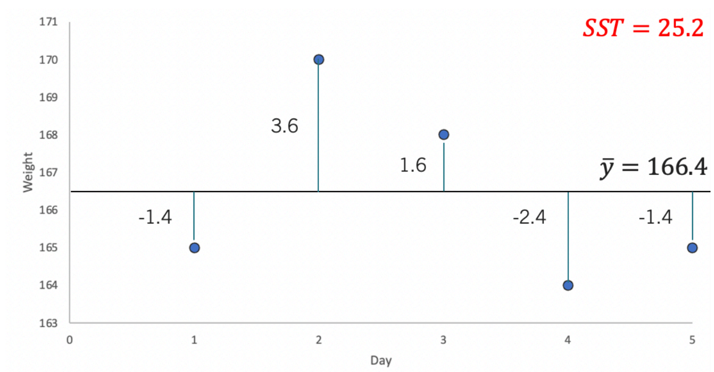
- SSR (Sum of Squares due to Regression) is the part of the variation in y explained by the model. Mathematically, $SSR = \sum (\hat{y}_i - \bar{y})^2$.
- SSE (Sum of Squares due to Error) is the part of the variation in y that is unexplained by the model. Mathematically, $SSE = \sum (y_i - \hat{y}_i)^2$.
- SST (Sum of Squares Total) is the total variation of y with respect to the mean. Mathematically, $SST = \sum (y_i - \bar{y})^2$.
- Note that $SST = SSR + SSE$.

Example: Let's consider data on the Weight (y) and Exercise (x) of a particular person.

Weight (y)	Exercise (x)
165	45
170	10

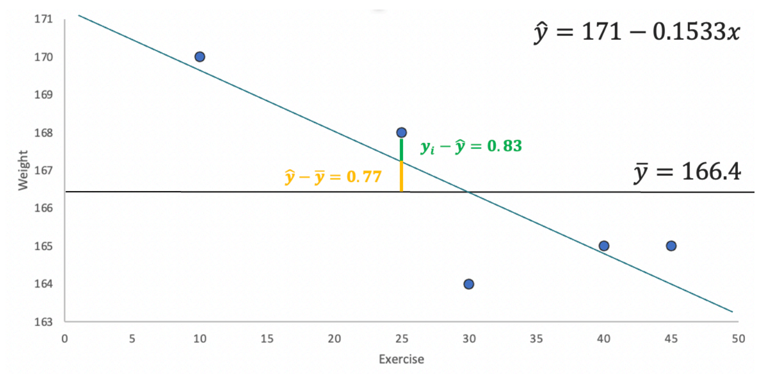
Weight (y)	Exercise (x)
168	25
164	30
165	40

Now, if we were to predict the weight of this person, we could consider just using the mean value (i.e., 166.4). If this were our guess, then the mistakes (errors) we would have made with this prediction are quantified by the $SST = 25.2$. Below, you can see these mistakes visually.



For example, in day 1 the mean prediction is 166.4, but the actual measure is 165. We have made a mistake of -1.4 by using the mean as our predicted weight. To get a sense of the overall mistake we have made, we add up all of the squared errors made by using the mean as a prediction. Hence, $SST = (-1.4)^2 + (3.6)^2 + (1.6)^2 + (-2.4)^2 + (-1.4)^2 = 25.2$.

The idea behind regression is that another variable related to weight can help us make a better prediction. Generally speaking, you burn calories if you exercise, so your weight should be lower when exercising. Hence, adding information on how many minutes a person exercises in a day should allow us to predict someone's weight better. The SSE quantifies the mistakes made by a prediction generated from regression. To understand this, let's start by looking at the regression line for Weight and Exercise.



The image highlights how the regression line can reduce the errors in our prediction. Consider the point with y coordinate 168. The error using the mean is 1.6 (the green and yellow lines). However, if we update our prediction using the regression line, the mistake is reduced to $168 - 167.17 = 0.83$ (the green line). Squaring all these deviations from the line and adding them gives us $SSE = 0.81 + 0.29 + 0.69 + 5.76 + 0.02 = 7.57$. Note how using the regression line has helped us reduce the errors in our prediction.

If regression has helped us make smaller mistakes, it has also helped us explain more of the variation in Weight (y). In the graph, the gap between the mean and the 168 point has been reduced by the amount highlighted in yellow (0.77). If we square and add up all of the "improvements," we get $SSR = 5.29 + 9.4 + 0.59 + 0 + 2.35 = 17.63$. In sum, the regression line has closed the gap of the prediction errors made by using the mean by $R^2 = \frac{SSR}{SST} \approx 0.7$ or 70%.

7.3 Multiple Regression

If we were to predict weight we could use several other variables that are related to get a better prediction. *Multiple regression* is a technique used predict a variable using more than one independent variable. Mathematically, you would be estimating $\hat{y_i} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 ... + \hat{\beta}_k x_k$, where k is the total number of independent variables included in the regression model.

Example: Let's consider an additional variable in our Weight and Exercise example. The table below includes information on Calories (z).

Weight (y)	Exercise (x)	Calories (z)
165	45	1200
170	10	1260
168	25	1220
164	30	1180
165	40	1190

The regression line can be estimated using computer software and is given by $\hat{y_i} = 83.96 - 0.025x + 0.069z$. A few things can be concluded from the regression line. First, as the amount of exercise increases our weight tends to go down, but as we consume more calories the weight tends to go up. Second, the effectiveness of calories seems to be better that that of exercise. For every minute exercise our weight goes down on average by 0.025 pounds. However, if we reduce our calorie consumption by 1, then our weight goes down by 0.069 pounds.

Anova

We can further explore the importance of each variable using the Anova table. This table helps us understand how well our regression model explains the dependent variable. For now, we will use this table to show the SSR generated by each variable as well as to track the errors. Using computer software we get the following table:

Source	Sum Squares
x	17.63
z	6.55
Residuals	1.01

The table shows that x "closes the gap" with respect to the mean prediction by 17.63. If we add calories to our model, we further decrease the gap by another 6.55. In sum, using them together increases the SSR to about 24.18, which in turn reduced the SSE to only 1.01. Finally, the R^2 also increases from 0.7 to now 0.96, $R^2 = \frac{17.63+6.55}{25.2}$.

Adjusted R^2

The **adjusted** R^2 recognizes that the R^2 is a non-decreasing function of the number of independent variables included in the model. This metric penalizes a model with more explanatory variables relative to a simpler model. It is calculated by $1 - (1 - R^2)\frac{n-1}{n-k-1}$, where k is the number of explanatory variables used in the model and n is the sample size. For our weight example $adjustedR^2 = 1 - (1 - 0.96)\frac{5-1}{5-2-1} = 0.9196$.

Residual Standard Error

The **Residual Standard Error** estimates the average dispersion of the data points around the regression line. Recall, that we have calculated the squared deviation from the regression line by using SSE. Hence if we want to find an average deviation we can divide using the number of observations and then find a square root to go back to linear values. In particular you can use $s_e = \sqrt{\frac{SSE}{n-k-1}}$, where k is the number of independent variables used in your regression. For our example with only exercise in our model $s_e = \sqrt{\frac{7.57}{5-1-1}} = 1.59$. If we include both calories and exercise then $s_e = \sqrt{\frac{1.01}{5-2-1}} = 0.71$.

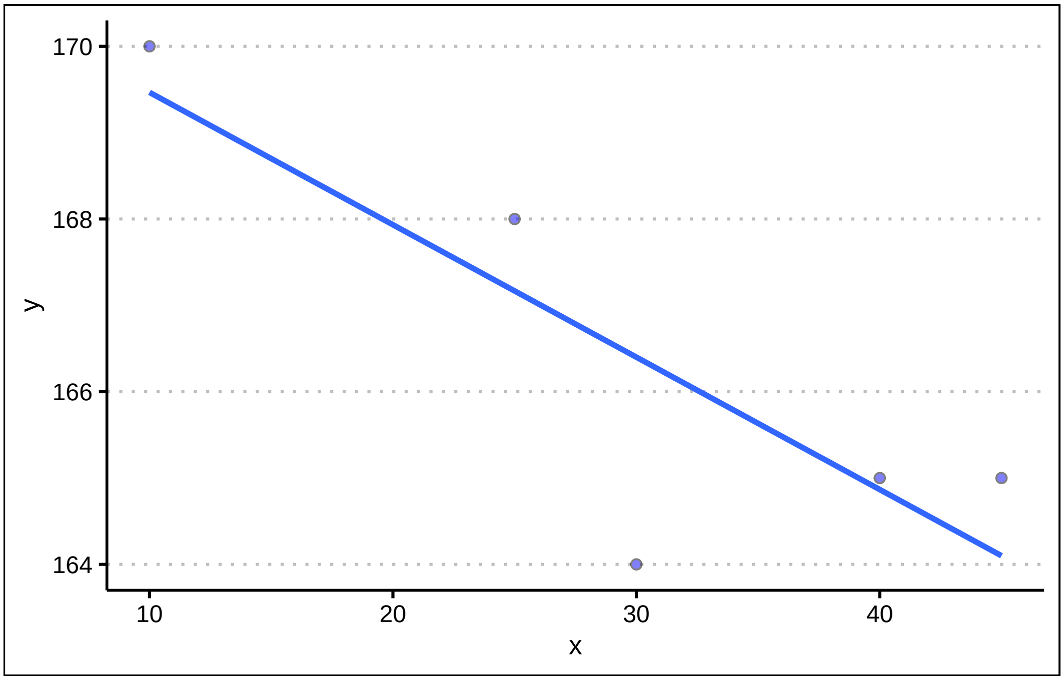
7.4 Regression in R

Below we conduct regression analysis in R, using the weight data. Let's start by creating a tibble to store the data.

```
library(tidyverse)
library(ggthemes)
d<-tibble(y=c(165,170,168,164,165),
          x=c(45,10,25,30,40),
          z=c(1200,1260,1220,1180,1190))
```

We can first visualize the relationship between x and y using `ggplot`.

```
d %>% ggplot() +
  geom_point(aes(y=y,x=x),
             pch=21, col="black", bg="blue", alpha=0.5) +
  theme_clean() +
  geom_smooth(aes(y=y,x=x),
              method="lm",se=F,formula=y~x)
```



The regression line is estimated using the `lm()` command as shown below:

```
fit<-lm(y~x, data=d)
```

We can retrieve the coefficients by either using the `coef()` function or the `summary()` function.

```
summary(fit)
```

```
Call:
lm(formula = y ~ x, data = d)

Residuals:
      1       2       3       4       5 
 0.9000  0.5333  0.8333 -2.4000  0.1333 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 171.00000    1.87913  91.000 2.93e-06 ***
x            -0.15333    0.05799  -2.644   0.0774 .  
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.588 on 3 degrees of freedom
Multiple R-squared:  0.6997,    Adjusted R-squared:  0.5996
F-statistic: 6.991 on 1 and 3 DF,  p-value: 0.07738
```

The summary function confirms our results for the R^2 and coefficients. If we wanted to make a prediction we can use the `predict()` function with the fit object as input. Below we predict y when x is equal to 4.

```r
predict(fit,newdata = tibble(x=c(4)))
```

```
       1
170.3867
```

Note that the values of x must be entered as a tibble in the function. We can run multiple regression by just adding z to our fit object.

```r
fit<-lm(y~x+z, data=d)
summary(fit)
```

```
Call:
lm(formula = y ~ x + z, data = d)

Residuals:
      1       2       3       4       5
-0.3375 -0.3375  0.7875 -0.3375  0.2250

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 83.96250   24.20433   3.469   0.0740 .
x           -0.02500    0.04413  -0.567   0.6281
z            0.06875    0.01911   3.598   0.0693 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7115 on 2 degrees of freedom
Multiple R-squared:  0.9598,    Adjusted R-squared:  0.9196
F-statistic: 23.89 on 2 and 2 DF,  p-value: 0.04018
```

Lastly, to obtain the Anova table we can use the `anova()` command.

```
anova(fit)
```

```
Analysis of Variance Table

Response: y
          Df  Sum Sq Mean Sq F value  Pr(>F)
x          1 17.6333 17.6333  34.831 0.02753 *
z          1  6.5542  6.5542  12.947 0.06931 .
Residuals  2  1.0125  0.5062
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Below we summarize the R functions: - The `lm()` function to estimates the linear regression model.

- The `predict()` function uses the linear model object to predict values. New data is entered as a data frame or tibble.
- The `coef()` function returns the model's coefficients.
- The `summary()` function returns the model's coefficients, and goodness of fit measures.
- The `Anova()` function creates the anova table for a regression. you must provide an object that contains the result of the `lm()` function.

7.5 Exercises

The following exercises will help you get practice on Regression Line estimation and interpretation. In particular, the exercises work on:

- Estimating the slope and intercept.
- Calculating measures of goodness of fit.
- Prediction using the regression line.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using R functions when possible.

1. Consider the data below. Calculate the deviations from the mean for each variable and use the results to estimate the regression line. Use R to verify your result. On average by how much does y increase per unit increase of x?

x	20	21	15	18	25
y	17	19	12	13	22

Answer

The regression lines is $\hat{y} = -4.93 + 1.09x$. For each unit increase in x, y increases on average 1.09.

Start by generating the deviations from the mean for each variable. For x the deviations are:

```
x<-c(20,21,15,18,25)
(devx<-x-mean(x))
```

```
[1]  0.2  1.2 -4.8 -1.8  5.2
```

Next, find the deviations for y:

```
y<-c(17,19,12,13,22)
(devy<-y-mean(y))
```

```
[1]  0.4  2.4 -4.6 -3.6  5.4
```

For the slope we need to find the deviation squared of the x's. This can easily be done in R:

```
(devx2<-devx^2)
```

```
[1]  0.04  1.44 23.04  3.24 27.04
```

The slope is calculated by $\frac{\sum_{i=i}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=i}^{n}(x_i-\bar{x})^2}$. In R we can just find the ratio between the summations of (devx)(devy) and devx2.

```
(slope<-sum(devx*devy)/sum(devx2))
```

```
[1] 1.087591
```

The intercept is given by $\bar{y} - \beta(\bar{x})$. In R we find that the intercept is equal to:

```
(intercept<-mean(y)-slope*mean(x))
```

```
[1] -4.934307
```

Our results can be easily verified by using the `lm()` *and* `coef()` *functions in R.*

```
fitEx1<-lm(y~x)
coef(fitEx1)
```

```
(Intercept)           x
  -4.934307    1.087591
```

2. Calculate SST, SSR, and SSE. Confirm your results in R. What is the R^2? What is the Standard Error estimate? Is the regression line a good fit for the data?

Answer

SST *is* 69.2, SSR is 64.82 and SSE is 4.38 (note that $SSR + SSE = SST$). The R^2 is just $\frac{SSR}{SST} = 0.94$ and the Standard Error estimate is 1.21. They both indicate a great fit of the regression line to the data.

Let's start by calculating the SST. This is just $\sum(y_i - \bar{y})^2$.

```
(SST<-sum((y-mean(y))^2))
```

```
[1] 69.2
```

Next, we can calculate SSR. This is calculated by the following formula $\sum(\hat{y}_i - \bar{y})^2$. To obtain the predicted values in R, we can use the output of the `lm()` function. Recall our fitEx1 object created in Exercise 1. It has fitted.values included:

```
(SSR<-sum((fitEx1$fitted.values-mean(y))^2))
```

```
[1] 64.82044
```

The ratio of SSR to SST is the R^2:

```
(R2<-SSR/SST)
```

```
[1] 0.9367115
```

Finally, let's calculate SSE $\sum(y_i - \hat{y}_i)^2$:

```
(SSE<-sum((y-fitEx1$fitted.values)^2))
```

```
[1] 4.379562
```

With the SSE we can calculate the Standard Error estimate:

```
sqrt(SSE/3)
```

```
[1] 1.208244
```

*We can confirm these results using the **summary()** function.*

```
summary(fitEx1)
```

```

Call:
lm(formula = y ~ x)

Residuals:
      1       2       3       4       5 
 0.1825  1.0949  0.6204 -1.6423 -0.2555 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept)  -4.9343     3.2766  -1.506  0.22916   
x             1.0876     0.1632   6.663  0.00689 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.208 on 3 degrees of freedom
Multiple R-squared:  0.9367,	Adjusted R-squared:  0.9156 
F-statistic:  44.4 on 1 and 3 DF,  p-value: 0.00689
```

3. Assume that x is observed to be *32*, what is your prediction of y? How confident are you in this prediction?

Answer

If $x = 32$ then $\hat{y} = 29.87$. The regression is a good fit, so we can feel good about our prediction. However, we would be concerned about the sample size of the data.

In R we can obtain a prediction by using the `predict()` *function. This function requires a data frame as an input for new data.*

```
predict(fitEx1, newdata = data.frame(x=c(32)))
```

```
       1
29.86861
```

Exercise 2

You will need the **Education** data set to answer this question. You can find the data set at https://jagelves.github.io/Data/Education.csv . The data shows the years of education (*Education*), and annual salary in thousands (*Salary*) for a sample of 100 people.

1. Estimate the regression line using R. By how much does an extra year of education increase the annual salary on average? What is the salary of someone without any education?

Answer

An extra year of education increases the annual salary about $5,300$ dollars (slope). A person that has no education would be expected to earn $17,2582$ dollars (intercept).

Start by loading the data in R:

```
library(tidyverse)
Education<-read_csv("https://jagelves.github.io/Data/Education.csv")
```

Next, let's use the `lm()` *function to estimate the regression line and obtain the coefficients:*

```
fitEducation<-lm(Salary~Education, data = Education)
coefficients(fitEducation)
```

```
(Intercept)   Education
  17.258190    5.301149
```

2. Confirm that the regression line is a good fit for the data. What is the estimated salary of a person with 16 years of education?

Answer

The R^2 is 0.668 and the standard error is 21. The line is a moderately good fit. If someone has 16 years of experience, the regression line would predict a salary of $102,000$ dollars.

Let's get the R^2 and the Standard Error estimate by using the `summary()` function and fitEx1 object.

```
summary(fitEducation)
```

```
Call:
lm(formula = Salary ~ Education, data = Education)

Residuals:
    Min      1Q  Median      3Q     Max 
-62.177  -9.548   1.988  15.330  45.444 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  17.2582     4.0768   4.233  5.2e-05 ***
Education     5.3011     0.3751  14.134  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.98 on 98 degrees of freedom
Multiple R-squared:  0.6709,	Adjusted R-squared:  0.6675 
F-statistic: 199.8 on 1 and 98 DF,  p-value: < 2.2e-16
```

Lastly, let's use the regression line to predict the salary for someone who has 16 years of education.

```
predict(fitEducation, newdata = data.frame(Education=c(16)))
```

```
       1 
102.0766 
```

Exercise 3

You will need the **FoodSpend** data set to answer this question. You can find this data set at https://jagelves.github.io/Data/FoodSpend.csv .

1. Omit any NA's that the data has. Create a dummy variable that is equal to 1 if an individual owns a home and 0 if the individual doesn't. Find the mean of your dummy variable. What proportion of the sample owns a home?

Answer

Approximately, 36% of the sample owns a home.

Start by loading the data into R and removing all NA's:

```
Spend<-read_csv("https://jagelves.github.io/Data/FoodSpend.csv")
```

```
Rows: 80 Columns: 2
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): OwnHome
dbl (1): Food

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
Spend<-na.omit(Spend)
```

To create a dummy variable for OwnHome we can use the ***`ifelse()`*** *function:*

```
Spend$dummyOH<-ifelse(Spend$OwnHome=="Yes",1,0)
```

The average of the dummy variable is given by:

```
mean(Spend$dummyOH)
```

```
[1] 0.3625
```

2. Run a regression with *Food* being the dependent variable and your dummy variable as the independent variable. What is the interpretation of the intercept and slope?

Answer

*The intercept is the average food expenditure of individuals without homes (*6417). The slope, is the difference in food expenditures between individuals that do have homes minus those who don't. We then conclude that individuals that do have a home spend about -2516 less on food than those who don't have homes.

To run the regression use the `lm()` *function:*

```
lm(Food~dummyOH,data=Spend)
```

```
Call:
lm(formula = Food ~ dummyOH, data = Spend)

Coefficients:
(Intercept)      dummyOH
       6473        -3418
```

3. Now run a regression with *Food* being the independent variable and your dummy variable as the dependent variable. What is the interpretation of the intercept and slope? Hint: you might want to plot the scatter diagram and the regression line.

Answer

The scatter plot shows that most of the points for home owners are below 6000. For non-home owners they are mainly above 6000. The line can be used to predict the likelihood of owning a home given someones food expenditure. The intercept is above one, but still it gives us the indication that it is likely that low food expenditures are highly predictive of owning a home. The slope tells us how that likelihood changes as the food expenditures increase by 1. In general, the likelihood of owning a home decreases as the food expenditure increases.

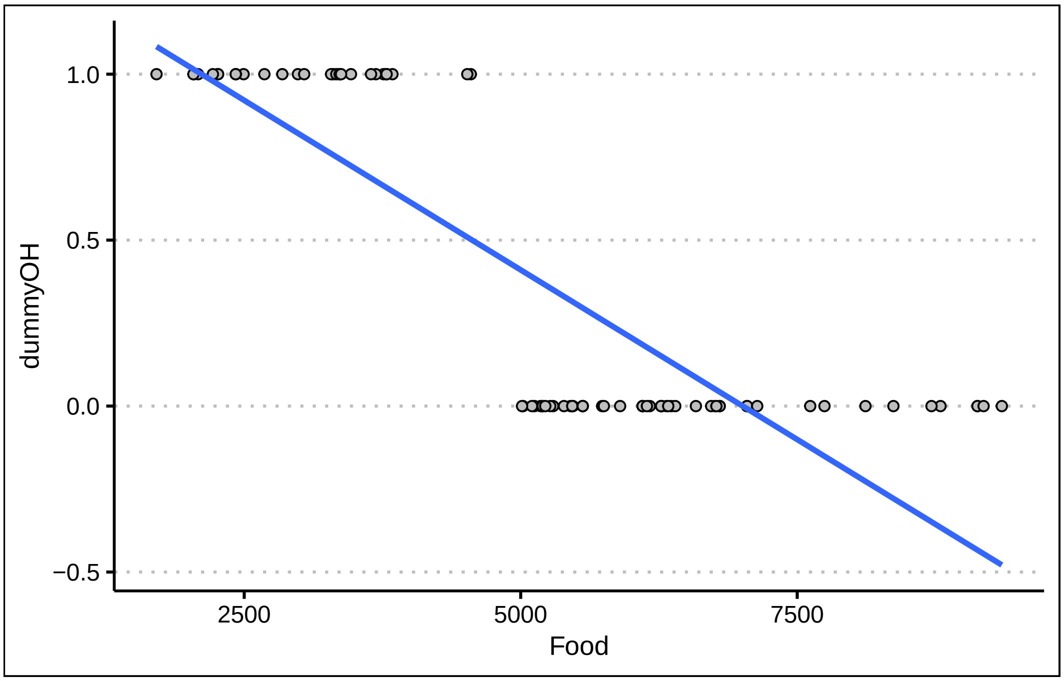
Run the `lm()` *function once again:*

```
fitFood<-lm(dummyOH~Food,data=Spend)
coefficients(fitFood)
```

```
  (Intercept)          Food
 1.4320766616 -0.0002043632
```

For the scatter plot use the following code:

```r
library(ggthemes)
Spend %>% ggplot() +
  geom_point(aes(y=dummyOH,x=Food),
             col="black", pch=21, bg="grey") +
  geom_smooth(aes(y=dummyOH,x=Food), method="lm",
              formula=y~x, se=F) +
  theme_clean()
```



Exercise 4

You will need the **Population** data set to answer this question. You can find this data set at https://jagelves.github.io/Data/Population.csv .

1. Run a regression of *Population* on *Year*. How well does the regression line fit the data?

Answer

If we follow the $R^2 = 0.81$ the model fits the data very well.

Let's load the data from the web:

```r
Population<-read_csv("https://jagelves.github.io/Data/Population.csv")
```

```
New names:
Rows: 16492 Columns: 4
-- Column specification
-------------------------------------------------------- Delimiter: "," chr
(1): Country.Name dbl (3): ...1, Year, Population
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

Now let's filter the data so that we can focus on the population for Japan.

```r
Japan<-filter(Population,Country.Name=="Japan")
```

Next, we can run the regression of Population against the Year. Let's also run the ***summary()*** *function to obtain the fit and the coefficients.*

```r
fit<-lm(Population~Year,data=Japan)
summary(fit)
```

```
Call:
lm(formula = Population ~ Year, data = Japan)

Residuals:
     Min       1Q   Median       3Q      Max
-9583497 -4625571  1214644  4376784  5706004

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -988297581   68811582  -14.36   <2e-16 ***
Year            555944      34569   16.08   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4871000 on 60 degrees of freedom
Multiple R-squared:  0.8117,    Adjusted R-squared:  0.8086
F-statistic: 258.6 on 1 and 60 DF,  p-value: < 2.2e-16
```

2. Create a prediction for Japan's population in 2030. What is your prediction?

Answer

The prediction for 2030 is about 140 million people.

Let's use the ***predict()*** *function:*

```
predict(fit,newdata=data.frame(Year=c(2030)))
```

```
        1
140268585
```

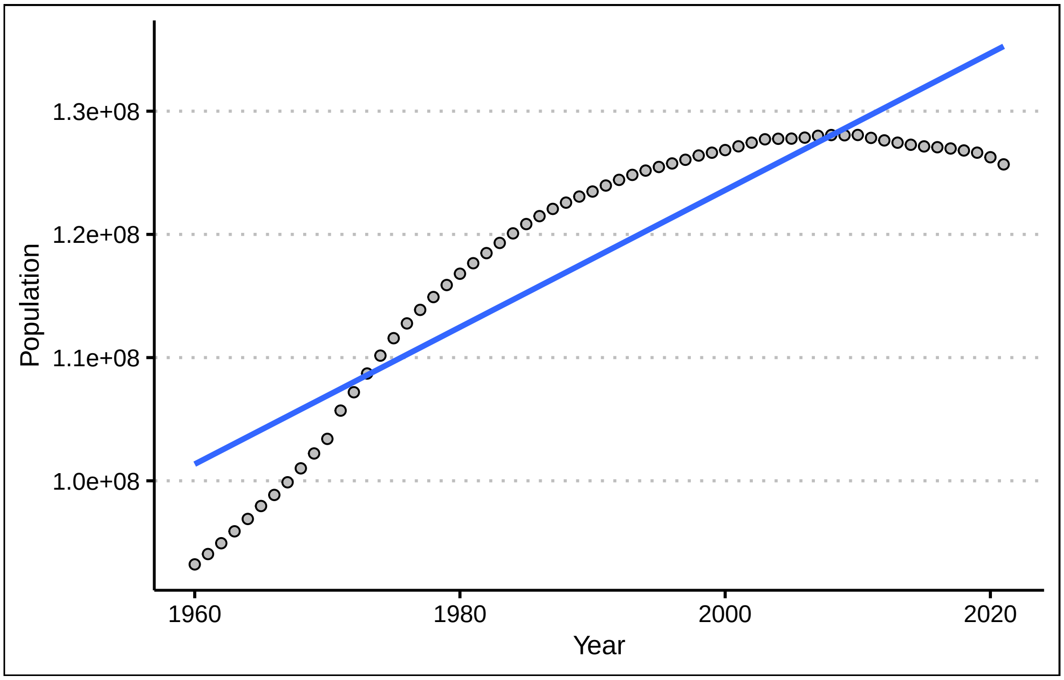
3. Create a scatter diagram and include the regression line. How confident are you of your prediction after looking at the diagram?

Answer

After looking at the scatter plot, it seems unlikely that the population in Japan will hit 140 million. Population has been decreasing in Japan!

Use the ***plot()*** *and* ***abline()*** *functions to create the figure.*

```
Japan %>% ggplot() +
  geom_point(aes(y=Population,x=Year),
             col="black", pch=21, bg="grey") +
  geom_smooth(aes(y=Population,x=Year),
              formula=y~x, method="lm", se=F) +
  theme_clean()
```



1.3e+08
1.2e+08
1.1e+08
1.0e+08
Population
1960
1980
2000
2020
Year

8 Probability I

Decision-making under uncertainty is at the core of business strategy, risk management, and forecasting. This section will study tools and concepts that will make us smarter when making data-driven decisions. Mastering probability in finance, marketing, supply chain, or operations helps professionals analyze risks, optimize strategies, and stay competitive in uncertain environments. We start with the basic concepts and ideas of probability below.

8.1 Frequentist Vs. Bayesian Statistics

There are two interpretations of probability. The **frequentist** interpretation assumes that probabilities represent proportions of specific events occurring over infinitely identical trials. In contrast, the **Bayesian** interpretation assumes that probabilities are subjective beliefs about the relative likelihood of events. Despite their philosophical differences, both approaches adhere to the core rules of probability —including the Addition Rule, Multiplication Rule, and Bayes' Theorem— which provide a consistent mathematical foundation for making probabilistic inferences.

Ex: A company might use frequentist statistics to determine the average return rate of a product, but apply Bayesian methods to adjust sales forecasts based on changing consumer behavior.

8.2 Experiments and Sets

The following concepts form the foundation of probability theory. They provide a structure that is useful when analyzing uncertainty and randomness.

- An **experiment** is a process that leads to one of several outcomes. *Ex: Tossing a Die, Tossing a Coin, Drawing a Card, etc.*
- An **outcome** is the result of an experiment. *Ex: A coin landing on heads, drawing the ace of spades.*
- The **sample space** (S) of an experiment contains all possible outcomes of the experiment. *Ex:* $S = \{1, 2, 3, 4, 5, 6\}$ is the sample space for tossing a die.

- An **event** is a subset of the sample space. *Ex:* $A = \{2, 4, 6\}$ is the event of tossing an even number when rolling a die. $B = \{1A, 1D, 1S, 1H\}$ is the event of drawing an ace from a deck of cards.

8.3 Basic Probability Concepts

A **probability** is a numerical value that measures the likelihood that an event occurs. To calculate **probabilities**, we find the ratio between favorable outcomes and total outcomes. Hence, the probability of an event is given by:

$$P(A) = \frac{Number\ of\ outcomes\ in\ A}{Number\ of\ outocomes\ in\ Sample\ Space}$$

The probability of A satisfies $0 \leq P(A) \leq 1$. Since A is a subset of S, the number of elements in A is at most that of the sample space S. Hence, if A=S, then P(A)=1. If A is a null event (i.e., has no elements) P(A)=0 and is impossible. We can conclude that P(A) ranges from 0 to 1, inclusive.

In many cases, we analyze how multiple events interact within the sample space, particularly when they either exclude each other or collectively cover all possible outcomes.**Mutually exclusive** events do not share any common outcomes. The occurrence of one event precludes the occurrence of others. **Exhaustive** events include all outcomes in the sample space.

Given these concepts if follows that the sum of the probabilities of a list of **mutually exclusive** and **exhaustive** events equals 1. Formally, we state: $\sum P(x_i) = 1$, where x_i is outcome i.

Example: Let's consider a standard deck of 52 playing cards to illustrate the key probability concepts.

If we randomly draw one card from a standard deck, the sample space S consists of all 52 cards. Suppose we define event A as *drawing a red card*. Since there are 26 red cards, the probability is:

$$P(A) = \frac{\text{Number of red cards}}{\text{Total number of cards}} = \frac{26}{52} = 0.5$$

Since $(0 \leq P(A) \leq 1)$, this follows the basic probability rule. Next let's consider mutually exclusive events. Let event B be drawing a heart and event C be drawing a spade. Since a single drawn card cannot be both a heart and a spade, these events are mutually exclusive.

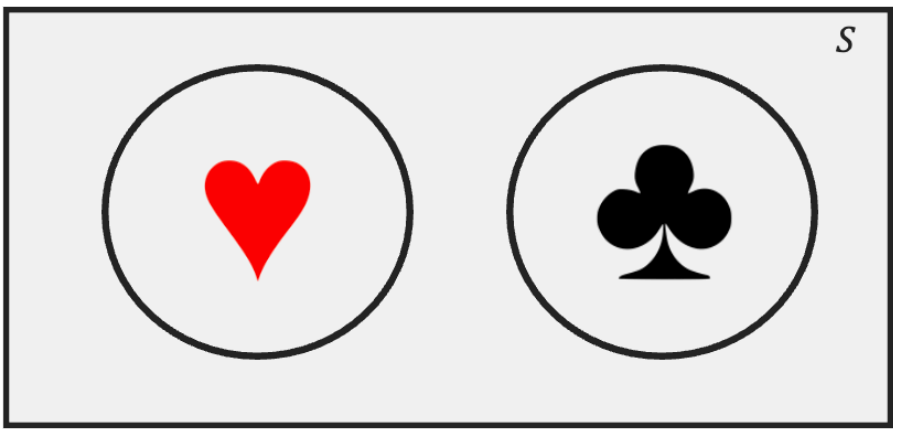
The four suits (*hearts, diamonds, clubs, spades*) form an exhaustive set, because every card belongs to exactly one suit. The sum of their probabilities equals 1:

$$P(\text{hearts}) + P(\text{diamonds}) + P(\text{clubs}) + P(\text{spades}) = \frac{13}{52} + \frac{13}{52} + \frac{13}{52} + \frac{13}{52} = 1$$

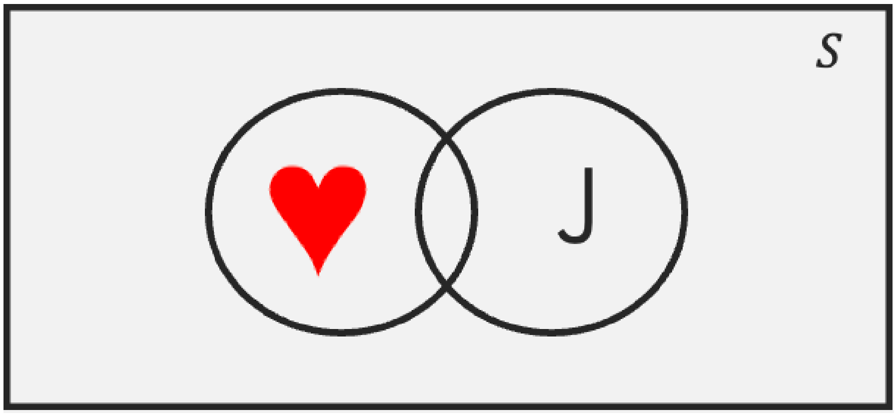
Note the this also satisfies the rule $\sum P(x_i) = 1$, where x_i represents each possible outcome (suit).

8.4 Venn Diagram

A Venn diagram is a visual representation of sets and their relationships. It consists of overlapping circles, each representing a set. Below is an example:



The rectangle enclosing the circles represents the sample set (i.e., the 52 playing cards). Each circle represents a set or event The heart circle contains all 13 hearts; the same holds for the clubs circle. The area outside the circle that has hearts represents its complement, meaning elements that are clubs, spades, or diamonds. Since the two sets do not overlap, they are mutually exclusive (i.e., if a card is a heart, it can't be a spade). The total area covered by both sets represents the **union** denoted by $\cup$. This is the set that contains all elements that belong to at least one of the sets. Formally, we would represent this set as $(Hearts \cup Spades)$.



The overlapping region of two sets contains elements that belong to both sets. This region is called the **intersection**. The intersection contains elements that are Hearts and Jacks (i.e., the jack of hearts). Formally, we denote the intersection by $(Hearts \cap Jacks)$.

8.5 Probability Rules

Probability rules help us quantify uncertainty, avoid cognitive biases, and make rational predictions based on available data. These rules allow us to make reliable decisions. Below we introduce some useful rules.

Complement Rule

The **Complement Rule** state that:

$$P(A^c) = 1 - P(A)$$

where A^c is the complement of A. This rule is useful when calculating probabilities for complex scenarios, where finding the probability of the complement is easier than computing the event itself.

Ex: The probability of drawing a spade, diamond or clubs can be estimated by just calculating the probability of drawing a heart. Since the probability of drawing a heart is $P(Heart) = \frac{13}{52}$, then using the complement rule, $P(Not\text{Heart}) = 1 - \frac{13}{52} = \frac{39}{52}$

Addition Rule

The **Addition Rule** states:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

where $\cap$ is intersection and $\cup$ is union. This formula prevents double-counting when events overlap by including the intersection of events.

Ex: Let's calculate the probability of drawing either a heart or a face card. If we let event A, be drawing a heart, and event B be drawing a face card, the probability is given by:

$$P(A \cup B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52}$$

Multiplication Rule

The **Multiplication Rule** states:

$$P(A \cap B) = P(A|B)P(B)$$

where $P(A|B)$ is the conditional probability. **Conditional probability** measures the likelihood of an event occurring given that another event has already happened. If we reorganize the formula above we can get:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Note that his formula focuses on cases where Event B has already occurred. We can then find the fraction the cases where Event A also happens.

Ex: Suppose we randomly draw two cards from a standard deck of 52 cards, without replacement. The probability that both cards are aces is given by:

$$P(Ace \cap Ace) = P(Ace|Ace) \times P(Ace) = \frac{4}{52} \times \frac{3}{51}$$

Lastly, if events A and B are independent, we can simplify the multiplication rule to:

$$P(A \cap B) = P(A)P(B)$$

Hence, we can define independence as: events A and B are independent if the occurrence (or non occurrence) of one does not affect the probability of the other occurring.

Ex1: Suppose we randomly draw a card from a standard deck, replace it, and then draw another card. The probability of drawing heart in the first draw is 13/52, the probability of drawing a spade in the second draw is also 13/52.

Ex2: Suppose that the weather is sunny, this has no effect on the outcome of a lottery. Hence, the weather and lottery outcomes are independent

Law of Total Probability

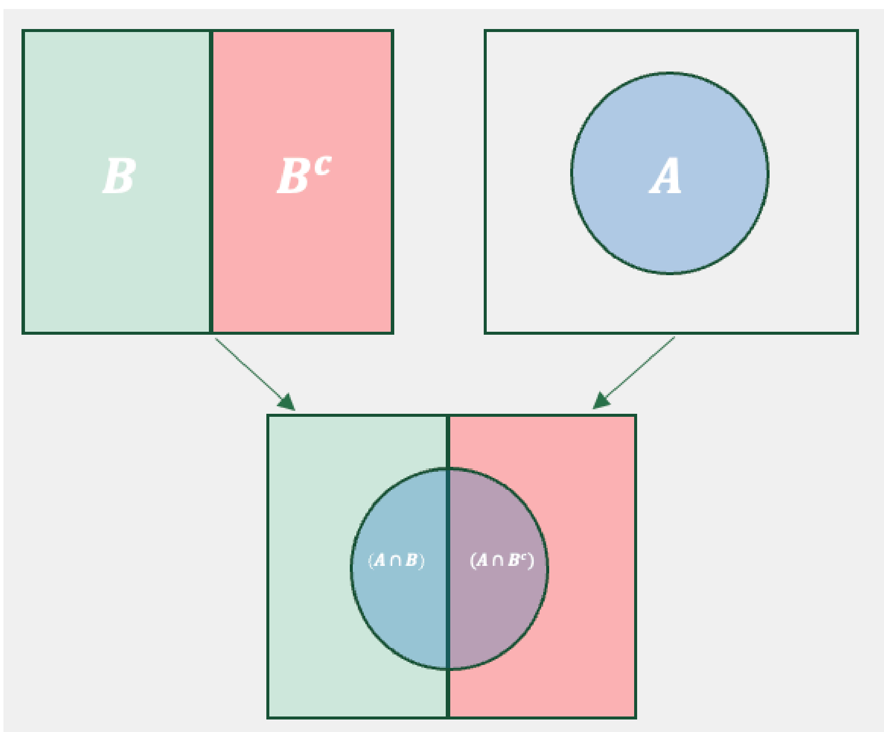
The **Law of Total Probability** states:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

The formula mainly states that If an event A can occur in multiple mutually exclusive scenarios (or conditions), then the total probability of A is the sum of its probabilities under each scenario, weighted by how often those scenarios occur. By using the multiplication rule we can alternately show the law of total probability as:

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

The Venn diagram helps us visualize the law of total probability.



The image shows how event A can occur. Essentially, when B occurs A is possible in the left semicircle. When B^c, then A is possible in the right semicircle. The addition of both semicircles, complete set A.

Ex: Consider the sets face cards and hearts. The entire set of hearts can be completed by taking face cards that are hearts, and non face cards (the complement of face cards) and hearts.

Bayes' Theorem

Bayes' Theorem states:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

In essence, Bayes' theorem is a way of updating probabilities in light of new evidence. The theorem tells us that the probability of an event A given that we have observed some evidence is equal to the prior probability ($P(A)$) times a factor that measures the informativeness of the observed evidence $P(B|A)/P(B)$.

Example: Assume that a rare form of cancer affects 1% of the population. That is, 1 out of 100 people will likely be affected by it. There is a test for the cancer, but it is imperfect. In particular, if you don't have cancer, the test says that you do 10% of the time (false positive rate). Also, if you do have cancer, it will tell you that you don't have it 5% of the time. Let's say you get tested, and your doctor calls you and says you have tested positive. How likely is it that you have this type of cancer?

We can use the Law of Total Probability and Bayes' theorem to answer this question. Let's first identify what we want to estimate. Since we need to figure out the likelihood of having cancer, given that we tested positive, we are looking for $P(C|+)$. Now, we can set up Bayes' rule to see what additional information we need to find our desired value. Bayes' rule states:

$$P(C|+) = \frac{P(+|C)P(C)}{P(+)}$$

Since the problem does not provide the $P(+)$ we must estimate it with the Law of Total Probability, mainly:

$$P(+) = P(+|C)P(C) + P(+|NC)P(NC) = (0.95 \times 0.01 + (0.1 \times 0.99) = 0.1085$$

We can now use this probability to find our new belief that we have cancer:

$$P(C|+) = \frac{(0.95 \times 0.01)}{0.1085} = 0.0875$$

In conclusion, if we before thought we had a 1 chance of having the rare form of cancer, once we were told that we tested positive, we should update our belief to about 8.75.

8.6 Contingency Tables

A **contingency table** (or cross-tabulation) summarizes the relationship between two or events (or variables). It is particularly helpful when analyzing frequencies of events and there relationship to other events.

The contingency table below shows the frequency of event A and B occurring/not occurring.

	B	B^C	Total
A	26	34	60
A^C	14	26	40
Total	40	60	100

We can see that event A occurs 60 times and that event B occurs 40 times. Most importantly, we can also see how the two events are not mutually exclusive, since A and B occur 26 times. The table can be easily translated to probabilities, by divinding each entry by the total amount of outcomes (i.e., 100). We call the table that includes probabilities the **joint probability table**.

	B	B^C	Total
A	0.26	0.34	0.60
A^C	0.14	0.26	0.40
Total	0.40	0.60	1

In the middle of the table we have the joint probabilities or intersections of events (i.e., $A \cap B$, $A \cap B^c$, etc.). The values in the totals columns or rows are the probabilities of each event happening (i.e., $P(A)$, $P(B^c)$, etc.). We call these probabilities **marginal probabilities** since they reside at the margin of the table.

8.7 Counting Rules

Counting rules are useful in probability as they allow us to determine the number of ways events can occur without listing out every possibility. Essentially, they facilitate the enumeration of possible outcomes for large sets.

Factorial

To illustrate our first counting rule, consider the number of ways that we can organize a group of three people (Ann, Beth, Charlie). The enumeration of all the possible orderings in this case is not that tedious and it can be shown in a table:

Ordering	Order
1	(Ann, Beth, Charlie)
2	(Ann, Charlie, Beth)
3	(Beth, Ann, Charlie)
4	(Beth, Charlie, Ann)
5	(Charlie, Ann, Beth)

	Ordering	Order
6		(Charlie, Beth, Ann)

The table shows that there are six possible ways to organize a group of three people. The **Factorial** counts how many ways a group of N objects can be organized. Formally, we write:

$$Factorial = N!$$

where the operator ! means to take the product of all positive integers from N down to 1. Additionally, 0! is equal to zero. If we then try to find in how many ways we can organize a group of three people, we can avoid enumeration by calculating $3! = 3 \times 2 \times 1 = 6$.

Ex: Let's assume that a manager has locked down the first five batters in a given game, but is undecided on the batting order. The number of ways that he can organize the five batters is given by $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Combinations

The **Combination** function counts the number of ways to choose x objects from a total of n objects. The order in which the x objects are listed does not matter. If repetition is not allowed use:

$$C(n, x) = \frac{n!}{(n-x)!x!}$$

Ex: Let's imaging visiting the Mexican restaurant Chipotle. We are choosing a bowl for lunch and we can choose from the following items: beans, chicken, rice, pico, guac, and beef. If we are only allowed to select four items out of the six, in how many ways can we construct a bowl? Below we use the combination formula to show that there are 15 possible bowls.

$$C(6, 4) = \frac{6!}{(6-4)!4!} = \frac{6 \times 5}{2} = 15$$

There are situations when we wish to count all possible combinations inclusive of repetition of elements. If repetition is allowed we use the following formula:

$$C(n + x - 1, x) = \frac{(x + n - 1)!}{(n-1)!x!}$$

Ex: Imagine you visit the ice cream store. You want a three scoop dish of ice cream. In how many ways can you choose your dish if the flavors are vanilla, chocolate, smores, brownie, and chips? Below we use the combination formula with repetition to show that the answer is 35.

$$C(7,3) = \frac{(5+3-1)!}{(5-1)!3!} = 7 \times 5 = 35$$

Permutations

The **Permutation** function also counts the number of ways to choose x objects from a total of n objects. However, the order in which the x objects are listed does matter. Note that since order matters, the permutation will always have a higher number of possible outcomes relative to the combination. When repetition is not allowed, permutations can be counted by:

$$P(n,x) = \frac{n!}{(n-x)!}$$

Ex: The Olympic medal finals for the 100 meters sprint has eight runners competing for three medals. How many medal podiums are possible? There are a total of 336. Below you can see the calculation.

$$P(8,3) = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$$

As with combinations, there might be situations were we wish to count when repetition is allowed. If this is this case, use:

$$P(n,x) = n^x$$

Ex: You are trying to guess the combination of a lock that has three possible numbers. Since there are 10 numbers to choose from, there are $10^3 = 1000$ possible permutations.

8.8 Probability in R

The `table()` function can be used to construct frequency distributions.

The `factorial()` function returns the factorial of a number.

The `gtools` package contains the `combinations()` and `permutations()` functions used to calculate combinations and permutations. Use the *repeats.allowed* argument to specify counting with repetition or no repetition. The *v* argument allows you to specify a vector of elements.

8.9 Exercises

The following exercises will help you practice some probability concepts and formulas. In particular, the exercises work on:

- Calculating simple probabilities.
- Applying probability rules.
- Using counting rules.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results with a calculator or R.

1. A sample space S yields five equally likely events, A, B, C, D, and E. Find $P(D)$, $P(B^c)$, and $P(A \cup C \cup E)$.

Answer

- $P(D) = 1/5 = 0.2$ since all events are equally likely.
- $P(B^c) = 4/5 = 0.8$
- $P(A \cup C \cup E) = P(A + C + E) = 3/5 = 0.6$.

2. Consider the roll of a die. Define A as {1,2,3}, B as {1,2,3,5,6}, C as {4,6}, and D as {4,5,6}. Are the events A and B mutually exclusive, exhaustive, both or none? What about events A and D?

Answer

- *Events* A and B are not mutually exclusive since they share some of the same elements.
- *The events are not exhaustive since the union of both doesn't create the sample space.*

3. A recent study suggests that 33.1% of the adult U.S. population is overweight and 35.7% obese. What is the probability that a randomly selected adult in the U.S. is either obese or overweight? Are the events mutually exclusive and exhaustive? What is the probability that their weight is normal?

Answer

- *The probability is* 68.8%.

- *The events are mutually exclusive. If someone is classified as obese, the person is not classified again as overweight.*
- *The events are not exhaustive since there are people in the U.S. that have a normal weight.*
- *The probability that the person drawn has normal weight is* 31.2%.

Exercise 2

For the following exercises, make your calculations by hand and verify results with a calculator or R.

1. Let $P(A) = 0.65$, $P(B) = 0.3$, and $P(A|B) = 0.45$. Calculate $P(A \cap B)$, $P(A \cup B)$, and $P(B|A)$.

Answer

From the multiplication rule,

$$P(A|B) * P(B) = P(A \cap B)$$

substituting values yields:

$$P(A \cap B) = 0.45 * 0.3 = 0.135$$

From the addition rule,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

substituting yields:

$$P(A \cup B) = 0.65 + 0.3 - 0.135 = 0.815$$

From the multiplication rule once again,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

substituting values yields,

$$P(B|A) = 0.135/0.65 = 0.2076923$$

2. Let $P(A) = 0.4$, $P(B) = 0.5$, and $P(A^c \cap B^c) = 0.24$. Calculate $P(A^c|B^c)$, $P(A^c \cup B^c)$, and $P(A \cup B)$.

Answer

From the complement rule we have that $P(A^c) = 0.6$ and $P(B^c) = 0.5$.
Using the multiplication rule,

$$P(A^c|B^c) = \frac{P(A^c \cap B^c)}{P(B^c)}$$

substituting yields:

$$P(A^c|B^c) = 0.24/0.5 = 0.48$$

From the addition rule,

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c)$$

substituting yields:

$$P(A^c \cup B^c) = 0.6 + 0.5 - 0.24 = 0.86$$

The event that has no elements of A or B is given by $P(A^c \cap B^c)$. Therefore $P(A \cup B) = 1 - 0.24 = 0.76$ has all the elements of A and B.

3. Stock A will rise in price with a probability of 0.4, stock B will rise with a probability of 0.6. If stock B rises in price, then A will also rise with a probability of 0.5. What is the probability that at least one of the stocks will rise in price? Prove that events A and B are (are not) mutually exclusive (independent).

Answer

In short, the problem states $P(A) = 0.4$, $P(B) = 0.6$, and $P(A|B) = 0.5$. Where A and B are events of stocks rising in price. The question asks for:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Using the multiplication rule,

$$P(A \cap B) = 0.5 * 0.6 = 0.3$$

Hence,

$$P(A \cup B) = 0.4 + 0.6 - 0.3 = 0.7$$

The events are not mutually exclusive since $P(A \cap B) = 0.3 \neq 0$. The events are also not independent since $P(A|B) = 0.5 \neq 0.4 = P(A)$.

Exercise 3

1. Create a joint probability table from the contingency table below. Find $P(A)$, $P(A \cap B)$, $P(A|B)$, and $P(B|A^c)$. Determine whether the events are independent or mutually exclusive.

	B	B^c
A	26	34
A^c	14	26

Answer

Below is the joint probability table. The $P(A) = 0.26+0.34 = 0.6$, $P(A \cap B) = 0.26$, $P(A|B) = 0.26/0.4 = 0.65$, and $P(B|A^c) = 0.14/0.4 = 0.35$. Events A and B are not independent since $P(A) \neq P(A|B)$. The events are not mutually exclusive since $P(A \cap B) = 0.26 \neq 0$.

	B	B^c	Total
A	0.26	0.34	0.6
A^c	0.14	0.26	0.4
Total	0.4	0.6	1

Exercise 4

You will need the **Crash** data set and R to answer this question. The data shows information on several car crashes. Specifically, if the crash was Head-On or Not Head-On and whether there was Daylight or No Daylight. You can find the data here: https://jagelves.github.io/Data/Crash.csv

1. Create a contingency table.

Answer

The probability of a Head-On crash is $(166 + 108)/4858 = 0.056$. The probability of a daylight crash is $(166+3258)/4858 = 0.70$. The probability that the car crash is Head-On given daylight is $166/(166 + 3258) = 0.048$.

Start by loading the data into R.

```
library(tidyverse)
Crash<-read_csv("https://jagelves.github.io/Data/Crash.csv")
```

To create a contingency table use the ***`table()`*** *command in R.*

```
(freq<-table(Crash$`Crash Type`,Crash$`Light Condition`))
```

```
             Daylight Not Daylight
Head-on           166          108
Not Head-On      3258         1326
```

This table is used to calculate probabilities. We can pass it through the ***prop.table()*** *function to get the contingency table.*

```r
round(prop.table(freq),2)
```

```
             Daylight Not Daylight
Head-on          0.03         0.02
Not Head-On      0.67         0.27
```

2. Find the probability that a) a car crash is Head-On, b) a car crash is in daylight c) a car crash is Head-On given that there is daylight.

Answer

The probability of a head-on crash is:

$$P(\text{Head-on}) = 0.03 + 0.02 = 0.05$$

The probability that the crash is in daylight is:

$$P(Daylight) = 0.03 + 0.67 = 0.70$$
$$P(Daylight) = \frac{P(\text{Head-on})}{P(Daylight)} = \frac{0.03}{0.70}$$

3. Show that Crashes and Light are dependent.

Answer

The two variables are dependent since $P(Head - On|Daylight) \neq P(Head - On)$, that is $0.048 \neq 0.56$.

Exercise 5

1. Use Bayes' Theorem in the following question. Let $P(A) = 0.7$, $P(B|A) = 0.55$, and $P(B|A^c) = 0.10$. Find $P(A^c)$, $P(A \cap B)$, $P(A^c \cap B)$, $P(B)$, and $P(A|B)$.

Answer

$P(A^c) = 1 - P(A) = 1 - 0.7 = 0.3$, $P(A \cap B) = P(B|A)P(A) = 0.55(0.70) = 0.385$, $P(A^c \cap B) = P(B|A^c)P(A^c) = 0.10(0.30) = 0.03$, $P(B) = P(A \cap B) + P(A^c \cap B) = 0.385 + 0.03 = 0.415$, and $P(A|B) = \frac{P(A \cap B)}{P(B)} = 0.385/0.415 = 0.9277$.

2. Some find tutors helpful when taking a course. Julia has a 40% chance to fail a course if she does not have a tutor. With a tutor, the probability of failing is only 10%. There is a 50% chance that Julia finds an available tutor. What is the probability that Julia will fail the course? If she ends up failing the course, what is the probability that she had a tutor?

Answer

Let the event of failing be F, the event of not failing be NF, the event of having a tutor be T, and the event of not having a tutor be NT. The probability of failing the course is 0.25. $(F) = P(F \cap T) + P(F \cap T^c) = P(F|T)P(T) + P(F|T^c)P(T^c) = 0.10(0.50) + 0.40(0.50) = 0.05 + 0.20 = 0.25$ The probability of not having a tutor, given that she failed the course is 0.2. $P(T|F) = \frac{P(F \cap T)}{P(F \cap T) + P(F \cap T^c)} = 0.05/0.25 = 0.20$.

Exercise 6

1. Calculate the following values and verify your results using R. a) 3!, b) 4!, c) C_6^8, d) P_6^8.

Answer

$3! = 3 \times 2 \times 1 = 6$, $4! = 6 \times 4 = 24$, $C_6^8 = 28$, and $P_6^8 = 20,160$

In R we can just use the factorial command. So 3! is:

```
factorial(3)
```

```
[1] 6
```

and 4! is:

```
factorial(4)
```

```
[1] 24
```

For combinations and permutations we can use the `gtools` package:

```
library(gtools)
C<-combinations(8,6)
nrow(C)
```

```
[1] 28
```

2. There are 10 players in a local basketball team. If we chose 5 players to randomly start a game, in how many ways can we select the five players if order doesn't matter? What if order matters?

Answer

If order doesn't matter, there are 252 ways. If order matters, then there are $30,240$ ways.

In R we can once more use the combination and permutation functions:

```
B1<-combinations(10,5)
nrow(B1)
```

```
[1] 252
```

```
B2<-permutations(10,5)
nrow(B2)
```

```
[1] 30240
```

9 Probability II

Studying discrete probability distributions is essential for understanding and predicting outcomes in uncertain scenarios, particularly in business, finance, and data science. These distributions model events with distinct, countable outcomes, such as customer purchases, defect counts, or market trends, enabling decision-makers to quantify risks and optimize strategies. Below, we introduce some popular distributions and their business application.

9.1 Random Variables

The study of probability distributions begins with a random variable. A **random variable** assigns a numerical value to each possible experimental outcome. The table below presents a collection of experiments along with their corresponding random variables.

Experiment	Random Variable (x)	Possible Values
Contact five customers	Number of customers who place an order	0, 1, 2, 3, 4, 5
Operate a restaurant for one day	Number of customers	0, 1, 2, 3, . . .
Selling a car	Result of Sale	0 No Sale; 1 Sale

For restaurant owners, the number of customers they attract is a crucial variable to monitor. Tracking this variable and its possible values helps in making informed business decisions. Below we will study how this type of experiment and several others behave, so that we can make better decisions. Note that since the outcomes are integer values, the resulting distribution is discrete. The **probability mass function** (PMF) summarizes the possible values generated by a discrete probability distribution.

9.2 Expected Value and Variance

When summarizing a random variable, we are mostly interested in the variable's central tendency (Expected Value) and dispersion (Variance).

The **expected value** (mean) is a measure of central location. For a discrete random variable it is given by:

$$E(x) = \mu = \sum x f(x)$$

where $f(x)$ is the probability mass function. For a continuous random variable it is given by $E(x) = \int_{-\infty}^{\infty} x f(x) dx$, where $f(x)$ is the probability density function.

The **variance** summarizes the deviation of the values of the random variable from the mean. It is calculated by:

$$var(x) = E[(x - E(x))^2] = E(x^2) - [E(x)]^2$$

Note that this formula can be used for both discrete and continuous random variables.

Example: Let's consider an incentive pay scheme for a company. The table below summarizes the scheme:

Bonus in ($1000's)	Performance Type	Probability
10	Superior	0.15
6	Good	0.25
3	Fair	0.4
0	Poor	0.2

If a worker receives a "Superior" rating, they earn a $10,000 bonus. Based on years of employee data, the company has determined that a "Superior" rating occurs 15% of the time. It also lists all of the possible ratings of its employees and their likelihood on the table.

To summarize this distribution we can calculate the mean bonus paid. Using the expected value formula we find that the company pays 4,200 dollars on average in compensation.

$$E(x) = (10 \times 0.15) + (6 \times 0.25) + (3 \times 0.4) + (0 \times 0.2) = 4.2$$

The variability of the compensation is given by:

$$var(x) = (100 \times 0.15) + (36 \times 0.25) + (9 \times 0.4) + (0 \times 0.2) - (4.2)^2 = 9.96$$

The expected value and the variance has helped us summarize the results of the company's incentive scheme. Mainly, the company pays on average an incentive of 4,200 dollars give or take 3,156 dollars.

9.3 Discrete Uniform Distribution

The **discrete uniform distribution** is a probability distribution that assigns equal probability to each outcome in a finite set of possible outcomes. In other words, each outcome in the set is equally likely to occur.

The **probability mass function** is given by:

$$f(x) = 1/n$$

where n is the number of elements in the sample space (all possible outcomes). The **expected value** is given by:

$$E(x) = \frac{\sum x_i}{n}$$

where x_i are the possible values, and n is the number of possible values. The **variance** is given by:

$$var(x) = \frac{\sum(x_i - E(x))^2}{n}$$

Example: Consider tossing a fair die. Since all outcomes are equally likely, the probability of the die landing on 6 is $f(x) = 1/6$. The expected value on a roll of a die is $E(x) = \frac{1+2+3+4+5+6}{6} = 3.5$ and the variance is $var(x) = \frac{(-2.5)^2+(-1.5)^2+(-0.5)^2+0.5^2+1.5^2+2.5^2}{6} \approx 2.92$.

9.4 Binomial Distribution

The binomial distribution is a probability distribution that describes the outcome of a sequence of n independent Bernoulli trials. In a Bernoulli trial, there are only two possible outcomes: "success" and "failure". The probability of success is denoted by p, and the probability of failure is denoted by $q = 1 - p$. In a sequence of n independent Bernoulli trials, the number of successes (x) is a random variable that follows a binomial distribution.

The **probability mass function** is given by:

$$f(x) = C_x^n (p^x)(1-p)^{n-x}$$

where n is the number of trials, x is the number of successes, p is the probability of success, and C_x^n is the number of ways there can be x successes in n trials.

The **expected value** of the binomial distribution is

$$E(x) = np$$

The **variance** of the binomial distribution is

$$var(x) = np(1-p)$$

Example: Consider the following scenario. Imagine your favorite team in the NBA Finals, trailing by 2 points with seconds left. Your star player, a 90% free-throw shooter, gets three shots. What's the chance your team takes the lead?

Since shooting free-throws can be thought as a binomial experiment, we can find the probability using the binomial probability mass function. To take the lead, the star player needs to make all three shots. We can substitute three into the function:

$$f(3) = C_3^3(0.9^3)(0.1)^0 = 0.9^3 = 0.729$$

The function indicates that there is a 72.9% chance of taking the lead.

9.5 The Hypergeometric Distribution

The **hypergeometric distribution** is a probability distribution that describes the outcome of drawing a sample from a population without replacement. It is used to calculate the probability of drawing a certain number of successes (x) in a sample of a given size (n), where the success or failure of each individual draw is dependent on the success or failure of other draws.

The **hypergeometric** experiment differs from the binomial since:

- trials are not independent.
- the probability of success changes from trial to trial.

The **probability mass function** is given by:

$$f(x) = \frac{C_x^r C_{n-x}^{N-r}}{C_n^N}$$

where n is the number of trials, x is the number of successes, r is the number of elements in the population labeled as success, and N is the number of elements in the population.

The **expected value** of the hypergeometric distribution is:

$$E(x) = n\frac{r}{N}$$

The **variance** of the hypergeometric distribution is:

$$var(x) = n\frac{r}{N}(1 - \frac{r}{N})(\frac{N-n}{N-1})$$

Example: Picture a box of 12 electric fuses from the manufacturer Ontario Electric. Five fuses are known to be defective. An inspector randomly picks 3 fuses to test. What is the probability that exactly one of the three chosen is defective?

We can use the hypergeometric probability mass function to solve this problem. From the problem we know that $N = 12$, $n = 3$, $r = 5$, and $x = 1$. Substituting these values into the PMF yields:

$$f(1) = \frac{C_1^5 C_4^7}{C_3^{12}} = 0.4772$$

There is a 47.72% chance that the inspector finds a defective fuse in his sample of three.

9.6 Poisson Distribution

The **Poisson distribution** estimates the number of successes (x) over a specified interval of time or space.

The **probability mass function** is given by:

$$f(x) = \frac{\mu^x e^{-\mu}}{x!}$$

where μ is the expected number of successes in any given interval and also the variance, and e is Euler's number (2.71828...).

An experiment satisfies a Poisson process if:

- The number of successes with a specified time or space interval equals any integer between zero and infinity.
- The number of successes counted in non-overlapping intervals are independent.

- The probability of success in any interval is the same for all intervals of equal size and is proportional to the size of the interval.

Example: Imagine a fast-food restaurant's drive-thru where, on average, 10 cars arrive in a 15-minute span. What's the likelihood of exactly 5 cars showing up in 15 minutes?

We can use the Poisson PMF function for this problem. The arrival rate is 10 cars every 15 minutes. So there are 10 successes in the 15 minute interval. We can substitute values into the PMF:

$$f(5) = \frac{10^5 e^{-10}}{5!} = 0.0378$$

There is a 3.78% chance of five cars arriving within a 15 minute interval.

9.7 Finding Probabilities in R

To calculate probabilities based on discrete random variables use the `dbinom()`, `dhyper()`, and `dpois()` functions. For the uniform distribution use the `extraDistr` package and the `dunif()` function. For example, to confirm the results of the dice example use the following code:

```
library(extraDistr)
ddunif(2,1,6)
```

```
[1] 0.1666667
```

The probability of rolling a 2 on a die is 0.167. The function's first argument is x, the second argument is the lower limit (i.e., 1) and the third argument is the upper limit (i.e., 6). To confirm the result of the basketball example, use the `dibinom()` function. Below is the code:

```
dbinom(3,3,0.9)
```

```
[1] 0.729
```

The first argument is x (number of successes), the second argument is n (the number of trials), and the last argument is p (the probability of success). To confirm the result of the fuse box problem we can use the `dhyper()` function in R:

```
dhyper(1,5,7,3)
```

```
[1] 0.4772727
```

The first argument is one more time x (number of successes in the sample), the second argument is r (the number of successes in the population), the third argument is N-r (the total number of failures), and the fourth argument is n (the sample size). Finally, for the drive-thru example we use the `dpois()` function to confirm that the probability of five arrivals is 3.78%.

```
dpois(5,10)
```

```
[1] 0.03783327
```

9.8 Cumulative Probabilities in R.

To calculate cumulative probabilities use the `pbinom()`, `phyper()`, `ppois()`, and `pdunif()` functions. For example, the probability of tossing a 3 or lower when tossing a die is 1/2. This is confirmed with the code:

```
pdunif(3,1,6)
```

```
[1] 0.5
```

Likewise, the probability of making at least 2 shots in our basketball example is:

```
pbinom(1,3,0.9, lower.tail = F)
```

```
[1] 0.972
```

Note the use of the *lower.tail* argument. This allows us to find the cumulative distribution to the right (i.e., upper tail of the distribution).

9.9 Finding quantiles in R

Quantiles can be calculated using the `qbinom()`, `qhyper()`, `qpois()`, and `qdunif()` functions. If we wanted to find the 70% percentile for the rolling die example we can write:

```
qdunif(0.7,1,6)
```

```
[1] 5
```

This means that about 70% of the values when rolling a die are less than or equal to 5. Note that 83.33% in fact of rolls are less than or equal to 5. R chooses the closes integer when calculating the quantile.

Let's recall the fuse box problem where an inspector samples some boxes and determines whether they are defective or not. The inputs for the problem are $N = 12$, $n = 3$, and $r = 5$. For this problem we can find the 80% percentile with the `qhyper()` function.

```
qhyper(0.8,5,7,3)
```

```
[1] 2
```

That is, 80% of the outcomes are at two or less defective items.

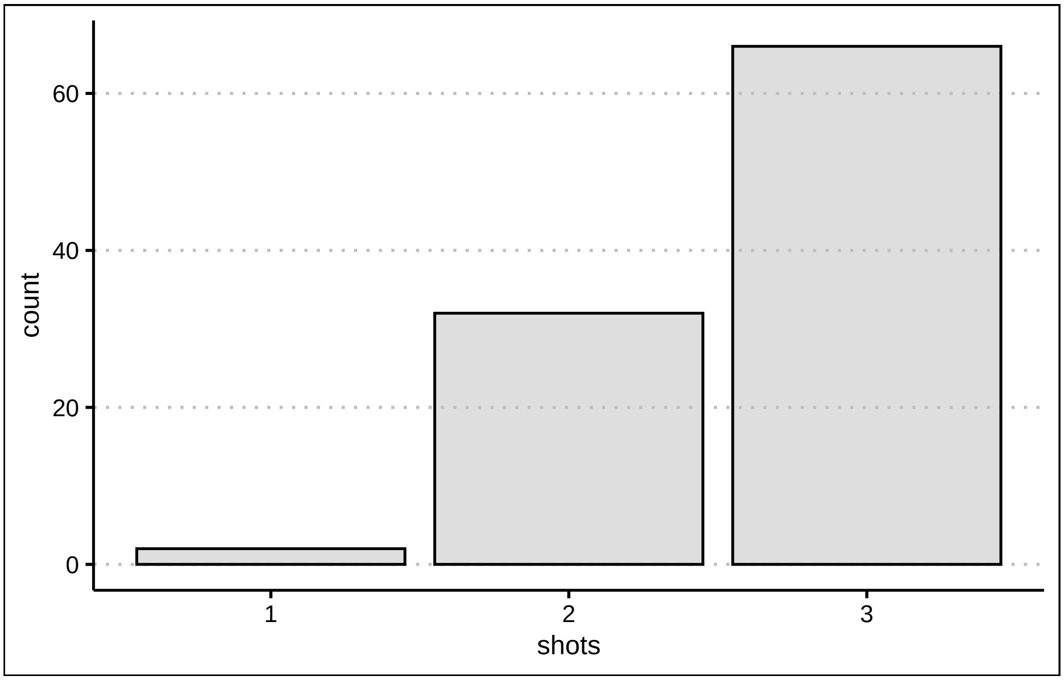
9.10 Generating Random numbers using R

To generate random numbers use the `rbinom()`, `rhyper()`, `rpois()`, and `rdunif()` functions. Let's generate 100 random numbers from the basketball example to look at the distribution of outcomes.

```
shots<-rbinom(100,3,0.9)
```

The first input in the function is the number of random numbers we want to generate from the binomial distribution. We can now plot using `ggplot`.

```
library(tidyverse)
library(ggthemes)
ggplot() + geom_bar(aes(shots), col="black",
                    bg="grey", alpha=0.5) +
  theme_clean()
```



The graph shows how the shooter makes all three shots about 72 times out of 100 shots.

9.11 Exercises

The following exercises will help you practice some probability concepts and formulas. In particular, the exercises work on:

- Calculating probabilities for discrete random variables.
- Calculating the expected value and standard deviation.
- Applying the binomial, Poisson and hypergeometric probability distributions.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results with a calculator or R.

1. Consider the table below. Calculate the mean and standard deviation. What is the probability that $x < 15$?

x	5	10	15	20
$P(X=x)$	0.35	0.3	0.2	0.15

Answer

The expected value is 10.75 and the standard deviation is 5.31. The probability of $x < 15$ is 0.65.

In R we can create vectors for both x and the probabilities $P(X=x)$.

```
x<-c(5,10,15,20)
px<-c(0.35,0.3,0.2,0.15)
```

The expected value is the sum product of probabilities and values. Formally, $\sum_{i=1}^{n} x_i p_i$ and in R:

```
(ex<-sum(x*px))
```

```
[1] 10.75
```

The standard deviation is given by $\sqrt{\sum_{i=1}^{n}(x_i-\mu)^2 p_i}$. We can calculate it in R with the following code:

```
(sd<-sqrt(sum((x-ex)^2*px)))
```

```
[1] 5.30919
```

2. Consider the table below. Calculate the mean and standard deviation. What is the probability that $x \geq -9$?

y	-23	-17	-9	-3
$P(Y=y)$	0.5	0.25	0.15	0.1

Answer

The expected value is -17.4 and the standard deviation is 6.86. The probability of is 0.25.

Let's create the vectors once more in R.

```
y<-c(-23,-17,-9,-3)
py<-c(0.5,0.25,0.15,0.1)
```

The expected value is given by:

```
(ey<-sum(y*py))
```

```
[1] -17.4
```

The standard deviation is given by:

```
(sdy<-sqrt(sum((y-ey)^2*py)))
```

```
[1] 6.858571
```

3. The returns on a couple of funds depends on the state of the economy. The economy is expected to be Good with a probability of 20%, Fair with probability of 50% and Poor with probability of 30%. Which fund would you choose if you want to maximize your return? What would you choose if you really dislike risk?

State of Economy	Fund 1	Fund 2
Good	20	40
Fair	10	20
Poor	-10	-40

Answer

Both funds have the same expected return of 6. The safest return comes from fund 1 since the standard deviation is only 11.14 vs. 31.05 for fund 2.

In R we can create a data frame with probabilities and the performance of the funds.

```
funds<-data.frame(probs=c(0.2,0.5,0.3),fund1=c(20,10,-10), fund2=c(40,20,-40))
```

Let's create a function for the expected value and standard deviation. For the expected value:

```
Expected_Value<-function(x,p){
  sum(x*p)
}
```

Now we can use the formula to calculate the expected value of fund1:

```
Expected_Value(funds$fund1,funds$probs)
```

```
[1] 6
```

and fund 2:

```
Expected_Value(funds$fund2,funds$probs)
```

```
[1] 6
```

For the standard deviation we can create another function:

```
Standard_Deviation<-function(x,p){
  sqrt(sum((x-Expected_Value(x,p))^2*p))
}
```

Using the function to get the standard deviation of fund 1 we get:

```
Standard_Deviation(funds$fund1,funds$probs)
```

```
[1] 11.13553
```

and for fund 2:

```
Standard_Deviation(funds$fund2,funds$probs)
```

```
[1] 31.04835
```

Exercise 2

1. Use the table below. A portfolio has 200,000 dollars invested in Asset X and 300,000 dollars in asset Y. If the correlation coefficient between the two investments is 0.4, what is the expected return and standard deviation of the portfolio?

Measure	X	Y
Expected Return (%)	8	12
Standard Deviation (%)	12	20

Answer

The expected return of the portfolio is 10.4 and the standard deviation is 14.60.

In R we can start by calculating the expected return. This is given by the formula $\alpha R_1 + \beta R_2$:

```
(ER<-(2/5)*8+(3/5)*12)
```

```
[1] 10.4
```

Next we can find the standard deviation with the formula $\sqrt{\alpha^2\sigma_1^2 + \beta^2\sigma_2 + \alpha\beta\rho\sigma_1\sigma_2}$:

```
(Risk<-sqrt(0.4^2*12^2 + 0.6^2*20^2+2*0.4*0.6*0.4*12*20))
```

```
[1] 14.59863
```

Exercise 3

1. Let Z be a binomial random variable with $n = 5$ and $p = 0.35$ use the binomial formula to find $P(Z = 1)$, $P(Z \geq 2)$. What is the expected value and standard deviation of Z?

Answer

$P(Z = 1) = 0.31$, and $P(Z \geq 2) = 0.57$. The expected value is $np = 1.75$ and the standard deviation is $\sqrt{np(1-p)} = 1.067$.

Let's use R and the ***dbinom()*** *function to find* $P(Z = 1)$.

```
dbinom(1,5,0.35)
```

```
[1] 0.3123859
```

We can now use ***pbinom()*** *to find the cumulative distribution. Since we want the right tale of the distribution, we will specify this with an argument.*

```r
pbinom(1,5,0.35, lower.tail=F)
```

```
[1] 0.571585
```

2. Let W be a binomial random variable with $n = 200$ and $p = 0.77$ use the binomial formula to find $P(W > 160)$, $P(155 \leq W \leq 165)$. What is the expected value and standard deviation of W?

Answer

$P(W > 160) = 0.14$, and $P(155 \leq W \leq 165) = 0.45$. The expected value is $np = 154$ and the standard deviation is $\sqrt{np(1-p)} = 5.95$.

Using the `pbinom()` function we find that $P(W > 160)$.

```r
pbinom(160,200,0.77, lower.tail = F)
```

```
[1] 0.136611
```

We make two calculations to find the probability. First, $P(W \leq 165)$ and then $P(W \geq 154)$. The difference between these two, gives us the desired outcome.

```r
pbinom(165,200,0.77, lower.tail=T)-pbinom(154,200,0.77, lower.tail=T)
```

```
[1] 0.4487104
```

3. Sixty percent of a firm's employees are men. Suppose four of the firm's employees are randomly selected. What is more likely, finding three men and one woman, or two men and one woman? Does your answer change if the proportion falls to 50%?

Answer

The probabilities are the same. Each event has a probability of 0.3456. If the probability changes to 0.5 now the event of two women and two men is more likely.

Let's calculate the probabilities in R. First, the probability of three men and one woman.

```r
dbinom(3,4,0.6)
```

```
[1] 0.3456
```

Now the probability of two men and two women.

```r
dbinom(2,4,0.6)
```

```
[1] 0.3456
```

Changing the probabilities reveals that:

```r
dbinom(3,4,0.5)
```

```
[1] 0.25
```

```r
dbinom(2,4,0.5)
```

```
[1] 0.375
```

Having two of each is the most likely outcome.

Exercise 4

1. Assume that S is a Poisson process with mean of $\mu = 1.5$. Calculate $P(S = 2)$ and $P(S \geq 2)$. What is the mean and standard deviation of S?

Answer

The $P(S = 2) = 0.25$ and $P(S \geq 2) = 0.44$. The expected value and the variance is 1.5.

In R we will make use of the `dpois()` *function:*

```r
dpois(2,1.5)
```

```
[1] 0.2510214
```

For the second probability we will use `ppois()`*:*

```r
ppois(1,1.5, lower.tail=F)
```

```
[1] 0.4421746
```

2. Assume that T is a Poisson process with mean of $\mu = 20$. Calculate $P(T = 14)$ and $P(18 \leq T \leq 23)$.

Answer

The $P(T = 14) = 0.039$ and $P(18 \leq T \leq 23) = 0.49$.

Using the ***dpois()*** *function once more:*

```
dpois(14,20)
```

```
[1] 0.03873664
```

For the second probability we will find the difference between two probabilities:

```
ppois(23,20, lower.tail=T)-ppois(17,20, lower.tail=T)
```

```
[1] 0.4904644
```

3. A local pharmacy administers on average 84 Covid-19 vaccines per week. The vaccines shots are evenly administered across all days. Find the probability that the number of vaccine shots administered on a Wednesday is more than eight but less than 12.

Answer

The probability of administering more than 8 but less than 12 shots is 0.3.

Let's first note that if 84 shots are administered on average weekly, then 12 are administered daily. Now we can use this average and the `ppois()` function to find the probability:

```
ppois(11,12)-ppois(8,12)
```

```
[1] 0.3065696
```

Exercise 5

1. Assume that X is a hypergeometric random variable with $N = 25$, $S = 3$, and $n = 4$. Calculate $P(X = 0)$, $P(X = 1)$, and $P(X \leq 1)$.

Answer

$P(X = 0) = 0.58$, $P(X = 1) = 0.37$, and $P(X \leq 1) = 0.94$.

In R we can use the ***dhyper()*** *function*

```r
dhyper(0,3,22,4)
```

```
[1] 0.5782609
```

once more for the second probability:

```r
dhyper(1, 3, 22, 4)
```

```
[1] 0.3652174
```

*For the last probability we can add the previous probabilities or use the **`phyper()`** function:*

```r
phyper(1, 3, 22, 4)
```

```
[1] 0.9434783
```

2. Compute the probability of at least eight successes in a random sample of 20 items obtained from a population of 100 items that contains 25 successes. What are the expected value and standard deviation of the number of successes?

Answer

The probability is 0.078.

*In R we use the **`dhyper()`** function once more:*

```r
phyper(7, 25, 75, 20, lower.tail = F)
```

```
[1] 0.07767093
```

3. For 1 dollar a player gets to select six numbers for the base game of Powerball. In the game, five balls are randomly drawn from 59 consecutively numbered white balls. One ball, called the Powerball, is randomly drawn from 39 consecutively numbered red balls. What is the probability that a player is able to match two out of five randomly drawn white balls? What is the probability of winning the jackpot?

Answer

The probability of matching two white balls is 5. Winning the jackpot is extremely unlikely! A probability of 0.00000000512. It is more likely to be struck by lightning according to the CDC.

*In R use the **`dhyper()`** function:*

```
dhyper(2, 5, 54, 5)
```

```
[1] 0.04954472
```

For the jackpot we first calculate the probability of getting all of the white balls.

```
options(digits = 5,scipen=999)
dhyper(5, 5, 54, 5)
```

```
[1] 0.00000019974
```

Now the probability of getting the Powerball.

```
dhyper(1, 1, 38, 1)
```

```
[1] 0.025641
```

Since the two events are independent, we can multiply them to find the probability of a jackpot.

```
dhyper(5, 5, 54, 5)*dhyper(1, 1, 38, 1)
```

```
[1] 0.0000000051217
```

10 Probability III

Studying continuous probability distributions is crucial for understanding and predicting outcomes in uncertain scenarios where variables can take on any value within a range, rather than being limited to distinct, countable outcomes. These distributions are widely used in fields such as engineering, economics, natural sciences, and machine learning to model phenomena like time to failure, stock prices, rainfall amounts, or human heights. By mastering continuous probability distributions, analysts and decision-makers can assess uncertainties, estimate probabilities of events, and develop data-driven strategies to address real-world challenges. Below, we introduce some popular continuous distributions and their practical applications.

10.1 Continuous Random Variables

Continuous random variables are characterized by their probability density function $f(x)$. The probability density function does not directly provide probabilities!

The probability of a continuous random variable assuming a single value is zero. Instead, probabilities are defined for intervals. These are calculated by areas under the PDF curve (integral).

10.2 Uniform Distribution

The **uniform** probability density function is given by:

$$f(x) = \frac{1}{b-a}$$

when $a \leq x \leq b$ and 0 otherwise. Here b is the upper limit of the distribution and a is the lower limit. The **expected value** of the uniform distribution is:

$$E(x) = \frac{a+b}{2}$$

The **variance** of the uniform distribution is

$$var(x) = \frac{(b-a)^2}{12}$$

Example: Consider the travel time between NY and CHI. Typically, this trips can take anywhere from 120-140 minutes. The specific time of a particular flight is unpredictable, but assumed to be anywhere between this interval. We assume a uniform distribution with upper limit of 140 and lower limit of 120. The probability that the flight take 122 minutes is zero since the interval has an infinite amount of possible values. The probability of the flight taking 130 minutes or less is 0.5.

$$f(x <= 130) = \frac{130 - 120}{20} = 0.5$$

We can confirm this result in R using the `punif()` function. Below is the code:

```
punif(130,120,140)
```

```
[1] 0.5
```

10.3 Normal Distribution

The **normal** PDF is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}(\frac{x-\mu}{\sigma})^2}$$

where μ is the mean, σ is the standard deviation, π is 3.1415... , and e is 2.7282... . The normal distribution has the following properties:

- It is symmetrical about the mean μ.
- The mean is at the middle and divides the area of the distribution into halves.
- The total area under the curve is equal to 1.
- The distribution is completely determined by its mean and standard deviation.

The **standard normal** has a mean of 0 and a standard deviation of 1. Otherwise, it has the exact same properties as the normal distribution.

Example: Dr Tires is planning on offering a mileage guarantee on each set of tires they sell. They are considering a 40,000 mile guarantee. Given that the mean tire mileage is 36,500 miles with a standard deviation of 5,000 miles, we can estimate that the probability that a tire lasts more than 40,000 miles 24.2%. We can confirm this using R.

```r
1-pnorm(40000,36500,5000)
```

```
[1] 0.2419637
```

10.4 Exponential Distribution

The **exponential distribution** is useful in computing probabilities for the time it takes to complete a task. It describes the time between events in a Poisson process.

The probability density function is given by

$$f(x) = 1/\mu \times e^{\frac{-x}{\mu}} = \lambda e^{-\lambda x}$$

Example: Let x represent the loading time for a truck at the Dock Dash loading dock. If the average loading time is 15 minutes, the probability that loading a truck will take 6 minutes or less is 32.97%.

```r
pexp(6,1/15)
```

```
[1] 0.32968
```

We can also estimate that the probability that it takes between 6 and 18 minutes is 36.91%.

```r
pexp(18,1/15)-pexp(6,1/15)
```

```
[1] 0.3691258
```

10.5 Triangular Distribution

The **triangular distribution** is characterized by a single mode (the peak of the distribution) and two boundaries. It is often used in situations where the lower and upper bounds of a potential outcome are known, but the exact likelihood of the outcome is uncertain.

The probability density function is given by

$$f(x) = \frac{2(x-a)}{(b-a)(c-a)}$$

for $a \leq x < c$

$$f(x) = \frac{2}{(b-a)}$$

for $x = c$ and

$$f(x) = \frac{2(b-x)}{(b-a)(b-c)}$$

for $c < x \leq b$, and $f(x) = 0$ otherwise. The **expected value** of the distribution is

$$E(x) = \frac{a+b+c}{3}$$

The **variance** of the triangular distribution is

$$var(x) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$$

Example: Bite Bliss is planning a new store in Williamsburg. It is estimated that the minimum weekly sales are 1000 and the maximum is 6000. They also estimate that the most likely outcome is around 3000. The probability that future sales will be between 2000 and 2500 is 12.5%.

```
library(extraDistr)
ptriang(2500,1000,6000,3000)-ptriang(2000,1000,6000,3000)
```

```
[1] 0.125
```

10.6 Useful R Functions

To calculate the density of continuous random variables use the `dunif()`, `dnorm()`, and `dexp()` functions. For the triangular distribution use the extraDistr package and the `dtriang()` function.

To calculate probabilities of continuous random variables use the `punif()`, `pnorm()`, `pexp()`, and `ptriang()` functions.

To calculate quartiles of continuous random variables use the `qunif()`, `qnorm()`,`qexp()`, and `qtriang()` functions.

To calculate generate random variables based on continuous random variables use the `runif()`, `rnorm()`, `rexp()`, and `rtriang()` functions.

10.7 Exercises

The following exercises will help you practice some probability concepts and formulas. In particular, the exercises work on:

- Calculating probabilities for continuous random variables.
- Calculating the expected value and standard deviation.
- Applying the uniform, normal, and exponential distributions.

Answers are provided below. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results with a calculator or R.

1. A random variable X follows a continuous uniform distribution with minimum of -2 and maximum of 4. Determine the height of the density function $f(x)$, the mean, the standard deviation, and calculate $P(X \leq -1)$.

Answer

The height of the density function $f(x) = 0.1667$, the mean is 1, standard deviation is 1.73, and $P(X \leq -1) = 0.1667$.

$f(x)$ can be easily estimated by using the formula of the continuous uniform random variable. $f(x) = \frac{1}{b-a}$. Using R as a calculator we find:

```
1/(4-(-2))
```

```
[1] 0.1666667
```

The mean is given by $\mu = \frac{a+b}{2}$. In R we determine that the mean is:

```
(-2+4)/2
```

```
[1] 1
```

The standard deviation is $\sigma = \sqrt{\frac{(b-a)^2}{12}}$. Using R we find:

```r
sqrt((4-(-2))^2/12)
```

```
[1] 1.732051
```

Finally, we can find the probability of Z being less than -1 by using the `punif()` function:

```r
punif(-1,-2,4)
```

```
[1] 0.1666667
```

2. Your internet provider will arrive sometime between 10:00 am and 12:00 pm. Suppose you have to run a quick errand at 10:00 am. If it takes 15 minutes to run the errand, what is the probability that you will be back before the internet provider arrives? What if you take 30 minutes?

Answer

The probability that you will arrive on time is 0.875. If the time of the errand is 30 minutes, then the probability goes down to 0.75.

There is a 120 minute interval in which the IP can arrive. The density function is given by $f(x) = 1/120$. Using R we can find $P(X > 15)$:

```r
punif(15,0,120,lower.tail=F)
```

```
[1] 0.875
```

Once more we can find $P(X > 30)$:

```r
punif(30,0,120,lower.tail=F)
```

```
[1] 0.75
```

Exercise 2

1. A random variable Z follows a standard normal distribution. Find $P(-0.67 \leq Z \leq -0.23)$, $P(0 \leq Z \leq 1.96)$, $P(-1.28 \leq Z \leq 0)$ and $P(Z > 4.2)$.

Answer

$P(-0.67 \leq Z \leq -0.23) = 0.158$, $P(0 \leq Z \leq 1.96) = 0.475$, $P(-1.28 \leq Z \leq 0) = 0.4$ and $P(Z > 4.2) \approx 0$.

Use the `pnorm()` function to find the probabilities. $P(-0.67 \leq Z \leq -0.23)$:

```
pnorm(-0.23)-pnorm(-0.67)
```

```
[1] 0.157617
```

$P(0 \leq Z \leq 1.96)$

```
pnorm(1.96)-pnorm(0)
```

```
[1] 0.4750021
```

$P(-1.28 \leq Z \leq 0)$

```
pnorm(0)-pnorm(-1.28)
```

```
[1] 0.3997274
```

$P(Z > 4.2)$

```
options(scipen=999)
pnorm(4.2,lower.tail = F)
```

```
[1] 0.00001334575
```

2. Let Y be normally distributed with $\mu = 2.5$ and $\sigma = 2$. Find $P(Y > 7.6)$, $P(7.4 \leq Y \leq 10.6)$, a y such that $P(Y > y) = 0.025$, and a y such that $P(y \leq Y \leq 2.5) = 0.4943$.

Answer

$P(Y > 7.6) = 0.005386$, $P(7.4 \leq Y \leq 10.6) = 0.0071$, a y such that $P(Y > y) = 0.025$ is 6.42, and a y such that $P(y \leq Y \leq 2.5)$ is -2.56.

Let's use once more the `pnorm()` *function in R.*

$P(Y > 7.6)$

```
pnorm(7.6,2.5,2,lower.tail = F)
```

```
[1] 0.005386146
```

$P(7.4 \leq Y \leq 10.6)$

```
pnorm(10.6,2.5,2)-pnorm(7.4,2.5,2)
```

```
[1] 0.007117202
```

y such that $P(Y > y) = 0.025$

```
qnorm(0.025,2.5,2,lower.tail = F)
```

```
[1] 6.419928
```

y such that $P(y \leq Y \leq 2.5) = 0.4943$. Note that 2.5 is the mean. Hence we are looking for a y that has $0.5 - 0.4943 = 0.0057$ on the left:

```
qnorm(0.0057,2.5,2)
```

```
[1] -2.560385
```

3. Assume that football game times are normally distributed with a mean of 3 hours and a standard deviation of 0.4 hour. What is the probability that the game lasts at most 2.5 hours? Find the maximum value for a game to be in the bottom 1% of the distribution.

Answer

The probability is 10.56%. A game lasting no more than 2.069 hours would be in the bottom 1%.

Let's use `pnorm()` *once more in R.*

```r
pnorm(2.5,3,0.4)
```

```
[1] 0.1056498
```

For the threshold we can use `qnorm()`*:*

```r
qnorm(0.01,3,0.4)
```

```
[1] 2.069461
```

Exercise 3

1. Random variable S is exponentially distributed with mean of 0.1. What is the standard deviation of S? What is $P(0.10 \leq S \leq 0.2)$?

Answer

The standard deviation is equal to the mean 0.1. $P(0.10 \leq S \leq 0.2) = 0.2325$

Let's use `pexp()` *in R:*

```r
pexp(0.2,rate = 10)-pexp(0.1,rate = 10)
```

```
[1] 0.2325442
```

2. A tollbooth operator has observed that cars arrive randomly at a rate of 360 cars per hour. What is the mean time between car arrivals? What is the probability that the next car will arrive within ten seconds?

Answer

The mean time between car arrivals is $1/360 = 0.002778$. The probability that the next car will arrive within the next 10 seconds is 0.6321.

Once more we use `pexp()` *in R:*

```r
pexp(1/360,360)
```

```
[1] 0.6321206
```

11 Inference I

Statistical inference is a cornerstone of modern business decision-making. It enables companies to extract meaningful insights from data amid uncertainty and complexity. Businesses can predict customer behavior, optimize marketing strategies, and assess risks by drawing reliable conclusions from sample data. In this chapter, we explore how we can draw insights from the population using sample data.

11.1 Statistical Inference

The power of statistical inference lies in allowing us to learn about a population by studying the properties of the sample. The process relies on estimating a sample statistic, like the sample mean (i.e., the sample statistic). The sample mean gives us a good guess about the population mean (i.e., the population parameter), as long as it comes from a fair and random selection. Certain qualities or properties of the sample mean enable us to make sound inference of the population parameter. Below, we study the properties of unbiasedness, consistency, and efficiency (asymptotic normality).

11.2 Unbiasedness

Simply put, unbiasedness states that expected value of the sample means is equal to the population mean. Formally,

$$E[\bar{X}] = \mu$$

To convince ourselves of this property, consider the data below as being the population.

x_1	x_2	x_3	x_4
3	12	18	30

In this example, we have access to the entire population, consisting of the four numbers {3, 12, 18, 30}. This allows us to calculate the population mean as:

$$\mu = \frac{3+12+18+30}{4} = 15.75$$

In practice, population data is rarely available, and we rely on samples to estimate population parameters. Consider taking a random sample of three elements from this population. The table below lists all possible samples of size three, along with their corresponding sample means.

Sample	Sample Outcome	$\bar{x}$
1	{3,12,18}	$\bar{x}_1 = 11$
2	{3,12,30}	$\bar{x}_2 = 15$
3	{3,18,30}	$\bar{x}_3 = 17$
4	{12,18,30}	$\bar{x}_3 = 20$

Upon examining the table, it is evident that none of the sample means $(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)$ equals the population mean ($\mu = 15.75$). This discrepancy is known as *sampling error*. Formally, sampling error is defined as the difference between a sample statistic, such as the sample mean, and the corresponding population parameter:

$$Sampling\ Error = \bar{x} - \mu$$

Sampling error is not the result of mistakes or biases in the sampling process; rather, it is a natural consequence of sampling variability, as illustrated by the varying sample means in the table. Intuitively, the sampling error would be zero if the sample included the entire population, but it could be substantial with a very small sample, such as a single element. Generally, sampling error tends to decrease as the sample size increases, as larger samples provide more accurate estimates of the population parameter.

> **Note**
>
> To explore how the sampling error changes with the sample size press here.

A critical insight from the table emerges when we calculate the mean of the sample means. Summing the sample means and dividing by the number of samples:

$$\frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \bar{x}_4}{4} = \frac{11+15+17+20}{4} = \frac{63}{4} = 15.75$$

Remarkably, this value is exactly equal to the population mean, $\mu = 15.75$. This is not a coincidence but a demonstration of a fundamental unbiasedness property. In statistical terms, an estimator is unbiased if its expected value equals the true population parameter. Once more, this is expressed as:

$$E[\bar{X}] = \mu$$

In this context, the mean of all possible sample means represents the expected value of the sample mean, and it precisely matches the population mean. This property underscores that, despite individual sample means deviating from μ due to sampling error, the sample mean is, on average, an accurate estimator of the population mean.

Far from being disconnected, the sample mean and population mean are strongly related through the concept of unbiasedness. While any single sample mean may differ from μ due to sampling variability, the unbiasedness property ensures that, over all possible samples, the sample mean centers on the true population mean. This relationship is the cornerstone of statistical inference, allowing us to use sample data to make reliable inferences about unknown population parameters. In practice, when population data is unavailable, the unbiasedness of the sample mean provides confidence that our estimates are, on average, correct, bridging the gap between sample statistics and population parameters.

11.3 Consistency

The property of consistency states that as the sample size increases, the sample mean converges in probability to the population mean. As we saw in the table above, different samples yield different sampling errors, with some errors being larger than others. This variability is random and depends on the specific sample drawn. Consistency ensures that larger samples produce more accurate estimates, meaning the probability that the sample mean deviates from the population mean μ by more than any fixed amount ϵ approaches zero as the sample size grows.

Example: Consider the population {3, 12, 18, 30}, with a population mean $\mu = 15.75$. To illustrate consistency, let's compare the sampling errors for samples of size two, size three, and size four (the entire population).

Below is a table listing all possible samples of size two, along with their sample means and sampling errors:

Sample	Sample Outcome	$\bar{x}$	$\bar{x} - \mu$
1	{3,12}	$\bar{x}_1 = \frac{3+12}{2} = 7.5$	-8.25
2	{3,18}	$\bar{x}_2 = \frac{3+18}{2} = 10.5$	-5.25
3	{3,30}	$\bar{x}_3 = \frac{3+30}{2} = 16.5$	0.75
4	{12,18}	$\bar{x}_4 = \frac{12+18}{2} = 15$	-0.75
5	{12,30}	$\bar{x}_5 = \frac{12+30}{2} = 21$	5.25
6	{18,30}	$\bar{x}_6 = \frac{18+30}{2} = 24$	8.25

The sampling errors range from −8.25 to 8.25, with an average absolute error of:

$$\frac{|-8.25|+|-5.25|+|0.75|+|-0.75|+|5.25|+|8.25|}{6}=\frac{8.25+5.25+0.75+0.75+5.25+8.25}{6}=4.75$$

Similarly, the average absolute error for a sample size of three is given by:

$$\frac{|-4.75|+|-0.75|+|1.25|+|4.25|}{4}=\frac{4.75+0.75+1.25+4.25}{4}=2.75$$

The sampling error for the population is:

$$\bar{x}-\mu=15.75-15.75=0$$

11.4 Precision

In simple terms, the precision property of the sample mean ensures that as the sample size grows, the sample mean becomes a more reliable and tightly clustered estimate of the population mean. This follows from the fact that the standard deviation of the sample means or standard error ($\sigma_{\bar{x}}=\sigma/\sqrt{n}$) is lower than the population standard deviation (σ).

Let's consider once more the population data below:

x_1	x_2	x_3	x_4
3	12	18	30

The population variance is given by:

$\sigma^2=\frac{(3-15.75)^2+(12-15.75)^2+(18-15.75)^2+(30-15.75)^2}{4}=96.1875$

For a sample size of $n=3$, the variance of the sample means, adjusted for a finite population of size $N=4$, is:

$\frac{\sigma^2}{n}\times\frac{(N-n)}{(N-1)}=\frac{96.1875}{3}\times\frac{(4-3)}{(4-1)}=32.0625\times\frac{1}{3}=10.6875$

Notice how the sample size n inversely affects the variance of the sample mean. As the sample size increases, the variance of the sample mean decreases, leading to more precise estimates. The equation above may not be immediately intuitive, so we verify it by calculating the variance using the sample means directly. Recall the sampling distribution data for samples of size $n=3$:

Sample	Sample Outcome	$\bar{x}$	$(\bar{x}-\mu)^2$
1	{3,12,18}	$\bar{x}_1 = 11$	22.5625
2	{3,12,30}	$\bar{x}_2 = 15$	0.5625
3	{3,18,30}	$\bar{x}_3 = 17$	1.5625
4	{12,18,30}	$\bar{x}_4 = 20$	18.0625

The variance of the sample means is:

$\text{Var}(\bar{x}) = \frac{22.5625+0.5625+1.5625+18.0625}{4} = 10.6875$

As a final note, most population data we consider is sufficiently large so that we don't have to adjust the measure of standard error. The standard error is then given by $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ and not $\frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{(N-n)}{(N-1)}}$.

In sum, the property of precision demonstrates that the variance of the sampling distribution is less than the population variance, and more importantly, as the sample size increases, the sample mean becomes an increasingly accurate estimator of the population mean.

11.5 Asymptotic Normality and the Central Limit Theorem

We have explored the desirable properties of estimators. However, to construct confidence intervals or perform hypothesis tests, we need to know (or approximate) the *shape* of the sampling distribution of an estimator, not just its center and spread. For small samples from non-normal (or binary) populations, this shape can be unpredictable and inconvenient.

This is where the **Central Limit Theorem (CLT)** emerges as one of the most profound and practical results in statistics. The CLT states that, regardless of the underlying population distribution (provided it has finite variance), the standardized sample mean becomes approximately normally distributed as the sample size (n) grows large:

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

This convergence to normality—building directly on the precision gains we have seen—enables us to use the familiar normal (or z) table for probabilistic statements, even when the original data are skewed, discrete, or non-normal. The same principle applies to the sample proportion ($\hat{p}$), where the sampling distribution of the sample proportion is also normally distributed for a sufficiently large (n).

$$\frac{\hat{p}-p}{\sqrt{p(1-p)/n}} \approx N(0,1)$$